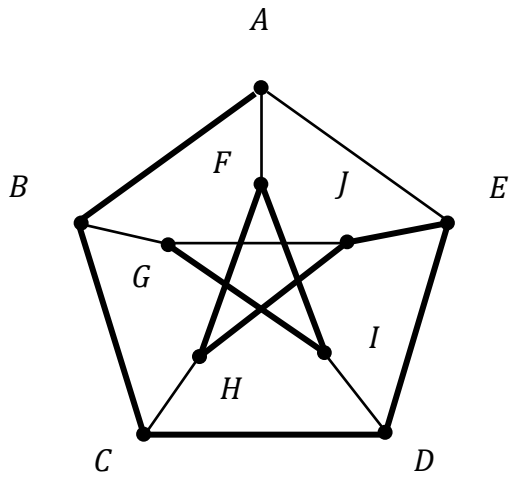
RZA BASHIROV

DISCRETE MATHEMATICS



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DISCRETE MATHEMATICS, SECOND EDITION
Bashirov, Rza

Includes bibliographical references and solutions to the chapter-ending exercises.

This textbook is appropriate for teaching a freshman-level course in discrete mathematics for students majoring in basic sciences, engineering and technology. The textbook is designed in accordance with the course material that we follow up for many years at the Eastern Mediterranean University (EMU) and has very simple language for readers to understand the core concepts of discrete mathematics such as sets, relations, functions, logic, mathematical induction, recursion, counting, graphs and trees. The end of the chapter exercises are selected from set of past examination papers and, as I expect, would help students understand a concept soon after learning it and get well-prepared to examinations.

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PREFACE TO THE FIRST EDITION

This is a concise textbook for a Discrete Mathematics course at an introductory level for students majoring in mathematics, computer science, engineering and technology. This textbook introduces mathematical structures and methods used to study discrete processes as opposed to continuous processes. Discrete Mathematics is a prerequisite of many courses ranging from abstract algebra, geometry, topology, combinatorics, logic and mathematical modeling to logic design and operational research. To increase readability for students specializing in science, engineering and technology the present textbook focuses on the interplay between disciplines. The material covers such concepts as basic set theory, functions, relations, recursion, counting, propositional logic and Boolean algebra, graphs and trees. The textbook also introduces mathematical methods of proof such as proof by induction, proof by truth table, proof by Venn diagram, proof by pigeonhole principle, etc.

This textbook is based on my lectures in Discrete Mathematics that I have given for many years at the Eastern Mediterranean University. The textbook consists of ten chapters presented according to the logical precedence of the topics. For a better understanding of the material, the textbook offers 47 examples, 67 solved problems and 161 chapter-ending exercises to be solved during or after the classes. Appendix B provides answers and solutions to the chapter-ending exercises presented in this textbook. Textbook contains over 80 figures and illustrations designed to stand for reinforcement of the provided content.

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CHANGES IN THE SECOND EDITION

The second edition is a review with changes based on feedback from students, instructors, and author insights. In the present edition, I corrected misprints and rephrased ambiguous explanations throughout the book. Over 30 new chapter-ending exercises, both routine and challenging, have been added to clarify material where students often have difficulty. Text on highly relevant applications from check for syntax errors in computer programs to organization of recursive algorithms, have been added. The result is a new edition that offers an improved content of topics which will help students more easily learn the material. I hope that instructors and students will closely examine this new edition to discover how it might meet their needs.

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ACKNOWLEDGMENTS

I want to thank my family for tolerating my incessant disappearances into my home office. This book benefited from the comments and suggestions of numerous individuals. I acknowledge with gratitude the present teaching staff of Discrete Mathematics for their comments on the text. I have deliberately included selected questions of past examinations in most exercise sets. In this sense, I am thankful to the teaching staff of Discrete Mathematics course who have really contributed to the design of examination questions over the past five years. I am grateful to members of the Eastern Mediterranean University Press Executive Board for their assistance and to the staff at EMU Printing House who have helped with this project. I thank those students who have read the draft of this text and reported on typos here and there throughout the text. I wish to thank anonymous reviewers invited on this project who gave me several helpful suggestions and criticism. I will sure find it extremely helpful to read your further comments and suggestions on various aspects of this book.

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Chapter 1 Sets

A set is an unordered collection of objects. Each object in a set is called an element of the set. The fact that a is an element of A can be expressed by writing $a \in A$ and the fact that a is not an element of A is expressed by writing $a \notin A$. Usually, we use capital English letters to represent sets and lower-case letters to introduce the elements.

One way to represent a set is by listing its elements. We include the elements of a set within a pair of parentheses. For instance, $\{a, b, c, d\}$ stands for a set of four elements being a, b, c and d.

Example 1.1 We can represent the set of vowels in English alphabet as follows:

$$V = \{a, e, i, o, u\}.$$

Example 1.2 We can write the set of odd positive integers that are less than 10 as follows:

$$O = \{1,3,5,7,9\}.$$

Example 1.3 Although sets are usually composed of objects of the same type, in general, a set is a collection of objects of different types. For example, the set $\{a, 2, \#, Montreal\}$ is made of a letter, an integer, a symbol, and a city name.

Sometimes we represent a set by a common property or properties of its elements. If this is the case, a common property must characterize all elements of the set. For instance, we can represent the set in Example 1.2 by a common property of its elements as follows:

$$O = \{x \in Z^{+} | x \text{ is odd integer}, x < 10\}$$

There are two advantages of representing by common properties. Firstly, this method allows us to represent a set in a compact way. Secondly, in such a way we can describe not only finite but also infinite sets, that is, sets with infinitely many elements.

The following sets play important role in discrete mathematics and its application:

$N = \{1,2,3,4, \dots\}$	natural numbers
$Z = \{\dots, -2, -1, 0, 1, 2, \dots\}$	integers
$Z^{-} = \{-1, -2, -3, \dots\}$	negative integers
$Z_0^{+} = \{0,1,2,3, \dots\}$	positive integers including 0
$Q = \left\{x \middle| x = \frac{p}{q}, q \neq 0, p \in Z, q \in Z\right\}$	rational numbers
R	real numbers

To define all real numbers properly requires considerable mathematical maturity. For our purposes, we think of real numbers as numbers that have decimal expansions of the form $a.a_1a_2a_3 \dots$ where a is an integer and $a_1, a_2, a_3, \dots$ are 0 to 9 integers. Any rational number in decimal format either terminates or repeats. For example, $\frac{10}{4} = 2.5$ and $\frac{10}{3} = 3.3(3)$. There are meanwhile real numbers for which these rules are not correct. Irrational numbers such as $\sqrt{2}$ and $\sqrt{5}$ are the ones whose decimal fractions neither terminate nor repeat. Set of real numbers is composed of infinite number rational numbers and infinite number of irrational numbers.

Below we remind interval notation for real numbers. For $a < b$; $a, b \in R$, the interval notation for real numbers is as follows:

$$[a, b] = \{x \in R | a \leq x \leq b\},$$
$$(a, b] = \{x \in R | a < x \leq b\},$$

$$[a, b) = \{x \in R | a \leq x < b\},$$
$$(a, b) = \{x \in R | a < x < b\}.$$

$[a, b]$ is a **closed** interval from a to b; (a, b) is **open** interval from a to b; $(a, b]$ and $[a, b)$ are **semi open** (or **semi closed**) intervals from a to b.

Definition 1.1 (Set equality) Two sets are equal if and only if these sets are composed of the same elements. The fact that A and B are equal sets is expressed by writing $A = B$.

Example 1.4 {1,3,5} and {3,5,1} are equal sets.

Definition 1.2 (Empty set) The set with no element in it is called empty set and written as $\emptyset$.

Definition 1.3 (Singleton) A set with a single element is called a singleton.

1.1 Venn Diagrams

We schematically illustrate sets by Venn diagrams, geometric shapes named after John Venn[1]. A set that contains all possible elements is called universal set, written U, and represented by a rectangle. We represent all sets within the universal set by circles embedded in rectangle.

---

[1] John Venn studied in London and received his degree in mathematics from Caius College, Cambridge in 1857. He was elected as a member of Caius College, and this membership continued until the end of his life. Venn's book on symbolic logic clarified the ideas originally put forward by Boole. In this book, Venn set forth the systematic development of a method using geometric figures, called Venn diagrams, that are used today in the analysis of mathematical theses and to show the relationships between sets. In addition to his work on symbolic logic, Venn contributed to probability theory and described these contributions in the widely used textbook on the subject.

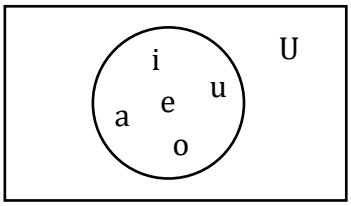


Figure 1.1 Set of vowels represented by Venn diagram.

Example 1.5 Consider a set of vowels in English alphabet. In this example, all English letters is the universal set and thereby we represent it by a rectangle, and the set of vowels is embedded in the universal set, which we represent by a circle. Figure 1.1 illustrates the idea behind of Example 1.5.

1.2 Subsets

Sometimes we need to deal with a small part of a large set. In such cases, it is more convenient to focus on local fragment of the set instead of dealing with the whole set.

Definition 1.4 (Subset) A is a subset of B, written $A \subseteq B$, if and only if, all elements of A are also elements of B.

Example 1.6 Set of odd positive integers is a subset of the set of positive integers, that is,

$$\{1,3,5,\dots\} \subseteq \{1,2,3,4,5,\dots\}.$$

Any non-empty set S has at least two subsets that are empty set and the set itself. Expressed in other words, $\emptyset \subseteq S$ and $S \subseteq S$. If A is a subset of B such that $A \neq B$ then we use proper subset notation, written $A \subset B$. If $A \subseteq B$ then either $A = B$ (A is improper subset of B) or $A \subset B$ (A is proper subset of B).

1.3 Set Cardinality

We often use sets in counting problems and therefore need to know the number of their elements.

Definition 1.5 (Cardinality number) Let S be a set. If S is a finite set composed of n objects then its cardinality number is n, written $|S| = n$. Two sets of the same cardinality if they have equal number of elements.

Example 1.7 Let A be the set of odd positive integers that are less than 10. Then the elements of A are 1, 3, 5, 7 and 9, from what we conclude that $|A| = 5$.

Example 1.8 Let T be the set of letters in Turkish alphabet. Since there are 29 letters in Turkish alphabet, $|T| = 29$.

Example 1.9 The empty set is the set with no element in it, that is, $|\emptyset| = 0$.

Example 1.10 Set of positive integers is an example to infinite set.

Problem 1.1 Check whether each of the following expressions is correct.

a) $\emptyset \in \emptyset$
b) $\emptyset \in \{\emptyset\}$
c) $\emptyset \subseteq \emptyset$
d) $\emptyset \subseteq \{\emptyset\}$
e) $\{\emptyset\} \in \{\emptyset\}$
f) $\{\emptyset\} \subseteq \{\emptyset, \{\emptyset\}\}$
g) $\{\emptyset\} \in \{\emptyset, \{\emptyset\}\}$

Solution

a) This is an incorrect statement since $\emptyset$ is not element in $\emptyset$.
b) This is a correct statement since $\emptyset$ is an element in $\emptyset$.

c) This is a correct statement since $\emptyset$ is a subset of any set.
d) This is a correct statement since $\emptyset$ is a subset of any set.
e) This is not correct statement since $\{\emptyset\}$ is not element of $\{\emptyset\}$.
f) This is a correct statement since $\{\emptyset\}$ is a subset of $\{\emptyset, \{\emptyset\}\}$.
g) This is a correct statement since $\{\emptyset\}$ is an element in $\{\emptyset, \{\emptyset\}\}$.

1.3 Power sets

Definition 1.6 (Power set) Given set S, its power set, written $P(S)$, is the set of all its subsets.
Problem 1.2 Write power set of $\{0,1,2\}$.
Solution $P(S) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0,1\}, \{0,2\}, \{1,2\}, \{0,1,2\}\}$
Power set of empty set has a single element which is empty set itself, that is, $P(\emptyset) = \{\emptyset\}$. If a set has n elements, then its power set contains 2^n elements.

1.4 Set operations

Definition 1.7 (Union of sets) Let A and B be two sets. The union of A and B, written $A \cup B$, is the set with elements either in A or in B or both in A and B.

If x is in the union of two sets A and B then x belongs to A or B or both of A and B and vice versa. Expressed in other way,

$$A \cup B = \{x | x \in A \text{ and/or } x \in B\}.$$

Let A and B be two sets. The union of A and B is represented by gray segment in Figure 1.2.

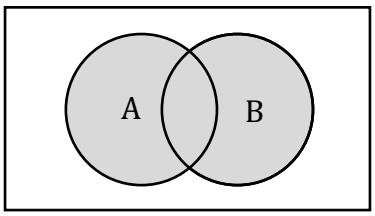


Figure 1.2 Venn diagram representation of $A \cup B$.

Example 1.11 The union of {1,3,5} and {1,2,3} is {1,2,3,5}.

Definition 1.8 (Intersection of sets) Let A and B be sets. The intersection of A and B, written $A \cap B$ is made of common elements of both sets. We write the intersection of A and B by $A \cap B = \{x | x \in A \text{ and } x \in B\}$ and represent it by gray segment in Figure 1.3.

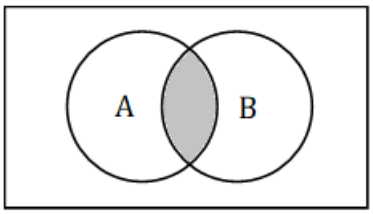


Figure 1.3 Venn diagram representation of $A \cap B$.

Example 1.12 Let $A = \{1,3,5,7,9\}$ and $B = \{2,4,6,8,10\}$. Then $A \cap B = \emptyset$. Such sets are called disjoint sets.

Definition 1.9 (Difference of sets) Let A and B be two sets. A difference B, written $A - B$, is the set of elements in A that are not in B. A difference B is also called complement of A relative to B.

Set difference between A and B can also be expressed by $A \backslash B$. Using set notation, we can write

$$A - B = \{x | x \in A \text{ and } x \notin B\}.$$

Example 1.13 $\{1,3,5\} - \{1,2,3\} = \{5\}$

Definition 1.10 (Set complement) Let U be the universal set. The complement of A, written $\bar{A}$, is the set of elements in U that are not in A. Consequently, the complement of A is $U - A$. It can be thought as $A \cup \bar{A} = U$ and $A \cap \bar{A} = \emptyset$. The complement of A is represented in Figure 1.4.

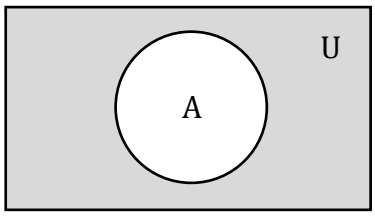


Figure 1.4 Venn diagram representation of $\bar{A}$.

Stated in another way, $x \in \bar{A}$ if and only if that $x \notin A$. We express this fact by writing

$$\bar{A} = \{x \in U | x \notin A\}.$$

In Figure 1.4, gray segment represents the complement of A

Definition 1.11 (Symmetric difference) The symmetric difference of A and B, written $A \oplus B$, is the set of elements in $A \cup B$ that do not belong to both A and B. Thus, $A \oplus B = (A - B) \cup (B - A)$. Gray segment in Figure 1.5 illustrates symmetric difference of A and B.

On the other hand, one can easily verify that $A \oplus B = (A \cup B) - (A \cap B)$. It must be noticed that $A \oplus B = B \oplus A$.

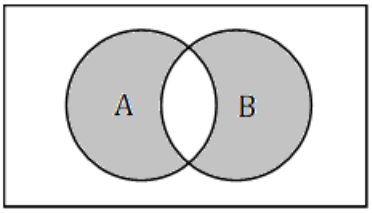


Figure 1.5 Venn diagram representation of $A \oplus B$.

1.6 Set identities

Table 1.1 Famous set identities	
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\bar{A})} = A$	Double complementation laws
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$\overline{A \cup B} = \bar{A} \cap \bar{B}$ $\overline{A \cap B} = \bar{A} \cup \bar{B}$	De Morgan's laws
$A \cup (A \cap C) = A$ $A \cap (A \cup C) = A$	Absorption laws
$A \cup \bar{A} = U$ $A \cap \bar{A} = \emptyset$	Complementation laws

Set identity establishes the equality between sets. We say two sets are equal if they have exactly same elements. We present the most famous set identities in Table 1.1.

We can prove set equalities or set identities using Venn diagrams.

Problem 1.3 Prove distributive law for sets.

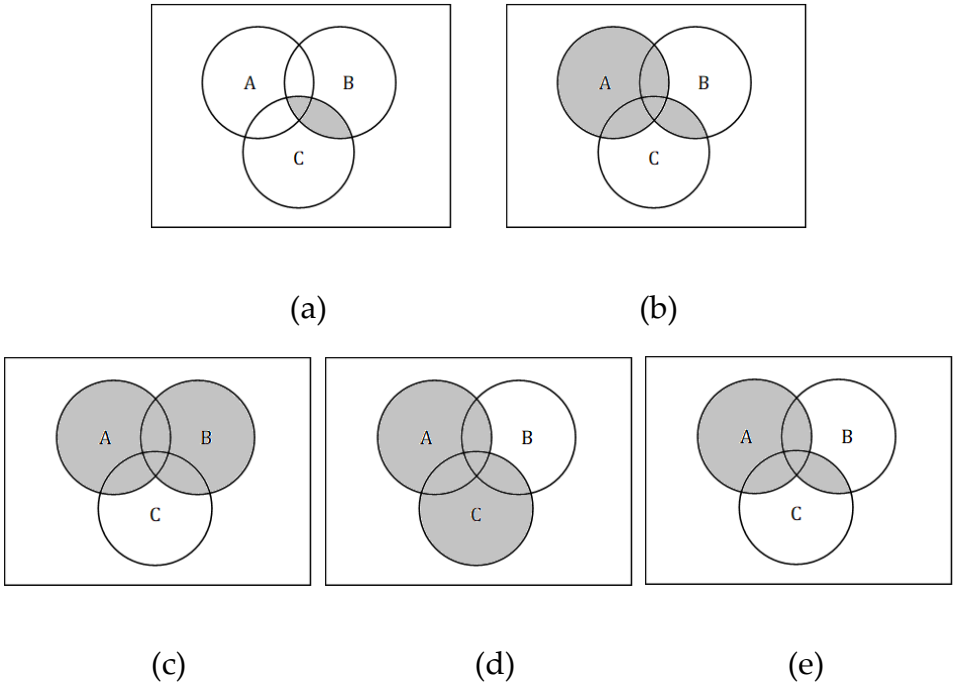


Figure 1.6 Proof of distributive law by use of Venn diagrams.

Proof Let A, B and C be sets. Let us show that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

The set identity has two sides: left and right. We create Venn diagram for the set represented in the left side and then for one represented in the right side. We show the sets $A \cup (B \cap C)$ and $(A \cup B) \cap (A \cup C)$ by Figure 1.6(b) and Figure 1.6(e), respectively. One can

easily see that gray segments in Figure 1.6(b) and Figure 1.6(e) are matching. Thus, the given set identity (distributive law) holds.

Exercises

1. Find the number of elements in $P(P(A))$ if A has 2 elements.
2. Let A be a set with 3 elements. Find the number of elements in $P(P(A))$.
3. Given C and D are sets, which of the following is equivalent to $C \backslash D$?

   a) $\bar{C} \cap D$
   b) $\bar{C} \cap \bar{D}$
   c) $C \cap \bar{D}$

4. Let $A = \{0,1,2,3,4,5,6,7,8\}$, $B = \{0,1,2,4,6,7,8,9\}$ and $C = \{2,5,7,8,9\}$. Find $(A - B) \oplus (B \cap C)$.
5. A is represented by interval notation as $(-3,100) \cap (-100,4)$. Find $\bar{A}$.
6. Which of the following is wrong?

   a) $\{\emptyset\} \in \{\emptyset\}$
   b) $\emptyset \in \{\emptyset\}$
   c) $\emptyset \subseteq \{\emptyset\}$
   d) $\{\emptyset\} \subseteq \{\emptyset\}$

7. Use Venn diagram to prove or disprove the set identity $(A \backslash B) \cap (C \backslash B) = (A \cap C) \backslash B$.
8. Use Venn diagrams to show that $A \cup (B \backslash A) = A \cup B$.

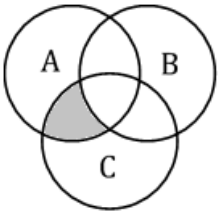
9. Let $U = \{1,2,3,\dots,12\}$ be the universal set and let $A = \{1,3,5,7,9,11\}$, $B = \{2,3,5,7,11\}$, $C = \{2,3,6,12\}$ and $D = \{2,4,8\}$ be the sets. Which of the following equals $\{1,9\}$?

   a) $(A \cup B) \cap C^c$
   b) $A \backslash B$
   c) $B \oplus D$
   d) $D \times C$

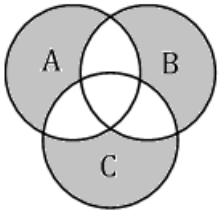
10. Let $U = \{1,2,3,4,5,6,7,8\}$ be the universal set and let $A = \{1,3,5,7\}$, $B = \{2,3,5,8\}$, $C = \{2,3,6,7\}$ be the sets. Which of the following is wrong?

   a) $(A\ U\ B) \cap \bar{C} = \{1,5,8\}$
   b) $P(A) + P(B) + P(C) = 48$
   c) $(A - C) \oplus (B \cap C) = \{1,2,3,5,7\}$
   d) $(A - B) \cap (C - B) = \{7\}$

11. Write the set which is represented by gray segment in the following figure.



12. Write the set which is represented by gray segment in the following figure.


A
B
C

Chapter 2 Relations

2.1 Cartesian Product

Definition 2.1 (Cartesian product of two sets) Cartesian product of the sets A and B, written $A \times B$, is a set of all ordered pairs (a, b) such that $a \in A$ ve $b \in B$. In other words, $A \times B = \{(a, b) | a \in A \text{ and } b \in B\}$. $A \times B$ is a set of ordered pairs and, therefore, it is important that first element appears in the first position and second element is in the second position. In general, for $a, b \in A$ (a, b) is not same as (b, a).

Problem 2.1 Let A be set of registered students in some university and B be set of courses offered in the same university. What is the meaning of $A \times B$?

Answer $A \times B$ is made of ordered pairs (a, b) where a and b stand for a student and a course, respectively. Hence, $A \times B$ represents all possible (student, course) assignments.

Problem 2.2 Given $A = \{1,2\}$ and $B = \{a, b, c\}$. Find $A \times B$.

Solution $A \times B$ is defined as follows:

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}.$$

Problem 2.3 Given A and B with $|A| = n$ and $|B| = m$. Find $|A \times B|$.

Solution $|A \times B| = n \times m$.

Definition 2.2 (Cartesian product of n sets) Let $A_1, A_2, \dots, A_n$ be the sets. Cartesian product of these sets is represented by $A_1 \times A_2 \times A_3 \times \cdots \times A_n$. The elements of the Cartesian product are ordered n-tuples $(a_1, \dots, a_n)$ where $a_i \in A_i$ for $i = 1,2, \dots, n$.

2.2 Binary Relations

Definition 2.3 (Binary relation from A to B) A binary relation R from set A to set B is a subset of $A \times B$.

Example 2.1 Let A be a set of students registered in certain university and B be the set of courses offered in the same university. If R is a relation defined from A to B then $(a, b) \in R$ is a collection of pairs of type (student, course). For instance, if Berk Öztürk is registered to COMP512 and Derya Aslan is registered to MATH163 then (Berk Öztürk, COMP512) and (Derya Aslan, COMP163) belong to R.

Example 2.2 Let $A = \{0,1,2\}$ and $B = \{a, b\}$ be sets. $\{(0, a), (0, b), (1, a), (2, b)\} \subseteq A \times B$ is a relation from A to B.

Definition 2.4 (Binary relation on A) A binary relation R on A is a relation from A to A. Stated in other words, a relation on A is a subset of $A \times A = A^2$.

Definition 2.5 (Reflexive relation) A binary relation R on A is reflexive if and only if $(a, a) \in R$ for all $a \in A$.

Definition 2.6 (Symmetric relation) A binary relation R on A is symmetric if and only if $(a, b) \in R$ then $(b, a) \in R$ for $a, b \in A$.

Definition 2.7 (Antisymmetric relation) A binary relation is antisymmetric if and only if both $(a, b) \in R$ and $(b, a) \in R$, then $a = b$ for $a, b \in A$.

Definition 2.8 (Transitive relation) A binary relation R on A is transitive if and only if both $(a, b) \in R$ and $(b, c) \in R$ implies $(a, c) \in R$ for $a, b, c \in A$.

Problem 2.4 Check whether each of the following relations is reflexive.

a) $R_1 = \{(x, y) \in R^2 | x \geq y\}$
b) $R_2 = \left\{(a, b) \in N^2 \middle| \frac{a}{b} \in N\right\}$
c) $R_3 = \{(x, y) \in R^2 | x^2 + y^2 > 0\}$

Solution

a) It is clear that $(x, x) \in R_1$ for all $x \in R$, since $x \geq x$ and therefore R_1 is reflexive.
b) It is obviously that $(a, a) \in R_2$ because $\frac{a}{a} = 1 \in N$ for all $a \in N$ and therefore R_2 is reflexive.
c) R_3 is not reflexive because $0^2 + 0^2 = 0 \ngtr 0$ and, thus, $(0,0) \notin R_3$.

Problem 2.5 Show that $R_1 = \{(x, y) \in R^2 | x^2 + y^2 = 1\}$ is a symmetric relation on R.

Solution The fact that $x^2 + y^2 = y^2 + x^2$ means that $(x, y) \in R_1$ implies $(y, x) \in R_1$ and, therefore, R_1 is symmetric.

Problem 2.6 Show that $R_2 = \{(x, y) \in Z \times Z | x - y \; is \; even\}$ is a symmetric relation on Z.

Solution One can easily observe that when $x - y$ is even $y - x$ is even as well. This is because $|x - y| = |y - x|$. Hence, $(x, y) \in R_2$ implies $(y, x) \in R_2$ and thereby R_2 is a symmetric relation.

Problem 2.7 Show that $R_3 = \{(x, y) \in R^2 | x^2 \geq y\}$ is not a symmetric relation.

Solution The proof is by providing a counter example. For example, $(2,1) \in R_3$ and $(1,2) \notin R_3$. This fact is good enough to claim that R_3 is not symmetric relation.

Problem 2.8 Prove that $R_4 = \{(x, y) \in R^2 | x \leq y\}$ is antisymmetric relation on R.

Solution If $x \neq y$ then either $x < y$ or $x > y$ which means that for two distinct x and y if $(x, y) \in R_4$, then $(y, x) \notin R_4$. Thus, R_4 is antisymmetric relation.

Example 2.3 $R = \{(1,2), (2,3), (3,3), (2,1)\}$ is not antisymmetric on $A = \{1,2,3\}$ because $(1,2) \in R$ and $(2,1) \in R$, but $1 \neq 2$.

Problem 2.9 Show that $\left\{(a, b) \in Z \times Z | \frac{a}{b} \in Z\right\}$ is transitive relation.

Solution For $a, b, c \in Z$ if $\frac{a}{b} \in Z$ and $\frac{b}{c} \in Z$ then $a = bn$ and $b = cm$ for $n, m \in Z$ and, therefore, $a = cmn$ and, consequently, $\frac{a}{c} \in Z$. Hence, this is a transitive relation.

2.3 Equivalence Relations

Two people can be of the same sex, be residents of the same region of a country, have the same eye color or live on the same street. We can continue with this list of rather natural relations between people. The above four relations are reflexive, symmetric and transitive.

Definition 2.9 (Equivalence relation) A binary relation R on A, which is reflexive, symmetric and transitive, is equivalence relation on A.

Example 2.4 Let A be the set of all people and let R_1 be the relation defined as

$$R_1 = \{(a, b) \in A \times A | a \text{ and } b \text{ have same parents}\}.$$

The above relation is reflexive, symmetric and transitive. R_1 is reflexive because any person has a pair of parents and therefore $(a, a) \in R_1$. R_1 is symmetric since if a and b have the same parents, then $(a, b) \in R_1$ and $(b, a) \in R_1$. R_1 is transitive because if $(a, b) \in R_1$ and $(b, c) \in R_1$ then a, b and c have the same parents, thus, $(a, c) \in R_1$.

Let R be an equivalence relation on a set A and $a, b \in A$. Notice that a and b are called equivalent elements. We interchangeably use notations aRb and $a \sim b$ to indicate that $(a, b) \in R$ for some equivalence relation R.

Example 2.5 Assume A is the set of all residents of a country, say the United States. a and b equivalent if and only if a and b are residents of the same state or district. Then aRb is an equivalence relation.

Example 2.6 The relation $\leq$ on the set of real numbers is not an equivalence relation on R. While it is reflexive, and transitive, it is not symmetric. Here is a counter example: $4 \leq 5$ and $5 \nleq 4$.

Definition 2.10 (Equivalence class) If ~ denotes an equivalence relation on a set A, the equivalence class of an element $a \in A$ is the set $\bar{a} = \{x \in A | x \sim a\}$.

Definition 2.11 (Partition) A partition of a set A is a collection of pairwise disjoint nonempty subsets of A whose union is A. Each subset is an equivalence class of A.

Example 2.7 Each of the following relations is an equivalence relation.

a) Turkey consists of 81 provinces.
b) Canada is partitioned into 10 provinces and 3 territories.
c) The human race is partitioned into classes by eye color.
d) A deck of 52 playing cards is partitioned into four suits.

Case a) is similar to Example 2.5, as a and b are equivalent if a and b are residents of the same province or territory. This is an equivalence relation, which defines partition of Canada into 10 provinces and 3 territories. In case b) we assume that a and b are equivalent, if a and b are residents of the same province. This is also equivalence relation, which determines a partition of Turkey into 81 provinces. In case c) relation is reflexive, symmetric and transitive and thereby it is equivalence relation. This relation defines a partition of human race into equivalence classes according to eye color. Finally, a deck of 52 playing cards is partitioned into four suits: spades, ♠, clubs, ♣, diamonds, ♢ and hearts, ♡, of 13 cards each.

2.4 Partial orders

Definition 2.12 (Partial order) Any reflexive, antisymmetric and transitive relation on A is a partial order on A. A pair of partial order $\preccurlyeq$ (notice that $\preccurlyeq$ is not same as $\leq$) and the set A, $(A, \preccurlyeq)$, is called is a partially ordered set or poset for short.

Example 2.8 Consider $\leq$ on the set of real numbers. This is a partial order on R since $a \leq a$ for all $a \in R$; $a \leq b$ and $b \leq a$ implies $a = b$; and $a \leq b$, $b \leq c$ implies $a \leq c$, thereby, this is reflexive, antisymmetric and transitive relation on R.

We often picture partial orders by means of a diagram named after Helmut Hasse[2]. We create Hasse diagram of a partially ordered set $(A, \preccurlyeq)$ according to the following rules.

- We associate a dot (or vertex) with each element of A;
- We position the dot for b higher than the dot for a if $a \preccurlyeq b$; and
- We draw a line from a to b if $a \prec b$ and there is no an intermediate c with $a \prec c \prec b$.

[2] Helmut Hasse (1898–1979) was born in Kassel, Germany. He began his university education at the University of Göttingen in 1918, moving to the University of Marburg in 1920 to work under the number theorist Kurt Hensel. During this time Hasse made fundamental contributions to algebraic number theory. He became Hensel's successor in Marburg, later becoming director of the famous mathematical institute in Göttingen in 1934, and taking a position at the University of Hamburg in 1950. He was noted for the openness and personal style of his lectures and made important contributions to number theory.

Example 2.9 Let $(A, |)$ be poset where $A = \{1,2,3,4,6,8,12\}$ and $|$ stands for operation "divides". Figure 2.1 illustrates Hasse diagram of this partial order.

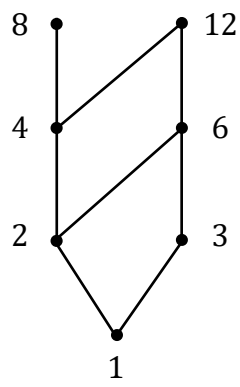


Figure 2.1 Hasse diagram of the partial order considered in Example 2.9.

Definition 2.13 (Poset, minimum and maximum) Let $(A, \preccurlyeq)$ be a poset. An element a of the given poset is maximum if and only if $b \preccurlyeq a$ for every $b \in A$ and it is minimum if and only if $a \preccurlyeq b$ for every $b \in A$.

If a poset has maximum, then this element is unique; similarly, a poset can have at most one minimum.

Definition 2.14 (Minimal and maximal elements) An element a of a poset $(A, \preccurlyeq)$ is maximal element if and only if $b \in A$ and $a \preccurlyeq b$ implies $b = a$ and a minimal element if and only if $b \in A$ and $b \preccurlyeq a$ implies $b = a$.

Thus, maximum is "bigger" than every other element in the set while a maximal element is a local maximum in some subset of the set of given elements while it is not "bigger" than all elements. A poset contains either maximum or maximal elements. If maximum exists, it is unique. Otherwise, poset contains multiple maximal

elements. Similarly, minimum is "smaller" than every other element of the set, while a minimal element is a local minimum in some subset, meaning that minimal element is not "smaller" than all elements. A poset has either minimum or minimal elements. If minimum exists, it is unique. Otherwise, poset contains at least two minimal elements.

Problem 2.10 For each of the following Hasse diagrams write corresponding partial order and determine maximum, minimum, maximal elements and minimal elements.

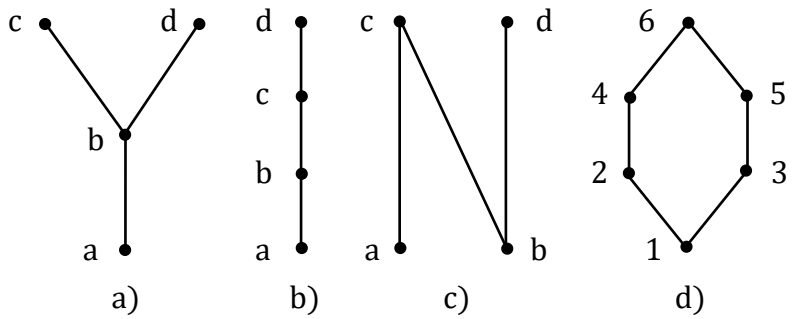


Figure 2.2 Four Hasse diagrams.

Solution

a) $R_1 = \{(a,a),(a,b),(a,c),(a,d),(b,b),(b,c),(b,d),(c,c),(d,d)\}$; minimum is a; c and d are maximal elements.

b) $R_2 = \{(a,a),(a,b),(a,c),(a,d),(b,b),(b,c),(b,d),(c,c),(c,d),(d,d)\}$; a is minimum; d is maximum.

c) $R_3 = \{(a,a),(a,c),(b,b),(b,c),(b,d),(c,c),(d,d)\}$; a and b are minimal elements, c and d are maximal elements.

d) $R_4 = \{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6),(1,2),(2,4),(1,4),(4,6)$

$(2,6), (1,3), (3,5), (1,5), (5,6), (3,6), (1,6)\}$; 1 is minimum; 6 is maximum.

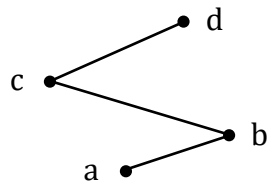
Exercises

1. A country is administratively separated into regions. Let R be the relation on the set of all citizens of the country defined by $a \sim b$ if and only if a and b are residents of the same region. Is R equivalence relation?
2. Consider relation $R = \{(a,a), (a,d), (b,b), (b,e), (c,c), (c,d), (d,a), (d,c), (e,e), (e,b)\}$ on the set $A = \{a,b,c,d,e\}$. Determine the maximum set of properties that this relation holds.
3. $\{(a,a), (b,b), (c,c)\}$ is a relation on $\{a,b,c\}$. Determine the maximum set of properties this relation holds.
4. Let $R_1 = \{(a,b) \in R^2 : a^2 - b^2 \leq 3\}$ be the relation. Determine the maximum set of properties that R_1 holds.
5. Which of the following is equivalence relations on $A = \{0,1,2,3,4\}$?

   a) $R = \{(x,y) \in A | x > y\}$
   b) $Q = \{(x,y) \in A | \mathrm{x} + \mathrm{y} = 4\}$
   c) $S = \{(x,y) \in A | x|y\}$, where $x|y$ stands for "x divides y"
   d) $T = \{(x,y) \in A | \gcd(x,y) = 1\}$,

6. Given relation $R_1 = \{(x,y) | x^2 = y^2;\ x,y \in R\}$. Determine the maximum set of properties that the relation holds.
7. Consider relation $R = \{(a,a), (a,d), (b,b), (b,e), (c,c), (d,a), (e,b), (d,d), (e,e)\}$ on the set $A = \{a,b,c,d,e\}$. Determine the maximum set of properties that this relation holds.

8. Let $X = \{2,3,4,5,8,12,24,25\}$ and let R be the partial order on X defined by $R = \{(m,n) | m \text{ divides } n\}$. Represent R by its Hasse diagram.
9. Write partial order on the set $\{a, b, c, d\}$, which is represented by the following Hasse diagram.



10. Which of the following is the relation that contains $(1,1), (1,2), (2,1)$ and $(1,-1)$.

    a) $R = \{(a,b) | a \leq b\}$
    b) $Q = \{(a,b) | a = b \text{ or } a = -b\}$
    c) $S = \{(a,b) | a = b + 1\}$
    d) $T = \{(a,b) | a + b \leq 3\}$

11. Let R be the relation on the set of all citizens within the European Union (EU) defined by $a \sim b$ if and only if a and b are residents of the same EU country. Determine all properties that R holds.
12. Let R be the relation on the set $A = \{1,2,3,4,5,6,7\}$ defined by the rule $(a,b) \in R$ if $(a - b)$ is divisible by 4. List the elements of R. Determine whether R is reflexive, symmetric, antisymmetric or transitive. Check whether R is equivalence relation, and if it is true then find equivalence classes. Check whether R is partial order.
13. Let $R = \{(1,1), (1,2), (3,2), (3,3), (2,3), (2,1), (2,2)\}$ be relation on $A = \{1,2,3\}$. Determine the maximum set of properties that the relation holds.

14. Which of the following relations on the set $A = \{1,2,3\}$ is both partial order and equivalence relation?

    a) $\{(1,1), (1,2), (2,1), (2,2), (3,3)\}$
    b) $\{(1,1), (1,2), (2,1), (2,2), (1,3), (3,3)\}$
    c) $\{(1,1), (2,2), (3,3)\}$

15. Given the relation "is a sibling of" on the set of all people. Determine the maximum set of properties that the relation holds.
16. Let mRn if $\gcd(m,n) > 1$ where $m, n \in N$ such that $m > 2, n > 2$. Determine the maximum set of properties that mRn holds.

Chapter 3 Functions

3.1 Basic Terminology

Definition 3.1 (Function) A function from a set A to a set B is a binary relation f from A to B which relates each input $a \in A$ to exactly one output $b \in B$, stated precisely, $(a, b) \in f$.

The key idea behind the above definition is that a function associates a unique element $b \in B$ with each element $a \in A$ is captured by viewing a function f as a subset of $A \times B$ under circumstance that there is precisely one pair (a, b) in f having first coordinate a for each $a \in A$. If $A = \{1,2,3\}$, $B = \{x, y\}$, and $f = \{(1, x), (2, y), (3, x)\}$, then f is just the rule that associates x with 1, y with 2, and x with 3.

There are two key points about the function. Firstly, elements in A must be the first component of ordered pairs while elements in B being the second component. Secondly, each element of A must be the first component of exactly one ordered pair. Function g from $A = \{1,2,3\}$ to $B = \{x, y\}$ defined by $g = \{(1, x), (3, y)\}$ is not a function at all because no ordered pair contains 2 as the first coordinate. Neither

$$h = \{(1, x), (2, x), (3, y), (2, y)\}$$

is a function from A to B since 2 is the first coordinate of two ordered pairs.

Problem 3.1 Suppose A is the set of surnames of people in a telephone directory. Check whether

$$f = \{(a, n) | a \text{ is on page } n\}$$

is a function from A to N.

Answer By definition of f, each element of A is the first coordinate of a pair $(a, n) \in f$, and that for each such a there is precisely one (a, n). Thus, f is a function from A to N.

Let f be a function from A to B. An image of element $a \in A$ is the unique element $b \in B$. This fact is expressed by $b = f(a)$. If it is so, we say that $a \in A$ is a preimage of $a \in A$. This particularly means that $(a, b) \in f$ is possible only if $b = f(a)$ and vice versa. It is customary to write $f: A \to B$ to mean that f is a function from A to B.

Definition 3.2 (Domain, codomain, range) Let f be a function from A to B.

- The domain of f, written dom f, is the set A.
- The target (or codomain) of f is the set B.
- The range or image of f, written rng f, is rng $f = \{b \in B | (a, b) \in f$ for some $a \in A\}$

For most problems, it is customarily easy to identify domain and codomain of a function as these two notions are mentioned implicitly or explicitly in a problem description. Furthermore, we find range of a function based on the givens of the problem.

Definition 3.3 (Injection/one-to-one property) The function is one-to-one (or injective) if and only if all elements of A have different images. Stated in other words, function f is one-to-one if and only if $a \neq b$ implies $f(a) \neq f(b)$.

Definition 3.4 (Surjection/onto property) The function f is onto (or surjective) if range of the function equals its target, rng $f = B$. That is, any element $b \in B$ is of the from $b = f(a)$ for some $a \in A$. Equivalently, for any $b \in B$, the equation $b = f(x)$ has a solution $x \in A$.

Definition 3.5 (Bijection/one-to-one correspondence property) The function is one-to-one correspondence (or bijective) if it is both one-to-one and onto (or both injective and surjective).

Example 3.1 Let $A = \{1,2,3,4\}$, $B = \{x, y, z\}$ be the sets and $f = \{(1, x), (2, y), (3, z), (4, y)\}$ be the function. Then, f is a function from A to B whose domain is A and target is B. Since rng $f = B$, f is onto. Because 2 and 4 have the same image, f is not one-to-one and consequently f is not one-to-one correspondence.

Example 3.2 Let $A = \{1,2,3\}$, $B = \{x, y, z, w\}$ be the sets and $f = \{(1, w), (2, y), (3, x)\}$ be the function. Then $f: A \to B$ is a function whose domain is A and range is $\{w, y, x\}$. Since rng $f \neq B$, f is not onto. This function is one-to-one since f maps 1, 2 and 3 to different images. However, f is not one-to-one correspondence because it is not onto.

Example 3.3 Let $f: Z \to Z$ be defined by $f(x) = 2x - 3$. Then dom $f = Z$. To find rng f, note that $b \in$ rng $f \leftrightarrow b = 2a - 3$ for some integer a. Range of the function covers set of odd integers and, therefore, this function is not onto. Consequently, this function is not one-to-one correspondence. Let us now show that f is one-to-one. We are going to use proof by contradiction method. Assume $\exists x, y \in Z\ x \neq y$ and $f(x) = f(y)$. If it is true then $2x - 3 = 2y - 3$ and, therefore, $x = y$. This contradicts to the fact that $x \neq y$. Hence f is one-to-one.

Example 3.4 Let $f: N \to N$ be defined by $f(x) = 2x - 3$. This might look like a perfectly good function, as in the last example, but there is a difficulty. For $f(1)$ we obtain $f(1) = 2(1) - 3 = -1$ and $-1 \notin N$. Hence, f is not a function at all.

Problem 3.2 Define $f: Z \to Z$ by $f(x) = x^2 - 5x + 5$. Determine whether f is one-to-one and onto.

Solution To determine whether or not f is one-to-one, we consider the possibility that $f(x_1) = f(x_2)$. In this case,

$$x_1^2 - 5x_1 + 5 = x_2^2 - 5x_2 + 5$$
$$x_1^2 - x_2^2 = 5x_1 - 5x_2$$
$$(x_1 - x_2)(x_2 + x_1) = 5(x_1 - x_2)$$
$$x_2 + x_1 = 5$$

This equation has integer solution only when $x_1 \neq x_2$. Hence, f is not one-to-one.

Is f onto? Recalling that the graph of $f(x) = x^2 - 5x + 5, x \in R$ is a parabola with vertex $\left(\frac{5}{2}, -\frac{5}{4}\right)$, clearly any integer less than -1 is not in the range of f. Alternatively, it is easy to see that 0 is not in the range of f because $x^2 - 5x + 5 = 0$ has no integer solutions. Either argument shows that f is not onto.

Example 3.5 Let $f: R \to R$ be defined by $f(x) = 2x - 3$. The domain of f is R and rng $f = R$ since any real number y can be expressed by $y = 2x - 3$. Graphcally, this means that any horizontal line intersects the graph of $y = 2x - 3$. Since rng $f = R$, f is onto. It is also one-to-one, so being onto and one-to-one, it is one-to-one correspondence from R to R.

Example 3.6 Let $g: R \to R$ be defined by $g(x) = x^2$. The domain of g is R; the range of g is the set of nonnegative real numbers. Since this is a proper subset of R, g is not onto. Neither is g one-to-one since $g(3) = g(-3)$, but $3 \neq -3$.

Example 3.7 Define $h: [0, \infty) \to [0, \infty)$ by $h(x) = x^2$. This function is identical to the function g of the preceding example except for its domain and target. By restricting the domain and target of g to the nonnegative reals we have produced a function h which is both one-to-one and onto.

3.2 The Absolute Value Function

The absolute value of a real number x, denoted $|x|$, is defined by

$$|x| = \begin{cases} x & if\ x \geq 0 \\ -x & if\ x < 0 \end{cases}$$

Domain and range of the absolute value function are R and $[0, \infty) = \{y \in R | y \geq 0\}$, respectively. Because of $|x| = |-x|$ this function is not one-to-one.

3.3 The Floor and Ceiling Functions

For any real number x, the floor of x, written $\lfloor x \rfloor$, is the greatest integer less than or equal to x, that is, the unique integer $\lfloor x \rfloor$ satisfying

$$x - 1 < \lfloor x \rfloor \leq x.$$

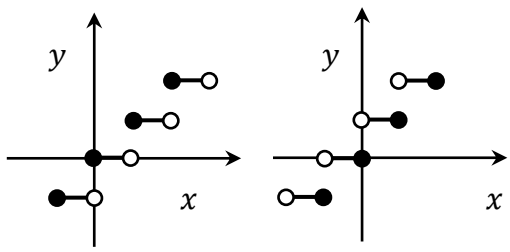


Figure 3.1 Graph of $\lfloor x \rfloor$ (in the left) and $\lceil x \rceil$ (in the right).

The ceiling of x, written $\lceil x \rceil$, is the least integer greater than or equal to x, that is, the unique integer $\lceil x \rceil$ satisfying

$$x \leq \lceil x \rceil < x + 1.$$

For instance, $\lfloor 2.01 \rfloor = 2$, $\lfloor 1.99 \rfloor = 1, \lfloor 15 \rfloor = 15, \lfloor -2.01 \rfloor = -3$ and $\lceil 2.01 \rceil = 3, \lceil 1.99 \rceil = 2, \lceil 15 \rceil = 15, \lceil -2.01 \rceil = -2$.

3.4 Inverse Functions and Compositions of Functions

Assume f is a bijective function from A to B. Given element $b \in B$, there exists $a \in A$ such that $f(a) = b$ since f is surjective function, and that such a is unique because f is injective function. Thus, for each $b \in B$, there is precisely one pair of the form $(a, b) \in f$. It follows that the set $\{(b, a) | (a, b) \in f\}$, obtained by reversing the ordered pairs of f, is a function from B to A.

Example 3.8 If $A = \{1,2,3,4\}$ and $B = \{x, y, z, w\}$, then

$$f = \{(1, x), (2, y), (3, z), (4, w)\}$$

is a one-to-one and onto function from A to B and, reversing its pairs, we obtain a function

$$B \to A: \{(x, 1), (y, 2), (z, 3), (w, 4)\}.$$

Definition 3.6 (Inverse function) A function $f: A \to B$ has an inverse if and only if the set that we obtain by reversing the ordered pairs of f is a function from B to A. If $f: A \to B$ has an inverse, the function

$$f^{-1} = \{(b, a) | (a, b) \in f\}$$

is called the inverse function of f.

Proposition A function $f: A \to B$ has an inverse $B \to A$ if and only if f is one-to-one and onto.

Example 3.9 If $f: R \to R$ is defined by $f(x) = 2x - 3$, then f is one-to-one and onto, so an inverse function exists. Accordingly, $y = f(x) = 2x - 3$. Thus, $x = \frac{1}{2}(y + 3)$. By changing the variables, we obtain $f^{-1}(x) = \frac{1}{2}(x + 3)$.

Definition 3.7 (Composition function) If $f: A \to B$ and $g: B \to C$ are functions, then the composition of g and f is the function $g \circ f: A \to C$ defined by $(g \circ f)(a) = g(f(a))$ for all $a \in A$.

Example 3.10 $A = \{a, b, c\}, B = \{x, y\}$, and $C = \{u, v, w\}$, and if $f: A \to B$ and $g: B \to C$ are the functions

$$f = \{(a, x), (b, y), (c, x)\}, g = \{(x, u), (y, w)\},$$

then

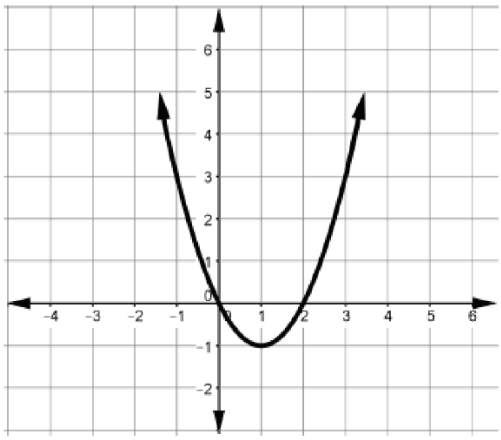
$$\begin{aligned}
&(g \circ f)(a) = g\big(f(a)\big) = g(x) = u,\\
&(g \circ f)(b) = g\big(f(b)\big) = g(y) = w,\\
&(g \circ f)(c) = g\big(f(c)\big) = g(x) = u,\\
&g \circ f = \{(a, u), (b, w), (c, u)\}.
\end{aligned}$$

Expressed in other words, to find $(g \circ f)(a)$ we first apply the function f to a to obtain $f(a)$ and then we apply the function g to $f(a)$ to obtain $(g \circ f)(a) = g\big(f(a)\big)$. Note that the composition $(g \circ f)$ exists if and only if rng $f \subseteq$ dom g.

The question arises under what circumstances two functions $f: A \to B$ and $g: C \to D$ are equal. Firstly, f and g must have same domain and same target, that is, $A = C$ and $B = D$. Secondly, $f = g$ if f and g contain same elements, that is, $f = g$ if and only if $(a, b) \in f$ implies $(a, b) \in g$ and vice versa.

Exercises

1. The graph of parabolic function f, whose vertex is at the point (1,-1), is shown in the following figure. Find range of f.



2. Suppose $f(x) = x^2 + 3$ and $g(x) = x - 2$ are functions from R to R. Find $(f \circ g)(x)$.
3. Let f and g be the functions from R to R such that $f(x) = x - 2$, $g(x) = \frac{2}{x^2+1}$.
4. Find $g \circ f$and f^{-1}.
5. Given that $f(x) = \frac{9}{5}x + 32$, find $f^{-1}\big(f(100)\big)$.
6. Let $f(x) = 4x^2$ be the function from R to R and let $g(x) = \sqrt{x} + 1$ be the function from $[0, \infty)$ to R. Find $g \circ f$ and determine its domain, codomain (target) and range.
7. Which of the following is invertible function from R to R?

   a) $f(x) = |x - 4|$
   b) $g(x) = (x + 4)^2$
   c) $h(x) = \frac{1}{(x-4)(x+4)}$
   d) $s(x) = (x - 4)^3$
   e) $w(x) = \lfloor x - 4 \rfloor$

8. Which of the following is a bijection from R to R?

   a) $f(x) = -3x^2 + 7$
   b) $g(x) = \lceil -3x \rceil$
   c) $h(x) = -3x + 4$
   d) $s(x) = |-3x|$

9. Let $A = \{1,3,5\}$, $B = \{2,4,6,8\}$ and $f: A \to B$ be defined by $f = \{(1,8), (3,2), (5,4)\}$. Which of the following is wrong for f?

   a) f is one-to-one
   b) f is onto
   c) rng $f \subseteq B$
   d) rng $f = \{2,4,8\}$

10. f and g are functions from R to R which is defined by $f(x) = 2x + 3$ and $g(x) = x^2 + 7$. Find x if $f(g(x)) = 25$.
11. Let $= [0, \infty)$, $B = [-1, \infty)$ and $f: A \to B$ be defined by $f(x) = x^2 - 1$. Check whether f is injective, surjective and bijective.
12. Let $f(x) = 1/(x - 4)$ be invertible function from R to R. Determine domain, target and range of this function. Find $f^{-1}(x)$.
13. Let $f: R \to R$. Which of the following functions is one-to-one?

   a) $f(x) = 2x^2 + 2$
   b) $f(x) = \lceil 4x \rceil$
   c) $f(x) = 9x^3 + 1$
   d) $f(x) = |2 - x|$

14. Which of the following is true for function $f: [0, \infty) \to [4, \infty)$ such that $(x) = (x + 2)^2$.

a) f is not injective and it is not surjective
b) f is injective but it is not surjective
c) f is not injective but it is surjective
d) f is bijective
e) f is not a function

Chapter 4 Propositional Logic

4.1 Background

Propositional logic studies the ways propositions interact with each other. According to Alonzo Church[3], an American mathematician and logician who contributed to logic a lot and who is one of the founders of computer science, propositions are "meanings" and "thoughts" expressed by declarative sentences.

Definition 4.1 (Proposition) Any declarative sentence is a proposition. A proposition has a definite truth or falsity value.

Example 4.1 Which of the following is a proposition?

a) It is sunny today.
b) Budapest is a capital of Czech Republic.
c) What time is it?
d) Good luck.
e) She walks to school.
f) $1 + 1 = 2$
g) $2 + 2 = 3$
h) $x + y = z$

[3] Alonzo Church (1903–1995) was born in Washington, D.C. He studied at Göttingen under Hilbert and later in Amsterdam. He was a member of the faculty at Princeton University from 1927 until 1967 when he moved to UCLA. Church was one of the founding members of the Association for Symbolic Logic. He made many substantial contributions to the theory of computability, including solution of the decision problem, invention of the lambda-calculus, and, of course, statement of what is now known as the Church–Turing thesis. Among Church's students were Stephen Kleene and Alan Turing. He published articles past his 90th birthday.

A paradox is a declarative sentence that is true and false at the same time – thus, a paradox is not a proposition at all because it has no definite truth value. Consequently, for Example 4.1, the sentence "she walks to school" is a paradox since we do not know who "she" is and therefore we cannot identify the truth or falsity of this statement. Likewise, x, y and z in "$x + y = z$" are variables that can be assigned any numeric value and thereby this sentence is also true and false at the same time – just another example to a paradox.

We distinguish between atomic and compound (or molecular) propositions. Atomic propositions are those having simple structure while compound ones are compound propositions that are made of two or more simple propositions.

Consider the following five simple declarative sentences:

A. $Art\ loves\ Betty.$
B. $Batman\ is\ left-handed.$
C. $Columbus\ discovered\ America.$
D. $Dynamite\ is\ dangerous.$
E. $Elephants\ have\ wings.$

Each of above sentences has truth/falsity value. For the sake of argument, we shall assume that each of these propositions is either true or false. The propositional logic has as its goal the discovery of valid arguments about propositions. In everyday life, we use "valid" in many different ways, e.g., "provide a valid excuse for not attending the examination," or "that is not a valid password". In logic, "valid" has a very specific meaning. Without diving deep into the details, we will note that an argument consists of a number of statements, called the premises, which together offer support for the belief that another statement, called the conclusion, is true. In a valid

argument, it is not possible that the conclusion is false when the premises are true or, stated in other words: in a valid argument, whenever the premises are true, the conclusion also has to be true.
Let us now consider the following compound proposition:

$$\text{“}If\ either\ Art\ loves\ Betty\ or\ Batman\ is\ left-handed,$$
$$and\ Batman\ is\ not\ left-handed, then\ Art\ loves\ Betty\text{”}.$$

Although it seems a meaningless sentence, it is logically correct, and, indeed, it is a special case of the following more respectable-sounding general proposition:

$$\text{“}If\ either\ A\ or\ B\ is\ true, and\ B\ is\ not\ true, then\ A\ is\ true\text{”}.$$

Definition 4.2 (Theorem/Tautology) The proposition, which is true regardless of the truth or falsity of its atomic propositions, is called a theorem (or tautology) of the propositional logic.
Definition 4.3 (Contradiction) A proposition, which is false regardless of the truth or falsity of its atomic propositions, is called a contradiction of the propositional logic.

Otherwise, a proposition is neither a theorem nor a contradiction. In such cases, we check the proposition for satisfiability, that is, identifying if there exists an interpretation that satisfies a given proposition. In other words, it asks whether the variables of a given proposition can be consistently replaced by the values true or false in such a way that the whole proposition evaluates to true.

Complex propositions are built from simpler ones by using five kinds of connectives: and, or, not, $implies$, and $if\ and\ only\ if$. For example, the compound propostion

$$\text{“}Art\ loves\ Betty\ and\ Columbus\ discovered\ America\text{”}$$

is built from propositions A and C using and connective. The and connective is represented by the symbol "∧" and in the notation of the propositional logic this compound proposition is represented as follows:

$$(A \wedge C), \text{ pronounced "}A \text{ and } C\text{"}$$

Left and right parentheses in the above expression are important. Similarly, the compound proposition "$either\ A\ or\ B$", which itself is shortening for "$either\ A\ is\ true, or\ B\ is\ true, or\ both$", is represented by the following symbol:

$$(A \vee B), \text{ pronounced "}A \text{ or } B\text{"}.$$

We represent the negation of a proposition by prefixing the proposition with the symbol "∼". Thus, we would represent the proposition "$Dynamite\ is\ not\ dangerous$" by

$$\sim D, \text{ pronunced "not } D\text{"}.$$

A proposition of if-then type uses the symbol "→" prounced "implies". Thus, the proposition

$$"If\ elephants\ have\ wings, then\ Batman\ is\ left-handed"$$

would be represented

$$(E \rightarrow B), \text{ pronounced "}E \text{ implies } B\text{"}.$$

Finally, we represent the connective $if\ and\ only\ if$ by the symbol "↔". For example, we denote the compound proposition

$$\text{“}Dynamite\ is\ dangerous\ if\ and\ only\ if\ Art\ loves\ Betty\text{”},$$

by

$(D \leftrightarrow A)$, pronounced "D if and only if A".

Starting with "atomic" propositions (like propositions A, B, C, D, E) and using the above five connectives, it is possible to form more complex propositions.

4.2 Well-Formed Propositions

We validate computer programs at different stages that these programs pass through. We usually distinguish between syntactic and semantic errors. Missing parenthesis or mismatched brackets refer to syntax errors that are detected at compilation stage, while incorrectly placed parentheses are categorized as semantic error. Semantic errors are detected when computer programs run to completion, but give incorrect results. Well-formed propositions provide a foundation of syntax check of computer programs.

Definition 4.4 (Well-formed proposition) A proposition, which is formed by successive application of the following six rules is called well-formed proposition (WFP).

Rules forming well-formed propositions:

- Variable letters like $A, B, C, \ldots$ are WFPs called atoms or atomic propositions.

- $(x \wedge y)$ is WFP.
- $(x \vee y)$ is WFP.
- $\sim x$ is WFP.
- $(x \rightarrow y)$ is WFP.
- $(x \leftrightarrow y)$ is WFP.

Problem 4.1 Which of the following are WFPs?

a) A
b) (A)
c) $\sim(A)$
d) $(A \leftrightarrow \sim A)$
e) $(A \wedge B \wedge C)$
f) $\big((A \wedge B) \wedge C\big)$
g) $\big((\sim(A \wedge B) \vee C) \rightarrow \sim A\big)$
h) $\Big(\Big(\big((A \rightarrow B) \rightarrow C\big) \rightarrow \sim(A \vee B) \leftrightarrow \sim\sim(C \wedge A)\Big)\Big)$

Answer

a) This is an atomic proposition and therefore it is WFP by Rule 1.
b) There is no rule that permits us to surround an atom with parentheses. Thus, this is not WFP.
c) It sees that we need to apply Rule 4 to the proposition (A). But in part b) we agreed that (A) is not WFP. Thus, this is not WFP.
d) This is WFP. We apply Rule 6 to the WFPs A and $\sim A$.
e) This is not WFP because there is no rule allowing us to combine three WFPs together using *and* connective!
f) This is WFP. We apply Rule 2 two times.
g) See Table 4.1
h) See Table 4.2

If a proposition is not large enough, it is rather easy to check and decide by eye whether it is WFP. For large propositions, it is not much easy to decide on this matter. The cases g) and h) present rather complex propositions. If a proposition is complex enough, in each step we apply one of the rules 1-6 to eliminate a fragment of the proposition by substituting that fragment by a specific symbol, say "*". If we obtain single * when the procedure terminates, then the proposition is WFP. Otherwise, it is not.

Table 4.1 Check for well formity	
Line 1	$\big((\sim(A \wedge B) \vee C) \rightarrow \sim A\big)$
Line 2	$\big((\sim(*\wedge*) \vee *) \rightarrow \sim *\big)$
Line 3	$\big((\sim *\vee *) \rightarrow *\big)$
Line 4	$\big((*\vee *) \rightarrow *\big)$
Line 5	$(*\rightarrow *)$
Line 6	$*$

Table 4.2 Check for well formity	
Line 1	$\Big(\big(\big((A \rightarrow B) \rightarrow C\big) \rightarrow \sim(A \vee B) \leftrightarrow \sim\sim(C \wedge A)\big)\Big)$
Line 2	$\Big(\big(\big((*\rightarrow *) \rightarrow *\big) \rightarrow \sim(*\vee *) \leftrightarrow \sim\sim(*\wedge *)\big)\Big)$
Line 3	$\big((*\rightarrow *) \rightarrow \sim * \leftrightarrow \sim\sim *\big)$
Line 4	$\big((*\rightarrow * \leftrightarrow \sim *)\big)$
Line 5	$\big((*\rightarrow * \leftrightarrow *)\big)$

Table 4.3 Truth table of five connectives						
x	y	$(x \wedge y)$	$(x \vee y)$	$\sim x$	$(x \rightarrow y)$	$(x \leftrightarrow y)$
F	F	F	F	T	T	T
F	T	F	T	T	T	F
T	F	F	T	F	F	F
T	T	T	T	F	T	T

Table 4.3 is a truth table, which summarizes truth/falsity of five connectives. To build truth table of a compound proposition we need to calculate truth/falsity values of all successive components as it is demonstrated in Problem 4.2.

Problem 4.2 For each of the following propositions calculate truth table

a) $(A \leftrightarrow \sim A)$
b) $\big((A \rightarrow B) \wedge \sim A\big)$
c) $\Big(\big((A \vee B) \wedge \sim B\big) \rightarrow A\Big)$
d) $\big((\sim (A \wedge B) \vee C) \rightarrow \sim A\big)$

Solution

A	$\sim A$	$(A \leftrightarrow \sim A)$
F	T	F
T	F	F

a)

A	B	$(A \rightarrow B)$	$\sim A$	$\big((A \rightarrow B) \wedge \sim A\big)$
F	F	T	T	T
F	T	T	T	T
T	F	F	F	F
T	T	T	F	F

b)

A	B	$(A \vee B)$	$\sim B$	$((A \vee B) \wedge \sim B)$	$(((A \vee B) \wedge \sim B) \rightarrow A)$
F	F	F	T	F	T
F	T	T	F	F	T
T	F	T	T	T	T
T	T	T	F	F	T

c)

A	B	C	$(A \wedge B)$	$\sim (A \wedge B)$	$(\sim (A \wedge B) \vee C)$	$\sim A$	$((\sim (A \wedge B) \vee C) \rightarrow \sim A)$
F	F	F	F	T	T	T	T
F	F	T	F	T	T	T	T
F	T	F	F	T	T	T	T
F	T	T	F	T	T	T	T
T	F	F	F	T	T	F	F
T	F	T	F	T	T	F	F
T	T	F	T	F	F	F	T
T	T	T	T	F	T	F	F

d)

We list the basic theorems of propositional logic in Table 4.4. Basic theorems provide one means of reasoning about propositions. We can use these theorems to transform one proposition into an equivalent one. We usually apply basic theorems to convert propositions into various normal forms.

Table 4.4 Thirteen theorems of propositional logic	
$(x \vee \sim x)$	Excluded middle
$(x \leftrightarrow \sim\sim x)$	Double negative
$((x \wedge y) \rightarrow x)$	Separation
$(x \rightarrow (x \vee y))$	Joining
$(((x \rightarrow y) \wedge x) \rightarrow y)$	Modus ponens or detachment
$((x \rightarrow y) \leftrightarrow (\sim y \rightarrow \sim x))$	Contrapositive
$(((\sim x \rightarrow y) \wedge (\sim x \rightarrow \sim y)) \rightarrow x)$	Proof by contradiction

$\big((x \vee y) \wedge \sim y\big) \rightarrow x\big)$	Disjunctive syllogism
$\big((x \vee y) \leftrightarrow (\sim x \rightarrow y)\big)$	Switcheroo
$\big((x \rightarrow y) \wedge (y \rightarrow z)\big) \rightarrow (x \rightarrow z)\big)$	Deduction
$\Big(\big((x \rightarrow y) \wedge (z \rightarrow w)\big) \wedge (x \vee z)\Big) \rightarrow (y \vee w)\Big)$	Constructive dilemma
$\big(\sim (x \vee y) \leftrightarrow (\sim x \wedge \sim y)\big)$	De Morgan's law I
$\big(\sim (x \wedge y) \leftrightarrow (\sim x \vee \sim y)\big)$	De Morgan's law II

Problem 4.3 Use De Morgan's laws to express the negation of "Eileen has a cellphone and she has a laptop computer"

Solution Let p be "Eileen has a cellphone" and q be "Eileen has a laptop computer." Then "Eileen has a cellphone and he has a laptop computer" can be represented by $p \wedge q$. By the first of De Morgan's laws, $\overline{(p \wedge q)}$ is equivalent to $\bar{p} \vee \bar{q}$. Consequently, we can express the negation of our original statement as "Eileen does not have a cellphone or he does not have a laptop computer."

Definition 4.5 (Logically equivalent propositions) Two propositions are equal or logically equivalent if these propositions have the same truth values. That is, p and q are logically equivalent if p is true whenever q is true, and vice versa, and if p is false whenever q is false, and vice versa. If p and q are logically equivalent, we write $p = q$.

Exercises

1. Which of the following is a proposition?

   a) Here is your book.
   b) Read the regulations.
   c) $x^2 + y^2 = z^2$
   d) If I knew, that he does not knows I would tell him.

e) Do you know that he knows what you know?

2. Which of the following is a proposition?

a) Will it be raining tomorrow?
b) $1 + 7 = x$
c) $x < y$
d) Drive slowly.
e) $2 + 4 = 5$
f) God bless you.

3. Which of the following sentences is a proposition?

a) Get me that book.
b) I do recommend you to prepare yourself for Tuesday evening interview.
c) Is it snowing?
d) The only even prime number is 5.

4. Which of the following are propositions?

a) $\frac{3+17}{2} > 10$
b) May God help you.
c) Eucalyptus is a tree.
d) Milk is black.

5. If P and Q are propositions, which of the following is logically equivalent to $P \leftrightarrow Q$?

a) $P \leftrightarrow \sim Q$

b) $\sim P \leftrightarrow Q$
c) $\sim P \leftrightarrow \sim Q$

6. Which logic connective has the following truth table?

xy	?
00	1
01	0
10	0
11	1

7. Which of the following is a contradiction?

a) Three is a prime number.
b) It is raining or it is not raining.
c) I understand and I don't understand.
d) If $1 + 1 = 2$ then $3 \cdot 5 = 12$.
e) She is eating if and only if she is hungry.

8. Make sure whether the proposition "*If you are sick then you go to doctor, and if you go to doctor then you are sick*" is a tautology, a contradiction or neither a tautology nor a contradiction.
9. Check whether "*If she is sad or she is not sad, and she is sad, then she is sad*" is a tautology, a contradiction or neither a tautology nor a contradiction.
10. Which of the following is WFP?

a) $\big(((p \rightarrow q) \rightarrow \sim r) \leftrightarrow s\big) \vee (t \wedge u)$
b) $((q \rightarrow \sim p) \leftrightarrow (p \leftrightarrow q))$

11. Which of the following is not WFP?

a) $((A \rightarrow B) \wedge \sim C)$
b) $\sim\big((\sim(A \vee B) \wedge C) \vee D\big)$
c) $(A \wedge B) \vee (A \wedge C)$
d) $\sim(\sim(\sim(A \vee B) \vee C) \vee D)$

12. Which of the following is a tautology?

a) $(\sim q \wedge p) \wedge q$
b) $(\sim q \wedge p) \wedge (p \wedge \sim q)$
c) $(\sim q \wedge p) \vee (p \vee \sim p)$
d) $(p \wedge q) \wedge \big(\sim(p \wedge q)\big)$
e) $(\sim q \wedge p) \vee q$

13. Which of the following is a contradiction?

a) $p \rightarrow (p \vee q)$
b) $(p \wedge q) \rightarrow p$
c) $(p \wedge q) \wedge (\sim p \vee q)$
d) $\sim (p \vee q) \wedge (p \wedge q)$
e) $(\sim p \wedge p) \rightarrow (\sim q \vee q)$

Chapter 5 Boolean Algebra

5.1 Basic Terminology

In the previous chapter, we studied the branch of symbolic logic called propositional logic. One of the earlier investigators of symbolic logic was George Boole[4], an English mathematician, who invented a systematic way of manipulating logic symbols, which later became known as Boolean Algebra. The striking feature of Boolean algebra, which distinguishes it from all other branches of algebra, is that all the variables and constants must assume just one of two values, which are usually called $true$ and $false$. It is also correct that computers and most other digital electronic devices do their work using only two symbols, which are one and $zero$. In 1938 Claude Shannon[5], an American mathematician and engineer,

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[4] George Boole (1815–1864) was born in Lincoln, England, in 1815. He became one of the most important mathematicians of the 1800s. While reading papers of the great French mathematician Lagrange, Boole made discoveries in the calculus of variations, the branch of analysis dealing with finding curves and surfaces by optimizing certain parameters. In 1848 Boole published The Mathematical Analysis of Logic, the first of his contributions to symbolic logic. In 1849 he was appointed professor of mathematics at Queen's College in Cork, Ireland. In 1854 he published The Laws of Thought, his most famous work. In this book, Boole introduced what is now called Boolean algebra.

[5] Claude Elwood Shannon (1916–2001) was born in Petoskey, Michigan. Shannon attended the University of Michigan, graduating in 1936. He continued his studies at MIT. Shannon's master's thesis, written in 1936, presents the first application of Boolean algebra to the design of switching circuits, which remains one of the most famous master's theses of the twentieth century. He received his Ph.D. from MIT in 1940. Shannon joined Bell Laboratories in 1940, where he worked on transmitting data efficiently.

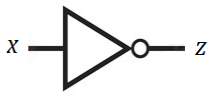
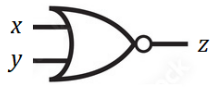
discovered that this is much more than just a coincidence, and that there is such a great similarity between the way logicians reason and the way electronic devices calculate, that Boolean algebra can be profitably applied to digital electronics. This chapter introduces the basics of Boolean algebra. The key idea is that of a Boolean function.

Definition 5.1 (Boolean function) A Boolean function $y = f(x_1, x_2, \dots, x_n)$ can assume either of the values 0 or 1, and that it is a function of n Boolean variables $x_1, x_2, \dots, x_n$ each of which also assumes either of the values 0 or 1.

5.2 Logic Gates and Logic Diagrams

We illustrate seven basic Boolean functions in Table 5.1. Each basic Boolean function has a name, is characterized by verbal expression, can be represented by truth table, has a functional expression and is represented by figure which is called the logic gate. We can further combine logic gates in many ways. Each such combination forms a more complex structure, which we call a logical circuit or logic circuit. In fact, we can build any Boolean function of any number of Boolean variables from basic logic gates.

He was one of the first people to use bits to represent information. At Bell Laboratories he worked on determining the amount of traffic that telephone lines can carry. Shannon made many fundamental contributions to information theory. In the early 1950s he was one of the founders of the study of artificial intelligence. He joined the MIT faculty in 1956, where he continued his study of information theory.

Table 5.1 Seven Boolean functions (digital logic gates)				
Name	Symbolic expression	Verbal expression	Truth table	Formula
AND		$x = 1$ and $y = 1$	x y z 0 0 0 0 1 0 1 0 0 1 1 1	$z = xy$
OR		$x = 1$ or $y = 1$	x y z 0 0 0 0 1 1 1 0 1 1 1 1	$z = x + y$
NOT		$x = 1$ is not true	x z 0 1 1 0	$z = \bar{x}$
NAND		($x = 1$ and $y = 1$) is not true	x y z 0 0 1 0 1 1 1 0 1 1 1 0	$z = \overline{xy}$
NOR		($x = 1$ or $y = 1$) is not true	x y z 0 0 1 0 1 0 1 0 0 1 1 0	$z = \overline{x + y}$
XOR		($x = 1$ or $y = 1$) but not both	x y z 0 0 0 0 1 1 1 0 1 1 1 0	$z = x \oplus y$
XNOR		($x = 1$ or $y = 1$ but not both) is not true	x y z 0 0 1 0 1 0 1 0 0 1 1 1	$z = \overline{x \oplus y}$

Problem 5.1 In the logic diagram shown below, c represents a Boolean function of three Boolean variables x, y, and z. Calculate truth table of c.

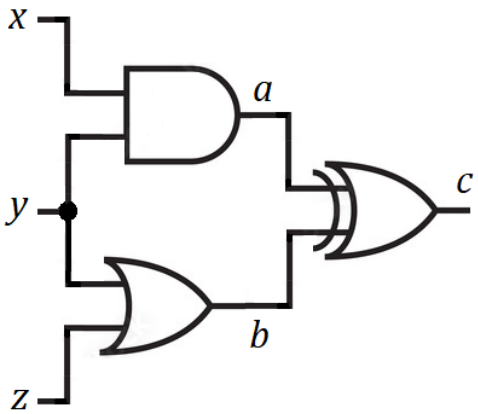


Figure 5.1 Logic diagram composed of AND, OR and XOR gates.

Solution Logic diagram in Figure 5.1 is a graphical representation of the Boolean function

$$c = a \text{ XOR } b = (x \text{ AND } y)\text{XOR } (y \text{ OR } z) = (xy)\oplus(y + z).$$

In terms of x, y and z we have the following truth table:

x	y	z	a	b	c
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	0	1	1
0	1	1	0	1	1
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	1	1	0
1	1	1	1	1	0

Problem 5.2 Show that the logic circuit in the left of Figure 5.2, which consists of three NAND gates, is equivalent to one OR gate.

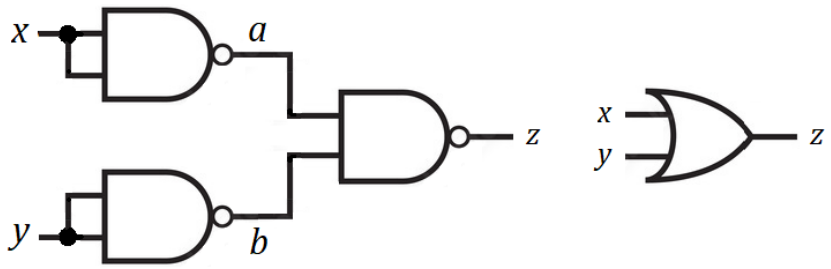


Figure 5.2 Logic diagram made of three NAND gates (in the left) and OR gate (in the right).

Solution The truth table of the logic diagram in Figure 5.2 is illustrated below.

xy	a	b	z
00	1	1	0
01	1	0	1
10	0	1	1
11	0	0	1

We summarize the above table as follows.

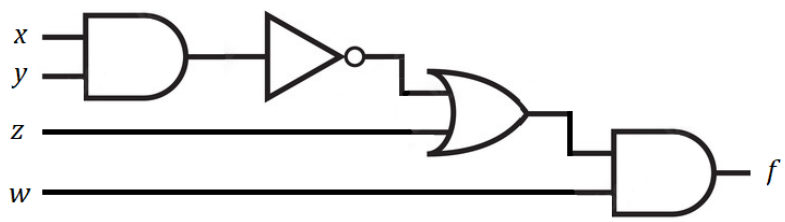
xy	z
00	0
01	1
01	1
11	1

What we obtained is nothing else than truth table of OR gate which proves the claim.

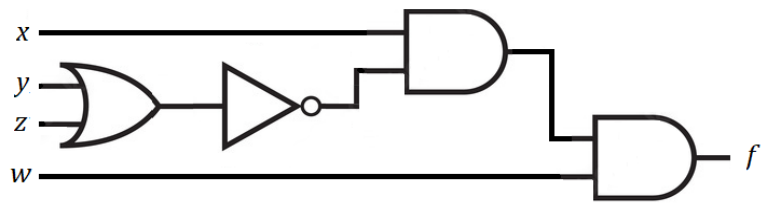
5.3 Minterm and Maxterm Expansions

Our goal in this section is to study relationship between Boolean functions, truth tables and logic diagrams, and show that any Boolean function can be built from the basic logic gates AND, OR, and NOT. We can represent Boolean functions in various ways. This is also true that we can simplify Boolean functions similar to the way we simplify arithmetic expressions. Given a Boolean function, finding its most simple form is the question of theoretical and practical interest.

Problem 5.3 Write Boolean algebra expression corresponding to the logic diagrams in Figure 5.3.



a)



b)

Figure 5.3 Two logic diagrams.

Solution Both functions made of two ANDs, one OR, and one NOT gates.

In the logic circuit a), the first operation is xy; this quantity is then complemented; then ORed with z; and finally ANDed with w. This sequence of operations is equivalent to the Boolean expression

$$f = \left((\overline{xy}) + z\right)w.$$

Similar deductions for the logic circuit b) yield

$$f = \left(x(\overline{y + z})\right)w.$$

Problem 5.4 Write Boolean functions shown in Figure 5.4.

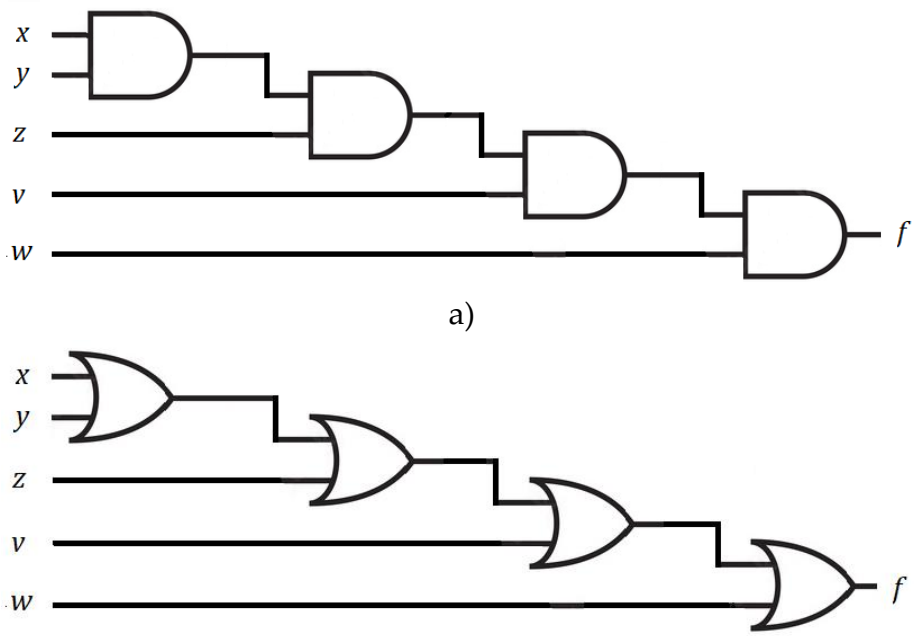


Figure 5.4 Two logic diagrams: a) cascade of four ANDs; b) cascade of four ORs.

Solution

In case a) we could proceed by calculating the truth tables for the two functions, by assigning the Boolean variables their $2^5 = 32$ different values. But these two Boolean functions are, on reflection, so simple that most of the work can be avoided. The f is a very simple Boolean function in a sense that $f = 0$ for 31 inputs while $f = 1$ only if $x = y = z = v = w = 1$. In Boolean algebra, f is represented by the function

$$f = \Big(\big((xy)z\big)v\Big)\,w.$$

In the above expression we can omit parentheses, as due to associative law parentheses are not really necessary. Thus, we can write

$$f = xyzvw.$$

Similarly, in the logic diagram composed of four OR gates, $f = 1$ for 31 inputs and $f = 0$ when $x = y = z = v = w = 0$. This function is expressed by

$$f = \Big(\big((x + y) + z\big) + v\Big) + w$$

which is due to associative law equivalent to

$$f = x + y + z + v + w.$$

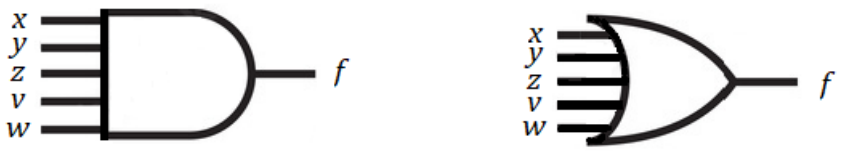


Figure 5.5 5-input AND gate (in the left) and 5-input OR gate (in the right).

These results lead to the idea that instead of logic diagram shown in case a) we could use a single AND gate with five inputs and single output as shown in Figure 5.5 (a gate in the left). This is equally true for the case b) - instead of logic diagram composed of four OR gates we could use single OR gate with five inputs and single output as is illustrated in Figure 5.5 (a gate in the right).

Ofcause, we can generalize the idea behind of Problem 5.4 to the case of more than five Boolean variables. By collecting $n-1$ AND gates in a similar tree, we can build a Boolean function of n variables, $x_1, x_2, \dots, x_n$, which is 1 if and only if $x_1 = x_2 = \cdots = x_n = 1$. Likewise, we can build an OR tree that will yield a Boolean function of n variables, $y_1, y_2, \dots, y_n$, which is 0 if and only if $y_1 = y_2 = \cdots = y_n = 0$.

In complex Boolean circuits, we can think of the output 1 as "acceptance" and 0 as "rejection" of particular input. The "big" AND gate with n inputs accepts only the input $\underbrace{1 \cdots 1}_{n}$, and "big" OR gate with n inputs rejects only $\underbrace{0 \cdots 0}_{n}$. In addition, we can use NOT gates "inverters" to build circuits that accept (or reject) a given input pattern, not just $\underbrace{1 \cdots 1}_{n}$ (or $\underbrace{0 \cdots 0}_{n}$).

Problem 5.5 Describe the Boolean functions represented by the following two logic circuits.

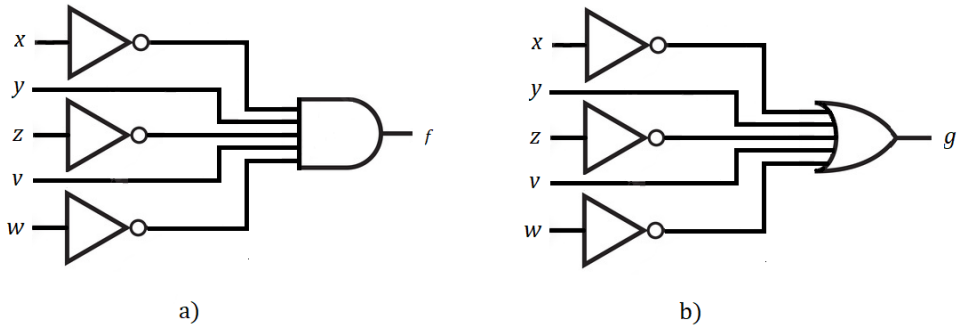


Figure 5.6 a) Logic circuit which accepts 01010 b) Logic circuit which rejects 10101.

Solution Obviously, $f = 1$ only if $x = 0, y = 1, z = 0, v = 1, w = 0$, that is, logic circuits in Figure 5.6 a) accepts 01010 only. On the other hand, $g = 0$, when $x = 1, y = 0, z = 1, v = 0, w = 1$, meaning that logic circuit in Figure 5.6 b) rejects 10101 only.

The concern is that we can build logic circuits that accept or reject the given collection of inputs. To build an accepter we use "big" AND gate, and invert 0s on input. To build a rejecter, we use "big" OR gate, and invert 1s on input. Figure 5.7 illustrates accepter of 01010 and rejecter of 10101. "Big" AND gate and "big" OR gate in Figure 5.7 are equivalent to logic circuits shown in Figure 5.6 (a) and Figure(b), respectively.

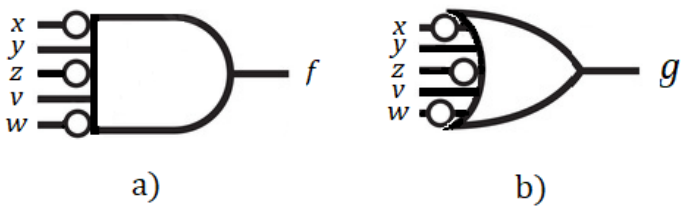


Figure 5.7 Accepter of 01010 (in the left) and rejecter of 10101 (in the right).

Let us now think of how a Boolean function can be constructed using ORs, ANDs, and NOTs. Below we illustrate this technique on

example of a Boolean function y of three variables, x_1, x_2, x_3, whose truth table is shown in Table 5.2.

Table 5.2 Truth table of Boolean function			
x_1	x_2	x_3	y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

The function y accepts three input patterns (or signals) 001, 100 and 101. To build y, we construct accepter for 001, 100 and 101. Then we connect the outputs of the three accepters to a three-input OR gate as shown in Figure 5.8 (diagram in the left). We represent this circuit by equation

$$y = \bar{x}_1\bar{x}_2x_3 + x_1\bar{x}_2\bar{x}_3 + x_1\bar{x}_2x_3.$$

The above expression for y is called the **minterm expansion** or **sum of the products** or **disjunctive normal form** for y. Minterm expansion of y is one way how this function can be represented. **Truth set** od Boolean function y consists of all inputs that turn y into 1. Truth set of Boolean function y is {001,100,101}. There exists another way to represent the same function. The following expression for y is called **maxterm expansion** or **product of the sums**

or **conjunctive normal form** for y. The following is maxterm expansion of y.

$$y = (x_1 + x_2 + x_3)(x_1 + \bar{x}_2 + x_3)(x_1 + \bar{x}_2 + \bar{x}_3)(\bar{x}_1 + \bar{x}_2 + x_3)$$
$$(\bar{x}_1 + \bar{x}_2 + \bar{x}_3)$$

We show corresponding logic diagram in Figure 5.8 (diagram in the right). Diagram in the left accepts signals 001,100,101, while the one in the right rejects signals 000,010,011,110,111.

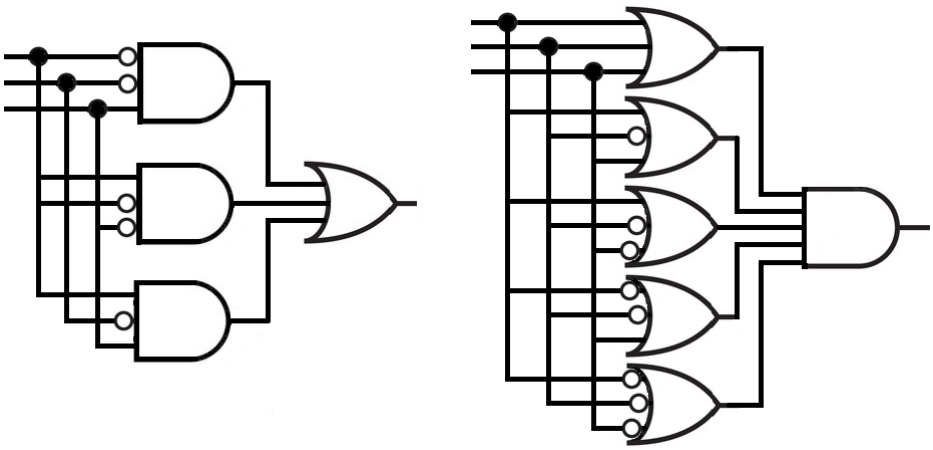


Figure 5.8 Accepter of 001, 100, 101 (in the left) and rejecter of 000, 010, 011, 110, 111 (in the right).

5.4 Laws of Boolean algebra

We have detailed several examples of logic circuits that indeed represent the same Boolean function. The last example of the previous section reveals that we can represent the Boolean function y in the form of minterm expansion as well as maxterm expansion.

Minterm expression of y can be further simplified using laws of Boolean algebra.

Definition 5.2 (Duality law) If in a valid Boolean identity, you replace all ANDs by ORs, ORs by ANDs, 0s by 1s and 1s by 0s, the result is a valid Boolean identity which is called dual form of the given identity.

Two logic circuits that we have detailed so far (see Figure 5.8) yield the same Boolean function.

Below we illustrate important laws of Boolean algebra both in primal and dual forms.

Table 5.3 Laws of Boolean algebra.

Name of Law	Primal Form	Dual Form
Associative	$x + (y + z) = (x + y) + z$	$x(yz) = (xy)z$
Commutative	$x + y = y + x$	$xy = yx$
Distributive	$x(y + z) = xy + xz$	$x + (yz) = (x + y)(x + z)$
De Morgan	$\overline{x + y} = \bar{x}\bar{y}$	$\overline{xy} = \bar{x} + \bar{y}$
Weak Absorption	$1 + x = 1$	$0 \cdot x = 0$
Strong Absorption	$y + xy = y$	$y(x + y) = y$
Involution	$\bar{\bar{x}} = x$	$\bar{\bar{x}} = x$
Idempotency	$x + x = x$	$x \cdot x = x$
Complementarity	$x + \bar{x} = 1$	$x \cdot \bar{x} = 0$
Redundancy	$x + \bar{x}y = x + y$	$x(\bar{x} + y) = xy$

Consensus	$xy + \bar{x}z + yz = xy + \bar{x}z$	$(x+y)(\bar{x}+z)(y+z) = (x+y)(\bar{x}+z)$
Identity	$x + 0 = x$	$x \cdot 1 = x$

Problem 5.6 The following is a valid Boolean identity

$$x(y + \bar{y}z) + \bar{y}z = xy + \bar{y}z.$$

Write this identity in dual form.
Solution By interchanging $\cdot$ and $+$, we receive

$$\Big(x + \big(y \cdot (\bar{y} + z)\big)\Big)(\bar{y} + z) = (x + y) \cdot (\bar{y} + z).$$

Problem 5.7 Prove De Morgan's law in primal form.
Solution The simplest and straightforward proof is by means of truth table.

x	y	$\bar{x}$	$\bar{y}$	$x+y$	$\overline{x+y}$	$\bar{x}\bar{y}$
0	0	1	1	0	1	1
0	1	1	0	1	0	0
1	0	0	1	1	0	0
1	1	0	0	1	0	0

The fact that columns for $\overline{x+y}$ and for $\bar{x}\bar{y}$ are matching approves the claim.

5.5 Karnaugh maps

We can further simplify minterm expansion using laws of Boolean algebra. As an illustration, consider Boolean function f of three variables whose truth set is

$$A = \{000, 001, 010, 011, 110, 111\}.$$

Minterm expansion of this function is as follows:

$$f = \bar{x}_1\bar{x}_2\bar{x}_3 + \bar{x}_1\bar{x}_2x_3 + \bar{x}_1x_2\bar{x}_3 + \bar{x}_1x_2x_3 + x_1x_2\bar{x}_3 + x_1x_2x_3.$$

This expression leads a logic circuit which requires 12 ANDs, 5 ORs, and nine NOTs. Now consider the following sequence of simplifications:

$$\begin{aligned}
f &= \bar{x}_1\bar{x}_2\bar{x}_3 + \bar{x}_1\bar{x}_2x_3 + \bar{x}_1x_2\bar{x}_3 + \bar{x}_1x_2x_3 + x_1x_2\bar{x}_3 + x_1x_2x_3 \\
&= \bar{x}_1\bar{x}_2(\bar{x}_3 + x_3) + \bar{x}_1x_2(\bar{x}_3 + x_3) + x_1x_2(\bar{x}_3 + x_3) \\
&= (\bar{x}_1\bar{x}_2 + \bar{x}_1x_2 + x_1x_2)(\bar{x}_3 + x_3) \\
&= \bar{x}_1\bar{x}_2 + \bar{x}_1x_2 + x_1x_2 \\
&= \bar{x}_1(\bar{x}_2 + x_2) + x_1x_2 \\
&= \bar{x}_1 + x_1x_2 \\
&= \bar{x}_1 + x_2.
\end{aligned}$$

The expression $\bar{x}_1 + x_2$ represents a circuit amazingly requiring one OR and one NOT and computes exactly same function as the long minterm expansion. This is only one example, but it shows that it can be well worth our while to try to simplify the minterm expansion using the laws of Boolean algebra. It requires creativity to simplify Boolean functions using laws of Boolean algebra. Our goal in this section is to describe a simple engineering method for simplifying minterm expansions, which is easily to adopt - the method of Karnaugh[6] maps. We can represent minterm expansion

[6] Maurice Karnaugh (1924 - 2022) was born in New York City, received his B.S. from the City College of New York and his M.S. and Ph.D. from Yale University. He was a member of the technical staff at Bell Laboratories

its by Karnaugh map. To do this, firstly, we can represent it by truth table. Here is truth table of the above function $f(x_1, x_2, x_3)$.

Table 5.4 Truth table of function f			
x_1	x_2	x_3	f
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

A Karnaugh map of this function is a 2×4 table in which rows stand for x_1 and its complement, and columns stand for all possible configurations of x_2x_3. Karnaugh map of f is as follows:

	00	01	11	10
0	1	1	1	1
1			1	1

The method of Karnaugh maps can be applied to Boolean functions of 2, 3, and 4 variables. Karnaugh maps of 2-, 3- and 4-variable functions are, respectively, 2×2, 2×4, and 4×4 tables as shown below.

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from 1952 until 1966 and Manager of Research and Development at the Federal Systems Division of AT&T from 1966 to 1970. In 1970 he joined IBM as a member of the research staff. Karnaugh has made fundamental contributions to the application of digital techniques in both computing and telecommunications.

	0	1
0	$\bar{x}_1\bar{x}_2$	$\bar{x}_1x_2$
1	$x_1\bar{x}_2$	x_1x_2

(a)

	00	01	11	10
0	$\bar{x}_1\bar{x}_2\bar{x}_3$	$\bar{x}_1\bar{x}_2x_3$	$\bar{x}_1x_2x_3$	$\bar{x}_1x_2\bar{x}_3$
1	$x_1\bar{x}_2\bar{x}_3$	$x_1\bar{x}_2x_3$	$x_1x_2x_3$	$x_1x_2\bar{x}_3$

(b)

	00	01	11	10
00	$\bar{x}_1\bar{x}_2\bar{x}_3\bar{x}_4$	$\bar{x}_1\bar{x}_2\bar{x}_3x_4$	$\bar{x}_1\bar{x}_2x_3x_4$	$\bar{x}_1\bar{x}_2x_3\bar{x}_4$
01	$\bar{x}_1x_2\bar{x}_3\bar{x}_4$	$\bar{x}_1x_2\bar{x}_3x_4$	$\bar{x}_1x_2x_3x_4$	$\bar{x}_1x_2x_3\bar{x}_4$
11	$x_1x_2\bar{x}_3\bar{x}_4$	$x_1x_2\bar{x}_3x_4$	$x_1x_2x_3x_4$	$x_1x_2x_3\bar{x}_4$
10	$x_1\bar{x}_2\bar{x}_3\bar{x}_4$	$x_1\bar{x}_2\bar{x}_3x_4$	$x_1\bar{x}_2x_3x_4$	$x_1\bar{x}_2x_3\bar{x}_4$

(c)

Problem 5.8 Use the templates to find the Karnaugh maps of the following minterm expansions.

a) $f(x_1, x_2) = \bar{x}_1\bar{x}_2 + \bar{x}_1 x_2 + x_1 x_2$.
b) $f(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2\bar{x}_3 + x_1\bar{x}_2\bar{x}_3 + x_1 x_2 x_3 + x_1 x_2 \bar{x}_3$.
c) $f(x_1, x_2, x_3, x_4) = \bar{x}_1 x_2 \bar{x}_3 x_4 + \bar{x}_1 x_2 x_3 x_4 + x_1 x_2 \bar{x}_3 x_4 + x_1 x_2 x_3 x_4 + x_1\bar{x}_2\bar{x}_3 x_4 + x_1\bar{x}_2 x_3 x_4$.

Solution This is indeed a matter of placing 1s in some cells and remaining empty other cells. Below we illustrate the results for the cases a) to c).

	0	1
0	1	1
1		1

a)

	00	01	11	10
0	1			
1	1		1	1

b)

	00	01	11	10
00				
01		1	1	
11		1	1	
10		1	1	

c)

We place all 1s in appropriate cells in the Karnaugh map and then group them in blocks. A block consists of 1s that are neighboring in a circular way row-wise and/or column-wise, but not diagonal-wise. We are allowed to group 1, 2, 4, 8, and 16 number 1s in a block. That is, a block is a collection of $2^k, k = 0,1,2,3,4$, number 1s. In general, we say that any product function, whose minterms form a subset of the minterms of a given Boolean function, is an **implicant**. Implicant that covers as many cells as possible we call a **prime implicant**. Prime implicant is the one, which is not fully embedded into another implicant. A prime implicant that covers at least one 1 (or minterm), which is not covered by any other prime implicant, we call an **essential prime implicant**.

Problem 5.9 Write three-variable minterm expansions represented by the following Karnaugh maps.

	00	01	11	10
0				
1		1	1	

(a)

	00	01	11	10
0	1	1		
1	1	1		

(b)

	00	01	11	10
0			1	
1			1	

(c)

	00	01	11	10
0			1	1
1			1	1

(d)

	00	01	11	10
0	1			
1	1			

(e)

	00	01	11	10
0		1	1	
1				

(f)

	00	01	11	10
0	1			1
1				

(g)

	00	01	11	10
0		1	1	
1		1	1	

(h)

	00	01	11	10
0	1			1
1	1			1

(i)

	00	01	11	10
0	1	1	1	1
1				

(j)

	00	01	11	10
0		1		
1			1	

(k)

	00	01	11	10
0	1		1	
1		1		1

(l)

Solution

a) $f(x_1, x_2, x_3) = x_1\bar{x}_2x_3 + x_1x_2x_3$.
b) $f(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2\bar{x}_3 + \bar{x}_1\bar{x}_2x_3 + x_1\bar{x}_2\bar{x}_3 + x_1\bar{x}_2x_3$.
c) $f(x_1, x_2, x_3) = \bar{x}_1x_2x_3 + x_1x_2x_3$.
d) $f(x_1, x_2, x_3) = \bar{x}_1x_2x_3 + \bar{x}_1x_2\bar{x}_3 + x_1x_2x_3 + x_1x_2\bar{x}_3$.
e) $f(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2\bar{x}_3 + x_1\bar{x}_2\bar{x}_3$.
f) $f(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2x_3 + \bar{x}_1x_2x_3$.
g) $f(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2\bar{x}_3 + \bar{x}_1x_2\bar{x}_3$.
h) $f(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2x_3 + \bar{x}_1x_2x_3 + x_1\bar{x}_2x_3 + x_1x_2x_3$.
i) $f(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2\bar{x}_3 + \bar{x}_1x_2\bar{x}_3 + x_1\bar{x}_2\bar{x}_3 + x_1x_2\bar{x}_3$.
j) $f(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2\bar{x}_3 + \bar{x}_1\bar{x}_2x_3 + \bar{x}_1x_2x_3 + \bar{x}_1x_2\bar{x}_3$.
k) $f(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2x_3 + x_1x_2x_3$.
l) $f(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2\bar{x}_3 + \bar{x}_1x_2x_3 + x_1\bar{x}_2x_3 + x_1x_2\bar{x}_3$.

To simplify Boolean function which is present in the form of sum of products. We first draw and fill in the Karnaugh map. Each minterm (or product function) is present by 1 placed in appropriate cell of the Karnaugh map. After we place 1s in proper cells, we cover all 1s with as few blocks as possible. This is achieved by selecting all blocks corresponding to essential prime implicants. If there are 1s that are not covered by essential prime implicants, we cover uncovered 1s as few prime implicants as possible. In order to identify a product function, we collect variables whose value remain unchanged for all 1s within the block. For example, if $x = 0$ for all

cells within the block then $\bar{x}$ is present in the product function. Likewise, if $x = 1$ for all cells within the block then x is present in the product function identifying the given product function. If $x = 0$ for some cells and $x = 1$ for other cells within still same block then neither x not $\bar{x}$ presents in the product function.

We can easily observe that the number of variables in the product function decreases with increase of the block size. For instance, for a Boolean function of four variables, a cell is represented by a product function of four variables, while a block of four neighboring cells (all in the same row, or square block of four cells) is represented by a product function of two variables.

Problem 5.10 For each of above Karnaugh maps, write minterm that identifies corresponding block of all 1s.

	0	1
0		
1	1	1

(a)

	0	1
0	1	
1	1	

(b)

00	1	1		
01	1	1		
11				
10				

(c)

00		1		
01				
11				
10		1		

(d)

	00	01	11	10
00	1	1	1	1
01				
11				
10	1	1	1	1

(e)

	00	01	11	10
00		1	1	
01				
11				
10		1	1	

(f)

	00	01	11	10
00	1			1
01				
11				
10	1			1

(g)

Solution

a) x_1 b) $\bar{x}_2$ c) $\bar{x}_1\bar{x}_3$ d) $\bar{x}_2\bar{x}_3x_4$ e) $\bar{x}_2$ f) $\bar{x}_2x_4$ g) $\bar{x}_2\bar{x}_4$

Problem 5.11 Use Karnaugh map method to simplify the following six Boolean functions.

	0	1
0	1	1
1		1

(a)

	00	01	11	10
0	1			
1	1		1	1

(b)

	00	01	11	10
0	1			1
1				1

(c)

	00	01	11	10
0		1		1
1	1		1	

(d)

	00	01	11	10
00				
01		1	1	
11		1	1	
10		1	1	

(e)

	00	01	11	10
00	1			
01	1	1	1	1
11				1
10	1	1	1	

(f)

Solution

a) $\bar{x}_1 + x_2$.
b) $\bar{x}_2\bar{x}_3 + x_1x_2$.
c) $\bar{x}_1\bar{x}_3 + x_2\bar{x}_3$.
d) $\bar{x}_1\bar{x}_2x_3 + \bar{x}_1x_2\bar{x}_3 + x_1\bar{x}_2\bar{x}_3 + x_1x_2x_3$.
e) $x_2x_4 + x_1x_4$.
f) This is a bit difficult case. The problem is that it is not possible to cover all 1s by essential prime implicants only. We observe that $\bar{x}_1x_2$ is an essential prime implicant, since it is the only prime implicant, that covers the minterms 0101 and 0111. Likewise, $x_2x_3\bar{x}_4$ is also essential because it is the only one, which covers 1110. Similarly, $x_1\bar{x}_2x_4$ is essential prime implicant, because it is the only one, which covers 1011. Then we find that remaining two 1s (minterms 0000 and 1000) can be covered by three prime impicants, which are $\bar{x}_1\bar{x}_3\bar{x}_4$, $x_1\bar{x}_2\bar{x}_3$ and $\bar{x}_2\bar{x}_3\bar{x}_4$. Our intention is to cover remaining 1s with the minimum number of prime implicants. As far as $\bar{x}_2\bar{x}_3\bar{x}_4$ is the only prime implicant, which covers both 0000 and 1000 we add $\bar{x}_2\bar{x}_3\bar{x}_4$ to the simplied expression. Below we present simplified minterm expansion obtained by the above procedure.

$$\bar{x}_1 x_2 + x_2 x_3 \bar{x}_4 + x_1 \bar{x}_2 x_4 + \bar{x}_2 \bar{x}_3 \bar{x}_4.$$

Problem 5.12 Simplify Boolean function represented by the following Karnaugh map.

	00	01	11	10
00	1		1	1
01			1	1
11	1	1	1	1
10	1	1		

Solution We identify six prime implicants in the Karnaugh map, but surprisingly none of these prime implicants is essential. There may be multiple solutions. Below we show one way how all 1s can be covered by minimum number of prime implicants.

$$x_1 x_2 + x_1 \bar{x}_3 + \bar{x}_1 x_3 + \bar{x}_1 \bar{x}_2 \bar{x}_4.$$

However, this is only one of the solutions.

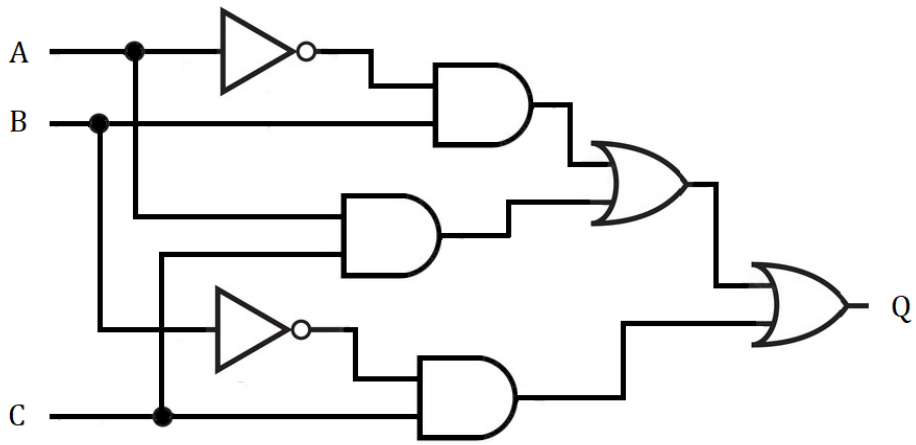
Exercises

1. Find the values of these expressions.

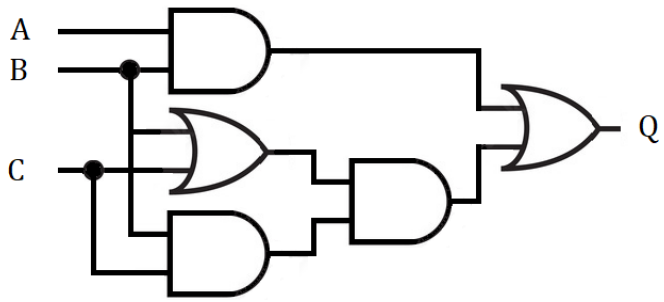
   a) $1 \cdot \bar{0}$ b) $1 + \bar{1}$ c) $0 \cdot \bar{0}$ d) $\left(\overline{1 + \bar{0}}\right)$

2. Write $x\bar{y} + 1$ in dual form.
3. Find the dual of $x(y + 0)$.

4. Construct the identity from $\bar{x} \cdot 1 + (\bar{y} + z)$ by taking duals.
5. Find the output of the following logic circuit.



6. Given the following logic diagram, determine its output.



7. Boolean function $f(x, y, z)$ is represented by truth set {000, 010, 011, 101, 110}. Write $f(x, y, z)$ in the form of maxterm expansion.
8. Write minterm expansion of the Boolean function which represented by the following truth table.

xyz	$f(x,y,z)$
000	1
001	0
010	1
011	0
100	1
101	0
110	1
111	0

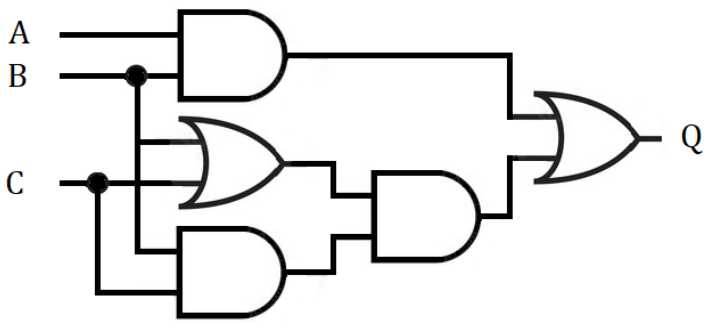
9. Boolean function is represented in the form of maxterm expansion as follows

$$(\bar{x}+y+\bar{z})(\bar{x}+\bar{y}+\bar{z})(x+y+\bar{z})(x+\bar{y}+\bar{z}).$$

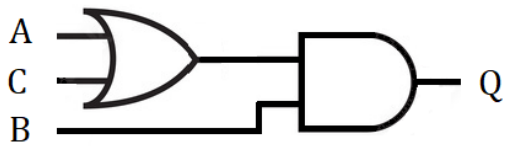
Write this Boolean function in the form of minterm expansion.

10. Given truth-set $\{0001, 0010, 0100, 0101, 0111, 0110, 1100, 1101, 1111, 1110, 1000, 1001, 1010\}$ of Boolean function $f(x_1, x_2, x_3, x_4)$, Write corresponding minterm expansion, draw Karnaugh map and simplify minterm expansion by Karnaugh map.

11. Check whether the following two logic diagrams are equivalent.

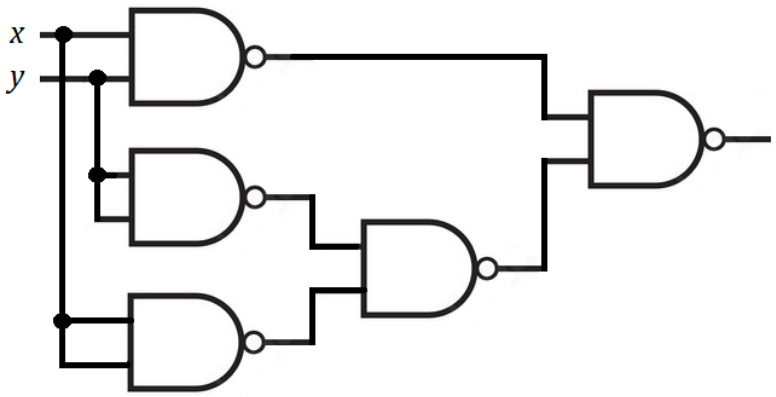


(a)



(b)

12. Which logic gate is logically equivalent to the logic circuit in the figure?



13. Simplify Boolean functions represented as follows:

	00	01	11	10
0				
1		1	1	

(a)

	00	01	11	10
0	1	1		
1	1	1		

(b)

	00	01	11	10
0			1	
1			1	

(c)

	00	01	11	10
0			1	1
1			1	1

(d)

	00	01	11	10
0	1			
1	1			

(e)

	00	01	11	10
0		1	1	
1				

(f)

	00	01	11	10
0	1			1
1				

(g)

	00	01	11	10
0		1	1	
1		1	1	

(h)

	00	01	11	10
0	1			1
1	1			1

(i)

	00	01	11	10
0	1	1	1	1
1				

(j)

	00	01	11	10
0		1		
1			1	

(k)

	00	01	11	10
0	1		1	
1		1		1

(l)

14. Simplify Boolean function represented as follows

	00	01	11	10
00			1	
01		1	1	
11	1	1	1	
10			1	

15. Simplify the Boolean function represented by the following Karnaugh map.
16. Given Karnaugh map, write the simplified minterm expansion.

	00	01	11	10
00			1	1
01	1	1		
11	1	1		
10			1	1

17. Simplify the Boolean function, which is represented by the following Karnaugh.

	00	01	11	10
00	1	1		
01	1	1	1	1
11	1			
10				1

Chapter 6 Mathematical Induction

Mathematical induction is a proof method of mathematics which establishes the correctness of a statement about natural numbers. The theme of mathematical induction can be expressed by the pronouncement "If you start out right and nothing can go wrong, then you will always be right". The context to which we apply this framework is the set of statements each of which is stated in terms of natural numbers. In this chapter, we will study both weak and strong forms of mathematical induction.

6.1 Weak Form

The Principle of Mathematical Induction (weak form) Given a statement P concerning the natural number n, suppose

- P is true for some particular integer n_0.
- If P is true for some particular integer $k \geq n_0$, then it is true for the next integer $k + 1$.

Then P is true for all integers $n \geq n_0$.

Problem 6.1 Prove that for any integer $n \geq 1$, the sum of the odd integers from 1 to $2n - 1$ is n^2.

Solution Expressed more precisely, we need to show that

$$1 + 3 + \cdots + (2n - 1) = n^2$$

for all integers $n \geq 1$.

Basic case: Substituting $n = 1$ in the above formula, we obtain $1 = 1$ which is correct. The basic step is done.

Induction case: Assume that $P(k)$ holds. Then

$$1 + 3 + \cdots + (2k - 1) = k^2.$$

Based on this hypothesis we will show that $P(k + 1)$ holds as well. Statement for $P(k + 1)$ has the following view

$$1 + 3 + \cdots + (2k - 1) + (2k + 1) = (k + 1)^2$$
$$[1 + 3 + \cdots + (2k - 1)] + (2k + 1) = k^2 + (2k + 1)$$

By subtracting $2k + 1$ from both sides of the above equation, we receive

$$1 + 3 + \cdots + (2k - 1) = k^2,$$

which is true by the induction hypothesis.

Problem 6.2 Prove that for any natural number $n \geq 1$,

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n + 1)(2n + 1)}{6}.$$

Solution

Basic case: For $n = 1$we observe that, the statement holds since

$$1^2 = \frac{1(2)(3)}{6} \Rightarrow 1 = 1$$

Induction case: In hypothesis phase, we suppose that $P(k)$ is true, that is

$$1^2 + 2^2 + \cdots + k^2 = \frac{k(k + 1)(2k + 1)}{6}$$

holds for some $k \geq 1$. By using this fact we will show that $P(k+1)$ is also true, meaning that

$$1^2 + 2^2 + \cdots + k^2 + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2 =$$
$$\frac{k(k+1)(2k+1) + 6(k+1)^2}{6} = \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} =$$
$$\frac{(k+1)[2k^2 + 7k + 6]}{6} = \frac{(k+1)(k+2)(2k+3)}{6}$$
$$= \frac{(k+1)(k+2)(2k+3)}{6}$$

which is just what we wanted to show. From what we conclude that by the Principle of Mathematical Induction the statement is true for all integers $n \geq 1$.

Problem 6.3 Prove that for any integer $n \geq 1$, $2^{2n} - 1$ is divisible by 3.

Solution

Basic case: We substitute n by 1 in the formula and obtain $2^2 - 1 = 3$, which is definitely divisible by 3.

Induction case: We assume that $2^{2k} - 1$ is divisible by 3. By using this fact, we will show that $2^{2(k+1)} - 1$ is divisible by 3. Further analysis shows that

$$2^{2(k+1)} - 1 = 2^{2k+2} - 1 = 4 \cdot 2^{2k} - 1 = (3+1)2^{2k} - 1$$
$$= 3 \cdot 2^{2k} + (2^{2k} - 1).$$

The first term, $3 \cdot 2^{2k}$, is definitely divisible by 3. The second term is divisible by the hypothesis. The proof is out here.

6.2 Strong Form

The Principle of Mathematical Induction (strong form) Given a statement P concerning integer n, suppose

- P is true for some integer n_0;
- If $k > n_0$ is any integer and P is true for all integers l in the range $n_0 \leq l < k$, then it is true also for k.

Then P is true for all integers $n \geq n_0$.

The two forms of the Principle of Mathematical Induction differ in the statement of the induction hypothesis. In the weak form, we assume the truth of the statement for just one particular integer and we had to prove it true for the next largest integer. In the strong form of mathematical induction, we assume that the statement is true for a given range and based on this assertion we show that then it is correct for that integer. When we first encounter mathematical induction, it seems to be the weak form, but not strong form, which is often used. This is a misconception. In the analysis of finite structures, we extensively employ the strong form of the principle. Recall that, Fundamental Theorem of Arithmetic states that every natural number greater than 1 is either prime or is a product of primes. The strong form of mathematical induction affords a very straightforward proof of this result.

Problem 6.4 (Fundamental Theorem of Mathematics) Use the strong form of mathematical induction to prove that every natural number $n \geq 2$ is either prime or the product of primes.

Proof The theorem is a statement about all integers $n \geq 2$. The first such integer is $n_0 = 2$, is prime, so that assertion of the theorem is true. Now let $k > 2$ and suppose that the claim remains correct for all positive integers $l, 2 \leq l < k$; that is, assume that every integer l

in the interval $2 \leq l < k$ is either prime or the product of primes. We must use the induction hypothesis to prove that k has this same property. There may one be two cases: either k is prime or it is composite number, that is, k is divisible by not only by 1 and by itself, but also by other natural numbers. There is nothing more to do if k is prime. If k is instead a composite number, then k can be factored into $k = ab$, where $2 \leq a, b < k$. According to the induction hypothesis, a and b are either prime or product of prime factors. From what we conclude that k is the product of primes, as the theorem claims. Hence, $n \geq 2$ is prime or the product of primes.

Exercises

1. Write the base case for the induction statement $7^n > n^3$ for $n \geq 3$.
2. Write the base case for $4^{n+1} > (n+1)^2$ with $n \geq 2$.
3. Given set A with $|A| = n$, use the principle of mathematical induction to prove that $|P(A)| = 2^n$.
4. Prove that $n^2 - 5n - 1 > 0$ holds for $n \geq 6$.
5. Prove that $n \leq 3^n$ for $n \in N$.
6. Prove that $n! > 2n$ for $n \geq 4$.
7. Prove that $n^3 + 2n$ is divisible by 3 for every natural number.
8. Let A be a set with n elements. Prove that A has 2^n subsets.
9. Prove that $5^n - 4n - 1$ is divisible by 16 for $n \geq 1$.
10. Prove that $3^n > 2n$ for $n \geq 1$.
11. Prove that $2^n \geq n + 10$ for $n \geq 4$.
12. Show that $11^n - 4^n$ is divisible by 7 for $n \geq 1$.
13. Show that $n^2 > n + 1$ for $n \geq 2$.
14. Show that $2^n \geq n^3$ for $n \geq 10$.
15. Prove that every third Fibonacci number is even.

16. Show that for $n \in N$

$$\left(1+\frac{1}{2}\right)^n \geq 1+\frac{n}{2}.$$

Chapter 7 Recursion

7.1 Recursively Defined Sequences

Suppose n is a natural number. How should we define 2^n? We could write

$$2^n = \underbrace{2 \times 2 \times \cdots \times 2}_{n}$$

or

$$2^1 = 2 \text{ and, for } k \geq 1,\ 2^{k+1} = 2 \cdot 2^k.$$

The latter statement is an example of a recursive definition. It explicitly defines 2^n when $n = 1$ and then, assuming 2^n has been defined for $n = k$, defines it for $n = k + 1$. By the Principle of Mathematical Induction, we know that $2^n = 2 \cdot 2^{n-1}$ remains correct for all integers $n \geq 1$.

Another expression that is most naturally defined recursively is $n!$. If we write

$$0! = 1 \text{ and, for } k \geq 0,\ (k+1)! = (k+1) \cdot k!$$

then it follows by the Principle of Mathematical Induction that $n!$ has been defined for every $n \geq 0$.

Some sequences of numbers can be defined recursively. A sequence of numbers can be represented as a function whose domain is infinite set of integers (often set of natural numbers) and whose range is the set of real numbers. For example, the sequence

1,4,9,16, ... is the function $f: N \to R$ defined by $f(n) = n^2$. We call the numbers in the list as **terms** of the sequence. We can define recursively the sequence 2,4,8,16, ... as follows:

$$a_1 = 2 \text{ and, for } k \geq 1,\ a_{k+1} = 2a_k\ .$$

The equation $a_{k+1} = 2a_k$, which defines one term of the sequence based on the previous term, is called a **recurrence relation**. We say $a_1 = 2$ is the **initial condition**.

Problem 7.1 Write the first six terms of the sequence defined by $a_1 = 1$, $a_{k+1} = 3a_k + 1$ for $k \geq 1$.

Solution The first six terms are

$$\begin{aligned}
a_1 &= 1 \\
a_2 &= 3a_1 + 1 = 3(1) + 1 = 4 \\
a_3 &= 3a_2 + 1 = 3(4) + 1 = 13 \\
a_4 &= 3a_3 + 1 = 3(13) + 1 = 40 \\
a_5 &= 3a_4 + 1 = 3(40) + 1 = 121 \\
a_6 &= 3a_5 + 1 = 3(121) + 1 = 364
\end{aligned}$$

Problem 7.2 A sequence is defined recursively by $a_0 = 1$, $a_1 = 4$ and $a_n = 4a_{n-1} - 4a_{n-2}$ for $n \geq 2$. Find the first six terms of this sequence.

Solution The first six terms of the recurrence relation are as follows.

$$\begin{aligned}
a_0 &= 1 \\
a_1 &= 4 \\
a_2 &= 4a_1 - 4a_0 = 4 \cdot 4 - 4 \cdot 1 = 16 - 4 = 12 \\
a_3 &= 4a_2 - 4a_1 = 4 \cdot 12 - 4 \cdot 4 = 48 - 16 = 32 \\
a_4 &= 4a_3 - 4a_2 = 4 \cdot 32 - 4 \cdot 12 = 128 - 48 = 80 \\
a_5 &= 4a_4 - 4a_3 = 4 \cdot 80 - 4 \cdot 32 = 320 - 128 = 192
\end{aligned}$$

7.2 Some Special Sequences

Definition 7.1 (Arithmetic sequence) The arithmetic sequence is the one whose first term and common difference are a and d, respectively, and which is defined recursively as follows

$$a_1 = a \text{ and for } k = 1,2,3, \dots\ a_{k+1} = a_k + d.$$

We can arrange the terms in an arithmetic sequence as follows:

$$a, a + d, a + 2d, a + 3d, \dots$$

Continuing the same logic, we derive a formula for finding the n'th term of arithmetic sequence:

$$a_n = a_{n-1} + d = a + (n - 1)d.$$

One can observe that sum of the terms that are in the same distance from both ends of arithmetic sequence have the same value. For example, $a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2} = \cdots$ and this rule remains same for all related pairs of the terms of arithmetic sequence. Based on above observation, we can devise a formula for sum of the first n terms in arithmetic sequence as follows:

$$S_n = \frac{n}{2}[2a + (n - 1)d].$$

Problem 7.3 Find sum of the first 100 terms of the arithmetic sequence -17, -12, -7, -2, 3, ...

Solution It can be easily seen that $a = a_1 = -17, d = -2 - (-7) = 5$. Then we substitute in the above summation formula a and d by their values and calculate the sum of the first 100 terms as follows

$$S = \frac{100}{2}[2(-17) + (100 - 1)5] = 50 \cdot (-34 + 99 \cdot 5)$$
$$= 50 \cdot (495 - 34) = 50 \cdot 461 = 23050.$$

Definition 7.2 (Geometric sequence) The geometric sequence is the one whose first term and common ratio are a and r, respectively, and which is recursively defined as follows:

$$a_1 = a \text{ and for } k \geq 1 \; a_{k+1} = ra_k.$$

The geometric sequence is of the form $a, ar, ar^2, ar^3, ar^4, \dots$, in which n'th term is $a_n = ar^{n-1}$. It is rather straightforward to show that the sum S_n of n terms of geometric sequence, provided $r \neq 1$, can be found by the formula

$$S_n = \frac{a(1 - r^n)}{1 - r}.$$

Problem 7.4 Find the sum of 29 terms of the geometric sequence with $a = 8^{12}$ and $r = 1/2$.
Solution In the above formula we substitute a and r by their values and find the sum as follows

$$S = \frac{8^{12}\left[1 - \left(\frac{1}{2}\right)^{29}\right]}{1 - \frac{1}{2}} = \frac{8^{12}[2^{29} - 1]2}{2^{29}} = \frac{2^{37}[2^{29} - 1]}{2^{29}} = 2^8[2^{29} - 1].$$

7.3 Recurrence Relations

In any recurrence relation, each term is defined in terms of previous terms. **Order** of recurrence relation is the number of successive previous terms used to determine the current term of the relation. Let us consider the following recurrence relation of second order

$$a_n = ra_{n-1} + sa_{n-2} + f(n).$$

Notice that, if $f(n) = 0$, then the recurrence relation is **homogenous**, and it is **inhomogenous** otherwise. If r and s are two constants then we say that **recurrence relation with constant coefficients** is under consideration. Later we will study how to find a general solution of second order homogenous recurrence relation with constant coefficients. In a **linear recurrence relation**, any term is a linear combination of previous terms.

Example 7.1 The following are second order linear recurrence relations with constants coefficients.

$a_n = a_{n-1} + a_{n-2}$ is the Fibonacci[7] sequence – a recurrence relation of the second order.

[7] Fibonacci (1170–1250) (short for filius Bonacci, or "son of Bonacci") was also known as Leonardo of Pisa. He was born in the Italian commercial center of Pisa. Fibonacci was a merchant who traveled extensively throughout the Mideast, where he came into contact with Arabian mathematics. In his book Liber Abaci, Fibonacci introduced the European world to Arabic notation for numerals and algorithms for arithmetic. Fibonacci also wrote books on geometry and trigonometry and on Diophantine equations, which involve finding integer solutions to equations.

$a_n = 5a_{n-1} - 6a_{n-2} + n$ is linear inhomogenous recurrence relation of the second order with constant coefficients.

However, $a_n = a_{n-1}a_{n-2} + n^2$ is not linear recurrence relation because a_n depends on the product of the previous two terms but not each of them.

Given a particular recurrence relation and certain initial conditions, you are asked to guess a formula for the nth term and prove that a guess is correct. Guessing is an important tool in mathematics and a skill that we can sharpen through practice. We now confess that, there is a procedure for solving most of the recurrence relations, which have encountered so far.

We describe a procedure for solving recurrence relations of the form

$$a_n = ra_{n-1} + sa_{n-2}$$

where r and s are constants. We call such recurrence relations **second order homogenous linear recurrence relation with constant coefficients**.

We associate the quadratic polynomial with this recurrence relation:

$$x^2 - rx - s = 0.$$

It is the **characteristic equation** of the recurrence relation. The roots are the **characteristic roots** of the recurrence relation. For example, the recurrence relation

$$a_n = 5a_{n-1} - 6a_{n-2}$$

has the following characteristic equation

$$x^2 - 5x + 6 = 0.$$

Method Let x_1 and x_2 be the two roots of the characteristic equation $x^2 - rx - s = 0$. Then the solution of the recurrence relation $a_n = ra_{n-1} + sa_{n-2}$, $n \geq 2$, is

$$a_n = \begin{cases} c_1 x_1^n + c_2 x_2^n, x_1 \neq x_2, \\ c_1 x^n + c_2 n x^n, x_1 = x_2 = x\,. \end{cases}$$

To determine constants c_1 and c_2 we use the initial conditions to write system of linear equations and then find simultaneous solution of the system of linear equations.

Problem 7.5 Solve the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$, $n \geq 2$ given that $a_0 = 1$, $a_1 = 4$.

Solution The characteristic equation in this case is presented by $x^2 - 5x + 6 = 0$. Let us represent the quadratic equation in the factored form as follows:

$$(x - 2)(x - 3) = 0.$$

Thus, the roots are $x_1 = 2$ and $x_2 = 3$. We represent the solution in the form:

$$a_n = c_1 x_1^n + c_2 x_2^n.$$

Then we take into account the initial conditions and find that

$$a_0 = c_1 + c_2 = 1,$$
$$a_1 = 2c_1 + 3c_2 = 4.$$

Simultaneously solving the above system, we obtain $c_1 = 1 - c_2$ and $2(1 - c_2) + 3c_2 = 4$ and, thefore, $c_2 = 2$. Thus, $c_1 = -1$. So the solution is

$$a_n = -2^n + 2 \cdot 3^n.$$

Problem 7.6 Solve the recurrence relation $a_n = 4a_{n-1} - 4a_{n-2}$, $n \geq 2$, with the initial conditions $a_0 = 1$, $a_1 = 4$.

Solution The characteristic equation $x^2 - 4x + 4 = 0$ and it has repeated root $x = 2$. Hence, the solution is $a_n = c_1 2^n + c_2 n 2^n$. The initial conditions yield $c_1 = 1$ and $2c_1 + 2c_2 = 4$, from which we obtain $c_2 = 1$. Thus, $a_n = 2^n + n2^n = (1 + n)2^n$.

Exercises

1. Consider arithmetic sequence from which we know that its second element is 15 and the sum of the second, third and fourth elements is 60. Find the first element a and the common difference d.
2. Find the sum of the first 71 positive integers.
3. Find the sum of the first 10 elements of arithmetic sequence whose 3rd, 4th and 5th elements are 3, 10 and 17, respectively.
4. The sum of the first 5 terms of arithmetic sequence is 5 and $a_1 = -7$. Find d.
5. Sum of the first 7 terms of an arithmetic sequence is 119 and the common difference is 3. Find the first term.
6. Consider the arithmetic sequence with $a_1 = 2$ and $a_2 = -2$. Find the value of a_{101}.
7. If $a_1 = 2$ and $d = 3$, then which term in the sequence does equal to 401?

8. Find the sum of the first 10 elements of arithmetic sequence whose 1st and 5th elements are -3 and 13, respectively.
9. Find the number of terms in the geometric sequence 32, 256, 2048, 16384, … , 2^{50}.
10. Find the 7th term of the geometric sequence -256, 128, -64, ...
11. Consider the geometric sequence with $a_1 = -2048$ and $a_4 = 256$. Find the common ratio.
12. Find the term at position 11 in the geometric sequence 32, 256, 2048, 16384, ...
13. Consider the geometric sequence with $a_1 = 4096$ and $a_3 = 1024$. Determine r, a_{10} and a_{20} and S_{12}.
14. Find the common ratio, if fourth and fifth elements of a geometric series are $a_4 = -96$ and $a_5 = 192$.
15. If the first term of a geometric sequence is 2 and its sixth term is 64, then what is the sum of first 100 terms?
16. What is the product of the 90th and 100th terms of a geometric sequence whose first three terms are 4, 12 and 36, respectively?
17. Find the term at position 10 in the geometric sequence in which 1st, 2nd and 5th terms are -5, 5/2 and -5/16, respectively.
18. Which of the following is a second order linear recurrence relation with constant coefficients?

    a) $a_k = -5a_{k-2} + n^2$
    b) $b_k = kb_{k-1} + b_{k-2}$
    c) $c_k = c_{k-1} \cdot c_{k-2}$
    d) $d_k = 3d_{k-2} + d_{k-3}$
    e) $r_k = 100\, r_{k-1}$

19. Which of the following is characteristic polynomial of the recurrence relation $a_n = -6a_{n-1} + 5a_{n-2}$?

a) $x^2 + 6x - 5$
b) $x^2 - 6x + 5$
c) $a_n - 6a_{n-1} + 5a_{n-2}$
d) $a_n + 6a_{n-1} - 5a_{n-2}$

20. Which of the following is homogeneous second order recurrence relation with constant coefficients?

a) $a_n = -6a_{n-1} + 5a_{n-3}$
b) $a_n = 24a_{n-1} - 15a_{n-2} + 2n$
c) $a_n = na_{n-1} + 2a_{n-2}$
d) $a_{n-4} = 7a_{n-5} - 11a_{n-6}$
e) $a_n = a_{n-2} + 27n^2$

21. Which of the following is a homogeneous second order linear recurrence relation with constant coefficients?

a) $a_n = 4a_{n-2}$
b) $a_n = 3a_{n-1}a_{n-2}$
c) $a_n = 2a_{n-1} + 10na_{n-2}$
d) $a_n = a_{n-1} + a_{n-2} + n$
e) $a_n = 7a_{n-2} + 15a_{n-3}$

22. Find a_5 of the recurrence relation $a_n = a_{n-1} + a_{n-2}$ if $a_0 = 2$ and $a_1 = 3$.
23. Solve recurrence relation $a_n = 2a_{n-2}$ if $a_0 = 2$ and $a_1 = 0$.
24. Solve recurrence relation $a_n = a_{n-1} + \frac{1}{4}a_{n-2}$ with $a_0 = a_1 = 1$.
25. Solve the recurrence relation $a_n = 10a_{n-1} - 25a_{n-2}$ where $a_0 = 3$ and $a_1 = 17$.

26. Find the solution of the second-degree linear homogeneous recurrence relation with constant coefficients $a_n = 7a_{n-1} - 10a_{n-2}$, $a_0 = 2$, $a_1 = 3$.
27. Solve the recurrence relation $a_n = 5a_{n-2}$, $a_0 = 1$, $a_1 = 1$.

Chapter 8 Inclusion-Exclusion

Of the 100 people who have already registered to the word processing course so far, three quoters are men and 80% own personal digital assistants (PDAs). Find a formula for the number of women in the course who do not have PDAs. How large might this number be? How small? How many of the men registered in this course could conceivably own PDAs?

The set induced in part a) is illustrated in the left of Figure 8.1, while the set induced in part b) is demonstrated in the right of Figure 8.1.

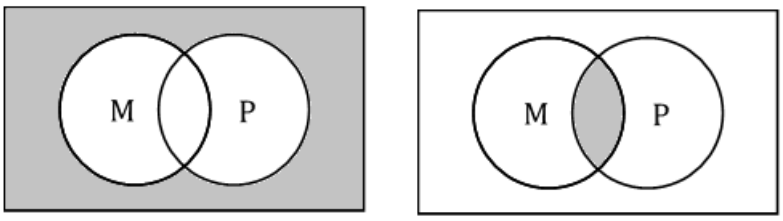


Figure 8.1 The gray segment in the left represents set of women who do not have PDAs and the gray segment in the right stands for set of men who own PDAs.

To answer the questions, we introduce the set U of all registrants, set M of male registrants, and set P of those registrants who own PDAs. We are given that $|U| = 100, |M| = 75$, and $|P| = 80$. The set of women who do not have PDAs is represented by $M^c \cap P^c$ and it is the size of this set with which part a) is concerned.

$$M^c \cap P^c = (M \cup P)^c$$

and so

$$|M^c \cap P^c| = |(M \cup P)^c| = 100 - |M \cup P|.$$

How big is $M \cup P$? Adding the number of men and the number of registrants with PDAs counts twice the men with PDAs. Thus,

$$|M \cup P| = |M| + |P| - |M \cap P| = 75 + 80 - |M \cap P| = 155 - |M \cap P|.$$

We can find the number of women without PDAs as follows:

$$|M^c \cap P^c| = 100 - (155 - |M \cap P|) = |M \cap P| - 55.$$

Since $M \cap P$ is a subset of M, a set of 75 people, $|M \cap P| \leq |M| = 75$ and, hence, $|M^c \cap P^c| \leq 75 - 55 = 20$. Consequently, $|M^c \cap P^c|$ is 0 if $|M \cap P| = 55$. The number of women without PDAs is between 0 and 20, inclusively.

Part b) asks about $M \cap P$. Observing that $|M \cap P| - 55 \geq 0$, we see that $|M \cap P| \geq 55$; that is, at least 55 men must own PDAs. As noted previously, the upper bound for $|M \cap P|$ is 75 and it is possible for this number to be realized.

Suppose that $|M \cap P| = 72$. Then

$$|M \cup P| = |M| + |P| - |M \cap P| = 75 + 80 - 72 = 83.$$

There are 83 people in the class, who are men or own a PDA. Surely, $75 - 72 = 3$;

$$|M \backslash P| = |M| - |M \cap P| = 75 - 72 = 3.$$

How many of the owners of PDAs would be women? Surely, $80 - 72 = 8$;

$$|P \backslash M| = |P| - |M \cap P| = 80 - 72 = 8.$$

How many of those registered are either men without PDAs or women with PDAs? We can answer this question, which asks for $|M \oplus P|$, in several ways. It is the number of men without PDAs plus the number of women with PDAs; that is,

$$|M \oplus P| = |M \backslash P| + |P \backslash M| = 3 + 8 = 11.$$

Definition 8.1 (Principle of Inclusion-Exclusion) Let $A_1, A_2, \dots, A_n$ be the finite sets, the number of elements in the union $A_1 \cup A_2 \cup \cdots \cup A_n$ is given by the formula

$$|A_1 \cup A_2 \cup \cdots \cup A_n| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots (-1)^{n+1} |A_1 \cap A_2 \cap \cdots \cap A_n|.$$

Problem 8.1 Out of 30 personal computers (PCs) 20 run Windows, 8 are supplied with 17-inch displays, 25 have CD-ROM drives, 20 have at least two of these features, and 6 have all three features. Find number of PCs that have at least one of three features. Find the number of PCs that have none of these features. Find the number of PCs that have exactly one feature.

Solution Let W be the set of PCs running under Windows, M the set of PCs with 17-inch Monitor, and C the set with CD-ROM drives. We are given $|W| = 20, |M| = 8, |C| = 25$,

$$|(W \cap M) \cup (W \cap C) \cup (M \cap C)| = 20,$$

and $|W \cap M \cap C| = 6$. By the Principle of Inclusion-Exclusion,

$$\begin{aligned} 20 &= |(W \cap M) \cup (W \cap C) \cup (M \cap C)| \\ &= |W \cap M| + |W \cap C| + |M \cap C| - 2|W \cap M \cap C|. \end{aligned}$$

Thus,

$$|W \cap C| + |W \cap M| + |M \cap C| = 20 + 2(6) = 32.$$

The number of PCs with at least one feature is

$$\begin{aligned} |W \cup M \cup C| &= |W| + |M| + |C| - |W \cap M| - |W \cap C| \\ &\quad -|M \cap C| + |W \cap M \cap C| \\ &= 20 + 8 + 25 - (|W \cap M| + |W \cap C| + |M \cap C|) + 6 \\ &= 59 - 32 = 27. \end{aligned}$$

It follows that $30 - 27 = 3$ PCs have none of the specified features. Since the number of computers with exactly one feature is the number with at least one feature less the number with at least two, the number with exactly one is $27 - 20 = 7$.

Problem 8.2

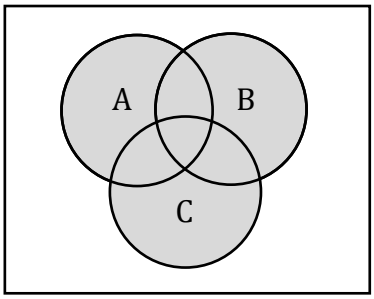
a) How many integers between 1 and 300, inclusively, are divisible by at least one of 3, 5, 7?

a) How many by 3 and by 5 but not by 7?

b) How many by 5 but neither 3 nor 7?

Solution Let A, B, and C be the sets of integers between 1 and 300 that are divisible by 3, by 5 and by 7, respectively. Stated in other words,

set of integers that are divisible by at least one of 3, 5, 7 is represented by gray segment in Venn diagram.



An integer that is divisible by 3 or 5 or 7 belongs at least one of the sets A, B and C. Thus, part a) asks us to find $|A \cup B \cup C|$. We know that

$$|A \cup B \cup C| = |A| + |B| + |C| - (|A \cap B| + |B \cap C| + |A \cap C|) + |A \cap B \cap C|$$

Let us calculate $|A|$, $|B|$, $|C|$, $|A \cap B|$, $|A \cap C|$, $|B \cap C|$ and $|A \cap B \cap C|$. The elements of A are 3,6,9,12, ... 300, that is, each third integer in the range 1 to 300. Thus, $|A| = 100$. For $|B|$ and $|C|$ we have

$$|B| = \left\lfloor \frac{300}{5} \right\rfloor = 60 \text{ and } |C| = \left\lfloor \frac{300}{7} \right\rfloor = 42.$$

Then we find $|A \cap B|$, the set of integers between 1 and 300 which are divisible by both 3 and 5. Since 3 and 5 are relatively prime numbers, any number divisible by each of them must be divisible by their product. Therefore, $A \cap B$ is just the set of integers between 1 and 300 which are divisible by 15 and, similarly, $A \cap C$, $B \cap C$ and $A \cap B \cap C$ are the sets of integers between 1 and 300 which are divisible by 21, 35 and 105, respectively. So we have

$$|A \cap B| = \left\lfloor \frac{300}{15} \right\rfloor = 20, |A \cap C| = \left\lfloor \frac{300}{21} \right\rfloor = 14,$$
$$|B \cap C| = \left\lfloor \frac{300}{35} \right\rfloor = 8, |A \cap B \cap C| = \left\lfloor \frac{300}{105} \right\rfloor = 2$$

Substituting unknowns by calculated values we obtain

$$|A \cup B \cup C| = (100 + 60 + 42) - (20 + 14 + 8) + 2 = 162.$$

Part b) asks to find the numbers in the range 1 to 300 that are divisible by 3 and 5, but not 7. These are precisely numbers in $(A \cap B)\backslash C$, a set of cardiality $|A \cap B| - |A \cap B \cap C| = 20 - 2 = 18$.
Part c) asks to fins the numbers of integers that are divisible by 5 but neither 3 nor 7. These are integers in $B\backslash(A \cup C)$, a set cardinality $|B| - |B \cap (A \cup C)|$. Since

$$B \cap (A \cup C) = (B \cap A) \cup (B \cap C),$$

Principle of Inclusion-Exclusion yields

$$|B \cap (A \cup C)| = |(B \cap A) \cup (B \cap C)|$$
$$= |B \cap A| + |B \cap C| - |A \cap B \cap C| = 20 + 8 - 2 = 26$$

and the number we seek is $|B| - 26 = 60 - 26 = 34$. It must be noticed that the following set identity is used in the above formula

$$(B \cap A) \cap (B \cap C) = A \cap B \cap C.$$

Exercises

1. Out of 100 people, 60 are men, 30 are young, and 10 are young men. How many are old women?

2. Out of 31 people, 19 are chestnut-eyed and 11 are black-haired. How many are both chestnut-eyed and black-haired if 3 people hold none of these properties?
3. Out of n students, 9 are enrolled for a course in Mathematics, 9 for History, 12 for Psychology, 2 for both Mathematics and History, 4 for both History and Psychology, 3 for both Mathematics and Psychology, 1 for all three courses and, finally, 2 for none of the courses. Find n.
4. Use the principle of inclusion-exclusion to find how many numbers in the range 1 - 300 (inclusively) are divisible by at least one of the integers 5 and 6.
5. How many integers between 1 and 1000 (inclusively) are divisible by at least one of the integers 2, 3 and 5?
6. Suppose there are 250 first-year students, 120 of which are taking Calculus and 100 of which are taking Algebra. If there are 70 first-year students that are taking both Calculus and Algebra, how many are taking neither course?
7. There are 20 cars to sale. 7 cars have CD players, 8 cars have cruise control and 10 cars have leather seats. 3 cars have CD players and cruise control, 4 cars have CD player and leather seats, 5 cars have cruise control and leather seats. If 3 cars have all of these features, how many cars have none of these features?
8. How many positive integers less than 1000 are divisible by 6? How many divisible by 6 but not 7? How many divisible by both 6 and 7? How many divisible by at least one of integers 6 and 7?
9. A coach of local teams buys volleyball and handball balls for both training and professional matches. Out of 100 balls he bought, 75 are handball balls and 80 are for training. Find a way to estimate the number of volleyball balls for professional matches. How large can be the number of volleyball balls for professional matches?

10. Out of 37 students, 18 are enrolled for a course in Mathematics, 16 for Physics, 17 for Chemistry, 7 for both Mathematics and Physics, 6 for both Mathematics and Chemistry, 5 for both Physics and Chemistry, 2 for all three courses. How many students are enrolled for none of these courses?

Chapter 9 Counting

9.1 Addition and Multiplication Rules

The Principle of Inclusion-Exclusion gives a formula for the elements in the union of a finite number of finite sets. If the sets are pairwise disjoint, then the formula takes a particularly simple form.
Given n pairwise disjoint finite sets, $A_1, A_2, \dots, A_n$, then

$$|A_1 \cup A_2 \cup \cdots \cup A_n| = |A_1| + |A_2| + \cdots + |A_n| = \sum_{i=1}^{n} |A_i|.$$

We use the general term **event** to mean the outcome of any process or experiment, for example, the course a student selects to complete his or her degree. Events are **mutually exclusive** if no two of them can occur together.

Notice that if n sets $A_1, A_2, \dots, A_n$ correspond to events with the elements of A_i representing the ways in which the corresponding event can occur, then events are mutually exclusive if and only if $A_i \cap A_j = \emptyset$ for all $1 \le i, j \le n$, that is, when the sets are pairwise disjoint. Thus, the Principle of Inclusion-Exclusion for disjoint sets translates into the following basic principle of counting.

Addition Rule The number of choices for mutually exclusive events to occur is the sum of the choices in which each event can occur.

Problem 9.1 How many are there choices to get a total of six when rolling a pair of dice?

Solution The event "get a total of six" is the union of the following mutually exclusive events.

A_1: "a 3 and a 3"
A_2: "a 2 and a 4"
A_3: "a 1 and a 5"

Although not mentioned explicitly, we distinguish between a pair of dice, that is, it matters whether first die shows 2 while second die showing 4 or vice versa. Notice that Event A_1 can occur in one way, while each of events A_2 and A_3 can occur in two ways. Hence, there are $1 + 2 + 2 = 5$ ways to get a six when rolling a pair of dice.

Multiplication Rule The number of choices for a sequence of events to occur is the product of the choices in which each individual event can occur.

Example 9.1 Assume that there are 2 roads from town A to town B, and suppose there are 3 roads from town B to town C. If there is no direct road from A to C then there are $2 \times 3 = 6$ choices to reach town C from town A.

Stated in another way, if there are n choices one event can occur and if there are m choices for each of n choices to occur then the total number choices in which both events can occur is given by the formula $n \times m$.

Problem 9.2 How many numbers in the range 1000-9999 do not have any repeated digits?

Solution Let us think of 4-digit template. There are 10 digits can used to compose 4-digit integers. Because the first digit cannot be 0, we can choose a digit for the first position in 9 different ways. There are 9 more ways to choose the second digit as the one used for the first position cannot be repeated. There are consequently 8 more ways for the third and 7 more ways for the fourth digit to be chosen. Thus, there are $9 \times 9 \times 8 \times 7$ 4-digit integers that have no repeated digits.

9.2 The Pigeonhole Principle

The Pigeonhole Principle is one of the fundamental counting principles. We use this principle to prove correctness of statements about natural number n as well as to count the number of samples in specific cases.

The Pigeonhole Principle If n objects are allocated into m boxes and $n > m$, then at least one box contains at least two of the objects.

Problem 9.3 Show that among 400 people at least 2 have birthdays in the same day.

Proof In this problem, people stand for pigeons while days for the pigeonholes. Obviously, there are 365 days in a regular year. To adopt The Pigeonhole Principle, we need to determine the values of the variables: $n = 400, m = 365$. We can see that $n > m$ and, therefore, we can use The Pigeonhole Principle. According to the principle there are at least 2 people who were born the same day. This is what we asked to prove and, hence, the proof is over.

We can formulate the same problem in a bit different way where we need to find minimum number of objects, which guarantees that at least one of the pigeonholes would contain certain number of objects.

Problem 9.4 Find minimum number of people, which guarantees that at least two of them were born the same day.

Solution In the worth-case scenario, up to 365 people can be equally allocated between days so that there will be no two people who were born the same day. However, if there 366 people, then it is worth to mention that even in the worth-case scenario there will be at least two people having birthdays the same day. In other words, we are given that $m = 365$ and asked to find minimum value for n which guarantees the claim, i.e. $n > m$. Hence, $n = 366$.

The Strong Form of the Pigeonhole Principle If n objects are put into m boxes and $n \gg m$, then some box must contain at least $\left\lceil \frac{n}{m} \right\rceil$ objects.

Although both forms of the principle look like similar, weak and strong forms of the principle differ in relationship between n and m. In the weak form, $m < n < 2m$, while in the strong form $n \geq 2m$.

Problem 9.5 Show that among 100 people at least 9 have birthdays in the same month.

Solution According to the strong form of the Pigeonhole Principle $n = 100, m = 12$. So the minimum number of people who were born the same month is calculated as $\left\lceil \frac{n}{m} \right\rceil = \left\lceil \frac{100}{12} \right\rceil = \lceil 8.3(3) \rceil = 9$.

Problem 9.6 Find minimum number of people, which guarantees that at least 9 of them were born the same month.

Solution Let us think in the following way. According to the worst-case scenario, we equally arrange the pigeons between pigeonholes. Hence, $8 \times 12 = 96$ people would not satisfy the condition of the problem, saying that at least 9 of the people must born the same month. However, adding one more to already found number $(8 \times 12 + 1 = 97)$ guarantees that at least 9 of them were born the same month.

9.3 Permutations

Definition 9.1 (r-permutation) For integers n and r, $n \geq 1, 0 \leq r \leq n$, the symbol $P(n, r)$ is defined by $P(n, 0) = 1$ and for $r > 0$,

$$P(n, r) = \underbrace{n(n-1)(n-2)\cdots(n-r+1)}_{r\ factors}.$$

Let us consider several well-coined counting problems.

Problem 9.7 The registrar of a school must schedule 6 examinations in an 8-day period and has promised the students not to schedule more than one examination per day. How many different schedules can we made?

Solution The solution is straightforward. We can schedule the first exam to any of 8 days, but once this day settled, we can schedule the second exam to one of remaining 7 days. Once the first two exams are scheduled, we have six days at our disposal. Continuing thinking the same way, we find that there are $8 \times 7 \times 6 \times 5 \times 4 \times 3$ possible ways to schedule all exams.

We can solve the same problem using permutation formula as follows

$$P(8,6) = \frac{8!}{(8-6)!} = \frac{8!}{2!}.$$

Problem 9.8 In how many ways can a blue, a white, and a red marble be put into 10 numbered boxes?

Solution If there is no limit on the number of marbles which can be put into a box, there are 10 possible boxes into which the blue marble can be placed and then 10 choices for the white one, and 10 more choices for the red one, giving $10^3 = 1000$ possibilities together. Suppose, on the other hand, we cannot to put more than one marble into a box. As before, there are 10 possible boxes for the blue marble, but now there are just 9 possible boxes to put the white marble and 8 to put the red marble. Altogether, there are $10 \times 9 \times 8 = 720$ possibilities.

As in the case of Problem 9.7, we apply permutation formula to find the number of possibilities under circumstance that a box can contain one marble at the most.

$$P(10,3) = 10 \times 9 \times 8 = \frac{10!}{(10-3)!} = \frac{10!}{7!}.$$

Problem 9.9 Find $P(5,3)$, $P(4,4)$ and $P(7,2)$.
Solution

$$P(5,3) = 5 \times 4 \times 3 = 60,$$
$$P(4,4) = 4 \times 3 \times 2 \times 1 = 24,$$
$$P(7,2) = 7 \times 6 = 42.$$

One can easily see that

$$P(n,r) = \underbrace{n(n-1)(n-2)\cdots(n-r+1)}_{r\ factors} = \frac{n!}{(n-r)!}.$$

Problem 9.10 Find the number of dance partners that we can organize from 12 women and 20 men.
Solution The first woman can be in dance pair with any of 20 men, the second woman with any of remaining 19 men, and so on. There are

$$P(20,12) = 20 \times 19 \times 18 \times \cdots \times 9$$

possible dance pairs.
Definition 9.2 A permutation of a set of distinct symbols is an arrangement of them in a line in some order.
Example 9.2 All possible arrangements of the elements in the set {1,2,3,4} are listed below:

1234 1243 1324 1342 1423 1432

2134	2143	2314	2341	2413	2431
3124	3142	3214	3241	3412	3421
4123	4132	4213	4231	4312	4321

Problem 9.11 Find the number of ways 7 people can be lined up. In how many ways can 7 people sit around round table?

Solution The first question refers to pure permutation formula. Thus, there are

$$P(7,7) = \frac{7!}{(7-7)!} = 7!$$

choices in which 7 people can be arranged in a line.

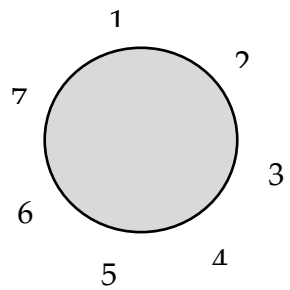


Figure 9.1 Illustration for 7 people sitting around round table.

However, the second question is about circle (or circular) permutations. Let us detail the difference between two cases. There is a head and a rear of a queue of people, but there is neither head nor rear of the objects that we arrange in a circular way (see figure). There is no first and last person among 7 people who sit around round table. Hence, there are $\frac{7!}{7} = 6!$ distinct ways to arrange 7 people around a round table.

Problem 9.12 A donkey, a dog, a cat and a rooster are walking together one after the other.

a) Find the number of ways this can happen.
b) Find the number of ways this can happen if the dog comes first.
c) Find the number of ways this can happen if the dog immediately follows the donkey.
d) Find the number of ways this can occur if the dog, and only the dog, is between the donkey and the cat.

Solution We solve the problem for each case under corresponding circumstance as follows.

$$P(4,4) = \frac{4!}{(4-4)!} = 4! = 24.$$

If the dog comes first, the other three members of the team can be arranged in $P(3,3) = 3! = 6$ distinct ways.

The donkey followed by the dog can be thought as one object, which when considered together with the cat and the rooster, leads to the formula $P(3,3) = 3! = 6$. Hence, this can occur in 6 different ways.

The object now is of the form donkey-dog-cat or cat-dog-donkey. In either case, by considering this triplet together with the rooster, we find that there are $2! + 2! = 4$ distinct ways in which this can occur.

9.4 Combinations

Consider the following well-coined problems. A customer wants to buy two different kinds of drinks out of 12 different kinds available

in a store. In how many ways can this happen? In how many ways 5 people can be chosen out of 12 women and 20 men to organize a party? In how many ways can exactly 5 heads turn up if a coin is thrown 8 times? All these questions of the same type and refer to combination formula. In the first question, we need to choose 2 different drinks out of 12 and the order in which those drinks are chosen does not matter. In the second problem, we are indeed asked to choose 5 people out of 32 in any composition. The way we choose them is not important. Thus, this is also a pure selection problem, in which order does not matter. Finally, in the third problem, we are asked to choose 5 objects out of 8 and the way they are arranged is not important. Indeed, we cannot arrange the objects because we are dealing with indistinguishable objects, i.e. five heads.

Definition 9.3 (r-combination) For integers r and n, $n \geq 1$ and $0 \leq r \leq n$, the combination formula, which is read "n choose r" is as follows

$$\binom{n}{r} = \frac{n!}{(n-r)!\,r!}.$$

The following identity holds:

$$\binom{n}{r} = \binom{n}{n-r}.$$

Obviously,

$$\binom{n}{r} = \frac{n!}{(n-r)!\,r!}$$

and

$$\binom{n}{n-r} = \frac{n!}{r!\,(n-r)!}.$$

Comparing the above two formulae, we find that $\binom{n}{r} = \binom{n}{n-r}$.
Now we are prepared to take the questions posed in the beginning of the section.

Problem 9.13 (revisited) A customer wants to buy two different kinds of drinks out of 12 different kinds available in a store. In how many ways can this happen?

Solution It is the combination formula that we will use to find the number of samples, because drinks selected are not arranged. Hence,

$$\binom{12}{2} = \frac{12!}{(12-2)!\,2!} = \frac{12!}{10!\,2!} = \frac{12 \times 11}{1 \times 2} = 66.$$

Problem 9.14 (revisited) In how many ways 5 people can be chosen out of 12 women and 20 men?

Solution Since membership of the selected persons is important, but not the order in which they are selected, there are

$$\binom{32}{5} = \frac{32!}{27!\,5!}$$

distinct ways in which a group of 5 people can be chosen.

Problem 9.15 (revisited) If a coin is tossed 8 times, in how many ways can exactly 5 heads turn up?

Solution There is no way to distinguish between heads, as all of them are same. It is also a combination formula that we use to solve the problem.

$$\binom{8}{5} = \frac{8!}{3!\,5!} = \frac{6 \times 7 \times 8}{1 \times 2 \times 3} = 56.$$

Problem 9.16 In how many ways can 20 people out of 32 be chosen to attend a music competition, if

a) Alex refuses to go to the competition?
b) Jasmin insists on going?
c) Eileen and Susan insist on going?
d) either Eileen or Susan (or both) go to the competition?
e) just one of Eileen and Susan go to the competition?
f) Alex and Michelle refuse to go together?

Solution

a) The answer is $\binom{31}{20}$ because Alex is out selection, and we need to select 20 people out of 31.
b) In this case, we need to select 19 people out of 31 since Jasmin is anyway in a team and therefore the number of ways is $\binom{31}{19}$.
c) Since two of them insists on going, then we need to choose 18 students from a group of 30 and, thus, the answer is $\binom{30}{18}$.
d) Let E be the set of teams of size 20 containing Eileen, and let S be the set of teams of size 20 containing Susan. We must determine $|E \cup S|$. By the Principle of Inclusion-Exclusion, we obtain $|E \cup S| = |E| + |S| - |E \cap S| = \binom{31}{19} + \binom{31}{19} - \binom{30}{18}$. An alternative method of obtaining this answer would be to count separately the cases – Eileen (but not Susan) goes to the competition, Susan (but not Susan) goes to the competition, both Eileen and Susan go to the competition. After adding the results we obtain

$$\binom{31}{19} + \binom{31}{19} - \binom{30}{18}.$$

e) Using the formula $|E \oplus S| = |E| + |S| - 2|E \cap S|$, we obtain

$$\binom{31}{19} + \binom{31}{19} - 2\binom{30}{18}.$$

f) The number of classes containing Alex and Michelle is $\binom{30}{18}$ by part c), so the number which do not contain both is $\binom{32}{20} - \binom{30}{18}$. Alternatively, observe that this part asks for $|(A \cap M)^c| = |A^c \cup M^c|$ (by one of the DeMargan laws). So the answer is

$$|A^c| + |M^c| - |A^c \cap M^c| = \binom{31}{20} + \binom{31}{20} - \binom{30}{20}.$$

9.4 Repetitions

In the previous two sections, we coined the number of ways of putting r marbles into n boxes. Let us consider a problem in wide spectrum of the cases. Consider the following marble-box problems.

A. How many are there ways to put r different colored marbles into n numbered boxes, if a box can accommodate one marble at the most?
B. How many are there ways to put r same-colored marbles into n numbered boxes, if a box can accommodate one marble at the most?
C. How many are there ways to put r different colored marbles into n numbered boxes, if a box can accommodate multiple marbles?

D. How many are there ways to put r same-colored marbles into n numbered boxes, if a box can accommodate multiple marbles?

For each of above cases we use different counting formula to find the number of ways.

A. It is a r-permutation formula that we use to find the number of ways: $P(n, r)$.
B. It is a r-combination formula that we use to find the number of ways: $\binom{n}{r}$.
C. It is a permutation formula with repetition that we need to use to find the number of ways: $n \times n \times \cdots \times n = n^r$.
D. It is a combination formula with repetition that we need to use to find the number of ways: $\binom{n+r-1}{r}$.

Problem 9.17 Drinks come in 30 different varieties and you want to buy a dozen. How many choices do you have?
Solution There are 30 varieties of drinks such as fanta, sprite, pepsi and so on. You want to buy 12 drinks in any composition. This problem is similar to putting $r = 12$ identical marbles into the boxes. There are

$$\binom{30+12-1}{12} = \binom{41}{12} = \frac{41!}{29!\,12!}$$

choices.
The following template can be useful for understanding when which formula we can use to find the number of samples of size r out of n possible.

Table 9.1 Permutations and combinations with and without repetition			
		repetition	
		without	with
order	important	$P(n,r)$	n^r
	not important	$\binom{n}{r}$	$\binom{n+r-1}{r}$

9.5 Permuting Indistinguishable Objects

In how many ways one can arrange the letters in the word ABRAKADABRA to form new meaningful or meaningless words. This and other similar problems deal with permutation in a set involving indistinguishable objects.

Definition 9.4 (Permutation of indistinguishable objects) Given n objects such that there are n_1 elements of type 1, n_2 elements of type 2, and so on, n_k elements of type k, the number of arrangements of n elements is calculated by the formula

$$\binom{n}{n_1, \dots, n_k} = \frac{n!}{n_1!\, n_2! \cdots n_k!}.$$

Coming back to the problem of arranging the letters in the word ABRAKADABRA, we now observe that there are

$$\binom{11}{5,2,2,1,1} = \frac{11!}{5!\, 2!\, 2!\, 1!\, 1!}$$

ways to create new words by arranging the letters.

Problem 9.18 How many new words (meaningful or meaningless) can we create by arranging the letters in the word ARRANGE?

Solution By using formula for permutation in the set with indistinguishable objects, we obtain

$$\binom{7}{2,2,1,1,1} = \frac{7!}{2!\,2!\,1!\,1!\,1!}.$$

9.6 Derangements

There are three houses on a street. An absent-minded postman arrives with one letter addressed to each house. In how many ways can he deliver the letters, one to each house, so that no letter arrives at the correct address? The person in charge of hats at the arts center loses all claim checks and returns hats randomly to the patrons. What are the chances that nobody gets own hat?

Definition 9.5 (Derangement) A derangement of n distinct objects, labelled from 1 to n, is a permutation in which no symbol is in its correct place. The number of derangements of n symbols is denoted by D_n. The number of derangements of $n \geq 1$ ordered symbols is

$$D_n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\right).$$

Problem 9.19 Find D_3 and D_4.

Solution

$$D_3 = 3!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!}\right) = 6\,\frac{3-1}{6} = 2,$$

$$D_4 = 4!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right) = 24\,\frac{(12-4+1)}{24} = 9.$$

Problem 9.20 (revisited) There are four houses on a street. An absent -minded postman arrives with one letter addressed to each house. In

how many ways can he deliver the letters, one to each house, so that no letter arrives at the correct address?

Solution We find the solution using derangement formula as follows:

$$D_4 = 4!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right) = 24\frac{(12-4+1)}{24}=9.$$

Problem 9.21 (revisited) The person in charge of 3 hats at the arts center loses all claim checks and returns hats randomly to 3 patrons. What are the chances that nobody gets their own hats?

Solution The solution is by the derangement formula as follows:

$$D_3 = 3!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!}\right) = 6\frac{3-1}{6}=2.$$

Problem 9.22 Hogwarts School of Magicians consists of four schools, each bearing the last name of its founder: Gryffindor, Slytherin, Ravenclaw and Hufflepuff. Harry Potter, Ron Weasley, Hermione Granger and Draco Malfoy are willing to study in different schools. The Sorting Hat at Hogwarts, that determines which of the four schools each new student belongs most to, allocates two students to the schools they want to study at, while other two students to the schools they do not want to. Find the number of ways The Sorting Hat can do this.

Solution The solution is by the formula:

$$\binom{4}{2} \cdot D_2 = \frac{4!}{2!\,2!} \cdot 2!\left(1 - \frac{1}{1!} + \frac{1}{2!}\right) = 6 \cdot 1 = 6.$$

Problem 9.23 Gandalf, Thranduil, Thorin, Rhosgobel and Azog love riding their favorite animals that are, respectively, White Horse,

Great Elk, Bighorn Sheep, Giant rabbits and Warg Matriarch. How many pairs can there be between the five characters of "The Hobbit" and "Lord of the Rings" and the five animals listed above, if only two of above personals got their favorite animals while the remaining three got animals they do not really prefer.

Solution The solution is by the formula:

$$\binom{5}{2} \cdot D_3 = \frac{5!}{3!\,2!} \cdot 3!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!}\right) = 10 \cdot 2 = 20.$$

9.7 The Binomial Theorem

The following triangular arrangement of integers is named after Blaise Pascal (1623-1662) because of the many applications in combinatorics and probability, which Pascal found for it. This famous triangle has several interesting properties.

Our major interest in Pascal[8]'s triangle concerns its connection with the binomial expansion $(x + y)^n$, which is called binomial because it is expression of two unknowns and bi is the translation of two into Latin. Binomial expansion $(x + y)^n$ for the numbers in row n of the triangle gives the requisite coefficients. Figure 9.2 shows binomial expansion for $n = 0,1,2,3,4,5$.

[8] Blaise Pascal (1623–1662) exhibited his talents at an early age, although his father, who had made discoveries in analytic geometry, kept mathematics books away from him to encourage other interests. At 16 Pascal discovered an important result concerning conic sections. At 18 he designed a calculating machine, which he built and sold. Pascal, along with Fermat, laid the foundations for the modern theory of probability. In this work, he made new discoveries concerning what is now called Pascal's triangle.

$$
\begin{aligned}
(a+b)^0 &= 1 \\
(a+b)^1 &= a+b \\
(a+b)^2 &= a^2+2ab+b^2 \\
(a+b)^3 &= a^3+3a^2b+3ab^2+b^3 \\
(a+b)^4 &= a^4+4a^3b+6a^2b^2+4ab^3+b^4 \\
(a+b)^5 &= a^5+5a^4b+10a^3b^2+10a^2b^3+5ab^4+b^5
\end{aligned}
$$

Figure 9.2 Illustration of binomial expansion for $n = 0, \dots, 5$.

$$
\begin{array}{c}
\binom{0}{0} \\
\binom{1}{0} \ \binom{1}{1} \\
\binom{2}{0} \ \binom{2}{1} \ \binom{2}{2} \\
\binom{3}{0} \ \binom{3}{1} \ \binom{3}{2} \ \binom{3}{3} \\
\binom{4}{0} \ \binom{4}{1} \ \binom{4}{2} \ \binom{4}{3} \ \binom{4}{4} \\
\binom{5}{0} \ \binom{5}{1} \ \binom{5}{2} \ \binom{5}{3} \ \binom{5}{4} \ \binom{5}{5}
\end{array}
$$

Figure 9.3 Illustration of binomial coefficients for $n = 0, \dots, 5$.

$$
\begin{array}{c}
1 \\
1 \quad 1 \\
1 \quad 2 \quad 1 \\
1 \quad 3 \quad 3 \quad 1 \\
1 \quad 4 \quad 6 \quad 4 \quad 1 \\
1 \quad 5 \quad 10 \quad 10 \quad 5 \quad 1 \\
1 \quad 6 \quad 15 \quad 20 \quad 15 \quad 6 \quad 1 \\
1 \quad 7 \quad 21 \quad 35 \quad 35 \quad 21 \quad 7 \quad 1
\end{array}
$$

Figure 9.4 Illustration of Pascal's triangle for $n = 0, \dots, 8$.

Coefficients needed to expand $(a+b)^n$ are determined by combination formula, which is illustrated in Figure 9.3 for $n = 0, \dots, 5$, while calculated coefficients are shown in Figure 9.4. We can

observe that binomial coefficients follow up a triangular shape. Hence the name Pascal's triangle.

Another interesting property of the Pascal's triangle is that any coefficient is the sum of two coefficients allocated in the previous row: one positioned immediately in the left of it and another immediately in the right.

This fact remains true for all rows and is expressed by the following formula

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

The proof of above formula is straightforward. First, we write formula for each of these combinations and, then, find sum of the last two formulae as follows.

$$\binom{n}{k} = \frac{n!}{(n-k)!k!} \qquad (1)$$

$$\binom{n-1}{k-1} = \frac{(n-1)!}{(n-k)!(k-1)!} \qquad (2)$$

$$\binom{n-1}{k} = \frac{(n-1)!}{(n-k-1)!k!} \qquad (3)$$

By finding sum of the right sides of (2) and (3), we obtain

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \frac{(n-1)!}{(n-k)!\,(k-1)!} + \frac{(n-1)!}{(n-k-1)!\,k!}$$

$$= \frac{k(n-1)! + (n-k)(n-1)!}{(n-k)!\,k!} = \frac{n!}{(n-k)!\,k!}.$$

The Binomial Theorem For any x and y and any natural number n,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$
$$= \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y^1 + \binom{n}{2} x^{n-2} y^2 + \cdots + \binom{n}{n} y^n.$$

Problem 9.24 Using the Binomial Theorem, expand and simplify $(3x^2+2y)^5$.

Solution

$$(3x^2+2y)^5 = (3x^2)^5 + \binom{5}{1}(3x^2)^4(2y) + \binom{5}{2}(3x^2)^3(2y)^2$$
$$+\binom{5}{3}(3x^2)^2(2y)^3 + \binom{5}{4}(3x^2)(2y)^4 + (2y)^5$$
$$= 243x^{10} + 5(81x^8)(2y) + 10(27x^6)(4y^2) + 10(9x^4)(8y^3)$$
$$+5(3x^2)(16y^4) + 32y^5$$
$$= 243x^{10} + 810x^8y + 1080x^6y^2 + 720x^4y^3 + 240x^2y^4 + 32y^5.$$

Problem 9.25 Using the Binomial Theorem, expand $\left(x - \frac{4}{x}\right)^6$ and simplify.

Solution

$$\left(x - \frac{4}{x}\right)^6 = x^6 + \binom{6}{1} x^5 \left(-\frac{4}{x}\right) + \binom{6}{2} x^4 \left(-\frac{4}{x}\right)^2 + \binom{6}{3} x^3 \left(-\frac{4}{x}\right)^3$$
$$+\binom{6}{4} x^2 \left(-\frac{4}{x}\right)^4 + \binom{6}{5} x \left(-\frac{4}{x}\right)^5 + \left(-\frac{4}{x}\right)^6$$
$$= x^6 + 6x^5\left(-\frac{4}{x}\right) + 15x^4\left(\frac{16}{x^2}\right) + 20x^3\left(-\frac{64}{x^3}\right) + 15x^2\left(\frac{256}{x^4}\right)$$
$$+6x\left(-\frac{1024}{x^5}\right) + \frac{4096}{x^6}$$
$$= x^6 - 24x^4 + 240x^2 - 1280 + \frac{3840}{x^2} - \frac{6144}{x^4} + \frac{4096}{x^6}.$$

Problem 9.26 Find the coefficient of x^{16} in the expansion of $\left(2x^2 - \frac{x}{2}\right)^{12}$.

Solution

By the Binomial Theorem we have

$$\binom{12}{k}(2x^2)^{12-k}\left(-\frac{x}{2}\right)^k = \binom{12}{k}2^{12-k}\left(-\frac{1}{2}\right)^k x^{24-k}.$$

From $24-k=16$ we find that $k=8$. Thus,

$$\binom{12}{8}2^4\left(-\tfrac{1}{2}\right)^8 = \tfrac{1}{16}\binom{12}{8}=\tfrac{495}{16}.$$

Exercises

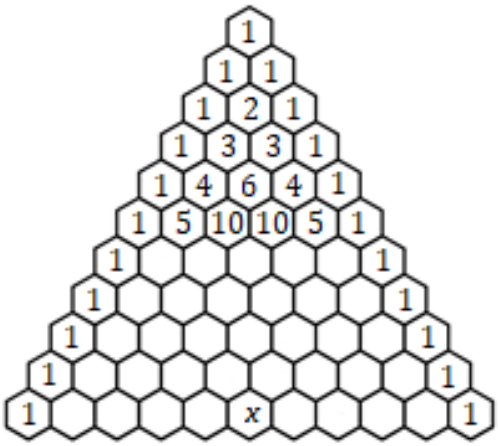
1. How many ways can get a sum of 5 when 3 indistinguishable dice are rolled?
2. A palindrome is a sequence of letters or 0 to 9 numbers, which reads the same backwards as it does when being read forward. How many palindromes of size 7 can be made if 1st and 3rd symbols are English letters while 2nd and 4th symbols being 0 to 9 numbers.
3. From a group of 12 girls and 9 boys how many committees of size 10 are possible a) if there is no additional restriction; b) at least 8 of them are girls.
4. Say the only clothes you got are two T-shirts, four pairs of jeans and six pairs of shoes. In how many combinations can you choose a T-shirt, a pair of jeans and a pair of shoes?
5. There are five flavors of ice cream: banana, chocolate, lemon, strawberry and vanilla. We want to have three scoops. How many variations will be there if there are enough amount from each flavor?

6. In how many ways can a head and a vice head of the department be selected out of 20 academic staff working in a university department?
7. A standard deck has 52 cards, divided into four suits (♡, ♠, ♢, ♣). There are 13 cards of each suit, 2, 3, . . . , 10, J (Jack), Q (Queen), K (King), A (Ace). How many shuffles are there of a deck of cards, such that A♡ is not directly on top of K♡, and A♠ is not directly on top of K♠?
8. How many are there four cards containing exactly two Queens and at least two Spades?
9. How many poker hands (5 cards from a standard deck of 52 cards) contain exactly 3 face cards (i.e., Jack, Queen or King)?
10. A sports store sales football, volleyball, basketball and handball balls and you want to buy 3 balls. In how many ways can you do this?
11. If the exam is graded from 0 to 100 points, find the number of students in a class, which guarantees that at least four of them receive same score on the final exam.
12. How many ways are there to select 11 students from 40 students if they will stand in line for a picture?
13. A team of 3 members is formed from 6 children, 7 teenagers, and 8 adults. How many are there teams with only one age group.
14. How many poker-hands (5 cards out of a deck of 52 cards) have three aces or two kings or both.
15. 1 red, 1 blue, 1 black, 1 yellow and 1 white car are all waiting at the traffic lights behind each other.
16. Find the number of ways cars can wait at the lights.
17. Find the number of ways cars can wait if the white car is the first in the line.

18. Find the number of ways the cars can wait, if the white car is immediately behind the yellow car.
19. Find the number of ways cars can wait if the black car and only the black car is between the white and yellow cars.
20. How many are there circular permutations of size 5?
21. There are 5 boys and 7 girls in a group. In how many ways 3 representatives can be chosen if there is at least 1 girl among them.
22. The lock uses 3 numbers from 1 to 9 that cannot be repeated in the combination. How many are there locker combinations?
23. How many new numbers can we compose by arranging the digits of 743347?
24. There are 7 different time periods, during which classes at a university can be scheduled. Find the minimum number of different rooms that we need to schedule 68 different classes.
25. In how many ways one can get a total of eight if two dice are rolled?
26. How many numbers in the range 400-999 do not have any repeated digits?
27. How many students must be in a class to guarantee that at least 4 students receive same score on the final exam, if the exam is graded on a scale from 0 to 100 points?
28. Use the strong form of the pigeonhole principle to decide at least how many of a set of 1473 people must have their birthday on the same day. Notice that there are only 366 possible birthdays to choose from (including February 29).
29. A drawer contains sixteen red and sixteen blue socks, all unmatched. A person takes socks out at random in the dark. How many socks must he take out to be sure that he has at least two blue socks?

30. There are three drawers. One contains 12 shirts, the second 7 neckties and the third 5 hats. How many are there ways to combine a shirt, a necktie and a hat?
31. Assume a drawer contains a mixture of ten red, eight blue, six green, five yellow, three black and two white socks, each of which can be worn on either foot, and that you are pulling a number of socks from the drawer without looking. What is the minimum number of pulled socks required to guarantee two pairs of socks of the same color?
32. In how many ways can we put r same-colored marbles into n boxes, if a box can contain multiple marbles?
33. Let us say a box contains 1 red marble, 1 green marble and 1 yellow marble. You pick randomly a marble at a time, check the color, and put it back in. In how many varieties you can select 6 marbles.
34. There are marbles of 5 different colors in a box and you want to select 3 marbles. How many choices do you have?
35. Trainer of a school football team wants to select 11 boys out of 40 for a football team. How many teams can he select if two boys refuse to be together in a team?
36. A bag contains exactly 6 red, 13 blue, 9 white, 10 green and 7 yellow marbles. Find the least number of balls to be chosen which will guarantee that there will be at least 4 balls of the same color.
37. Five friends meet at the local cafe, where each of them orders one soft drink. In how many ways can a waiter serve the drinks, one to each customer, if two of them receive drinks they ordered, while the other three receiving drinks they do not really order?
38. In how many ways can a red marble, a green marble and a yellow marble be put in 5 numbered boxes if a box can accommodate multiple marbles?

39. There are apples, pears, apricots and plums in a bucket and you want to get three fruits. How many ways are there to do this?
40. How many new words can be formed using all letters in the word OOPSSSNOOO.
41. In how many ways can you create new words by exchanging the letters in SEESHELL?
42. How many horizontally striped flags can we make with 7 stripes, if we have 2 white, 2 red and 3 green stripes?
43. We have four different dishes, two dishes of each type. In how many ways can these be distributed among 8 people?
44. If there are 5 letters and 5 envelops, one letter for one envelope, in how many ways 2 letters can be put in correct envelops while the other 3 in wrong envelops.
45. The person in charge of coats at the arts center loses all 10 claim checks and returns hats randomly to the patrons. In how many ways 4 persons get their own coats while other 6 do not.
46. How many terms are in the expansion of the binomial $(x + y)^{98}$?
47. Find the coefficient of x^7y^8 in expansion of $(3x - 4y)^{15}$.
48. Find the coefficient of x^7 in the expansion of $(x^3 + 2x^{-1})^5$.
49. Given the binomial $(x^2 - 2y^3)^{20}$, find the coefficient of $x^{10}y^{45}$.
50. Calculate x in the following Pascal's triangle.


1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 1
1 1
1 1
1 1
1 1
1 x 1

Chapter 10 Graphs and Trees

10.1 Introduction

In the beginning, we provide the reader with some experience by presenting several real-world problems and demonstrating the way we can formulate these problems by means of graphs.

Many practical problems require visualization of the problem to look at it from a different point of view. The Problem of Königsberg Bridges is of this type problem where one needs to see the picture behind of the problem to deal with the problem. This was a long-standing problem until Leonard Euler[9] solved it in 1736.

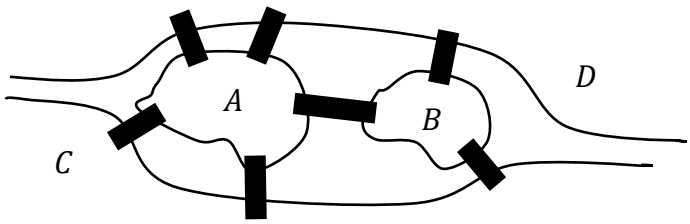


Figure 10.1 Illustration of The Problem of Königsberg Bridges.

[9] Leonhard Euler (1707–1783) was born in Basel, Switzerland. At 13 he entered the University of Basel, pursuing a career in theology. Euler obtained his master's degree in philosophy at the age of 16. In 1741 he moved to the Berlin Academy, where he stayed until 1766. He then returned to St. Petersburg, where he remained for the rest of his life. Euler was incredibly prolific, contributing to many areas of mathematics, including number theory, combinatorics, graph theory and analysis, as well as its applications to such areas as music and naval architecture. He was blind for the last 17 years of his life, but because of his fantastic memory this did not diminish his mathematical output.

Königsberg is old Prussian city. In XVIII century, it was the capital of East Prussia. The Pregel River flows through Königsberg and splits the city into two sides, C and D. Two islands in the river are labelled by A and B as is shown in Figure 10.1. Seven bridges link four land masses. The question arises if one can start on one of the land masses, walk over each bridge exactly once, and be back to the starting point without getting wet.

The Three Houses – Three Utilities Problem is another real-world problem, which we can model in terms of graphs. Each of three houses is to be connected to each of three utilities – water, electricity, and telephone – by means of underground pipes. Is it possible to make these connections without any crossovers?

10.2 Basic Terminology

After a short introduction to the graphs in the previous section, it is time now to define this term properly and to introduce the basic terminology to be used further in this chapter.

A **graph** is a pair (V, E) of sets, V nonempty and each element of E a set of two distinct elements of V. The elements of V are called **vertices**; the elements of E are called **edges**. Thus, if e is an edge, then $e = \{ v, w \}$, where v and w are different elements of V called the **end vertices** or **ends** of e. (Colloquially, we often say that the edge $e = \{ v, w \}$ **joins** vertices v and w). We usually abandon set notation and refer to the edge vw; this is, of course, the same as the edge wv. The vertices v and w are said to be **incident** with the edge vw; the edge vw is **incident** with each vertex. Two vertices are **adjacent** if they are the end vertices of an edge; two edges are **adjacent** if they have a vertex in common. The number of edges incident with a vertex v is

called the **degree** of that vertex and is denoted deg (v). If deg (v) is an even number, then v is said to be an **even vertex**; if deg (v) is an odd number, vertex v is **odd**. A vertex of degree 0 is said to be **isolated**. In this text, graphs will always be **finite**, meaning that the set of vertices and, hence, also the set of edges, are finite sets. Finally, when we say that $G = (V, E)$ is a graph, we mean that G is a graph with vertex set V and edge set E.

We usually draw a picture of a graph rather than presenting it formally as sets of vertices and edges. For instance, the graph G with vertices

$$V = \{v_1, v_2, v_3, v_4, v_5, v_6\}$$

and edges

$$E = \{v_1v_4, v_1v_6, v_2v_5, v_4v_5, v_5v_6\}$$

can be drawn as shown in Figure 10.2.

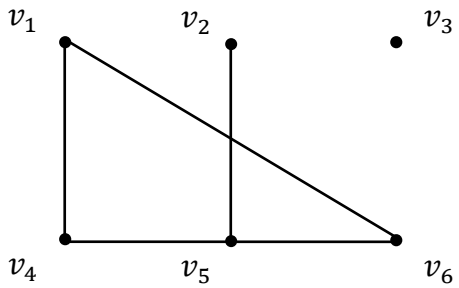


Figure 10.2 A graph.

Many of the graphs are not graphs at all according to the definition of a graph that we provided in the beginning of this section. The point is that if a graph has **multiple edges**, that is,

several edges incident with the same two vertices, and/or a **loop** at a vertex, that is, an edge that is incident with only one vertex, it is convenient to call it a **multigraph** or **pseudograph** rather a graph. For instance, in Figure 10.3 e_1 and e_2 are multiple edges and e_5 is a loop and thereby we refer to this drawing as multigraph.

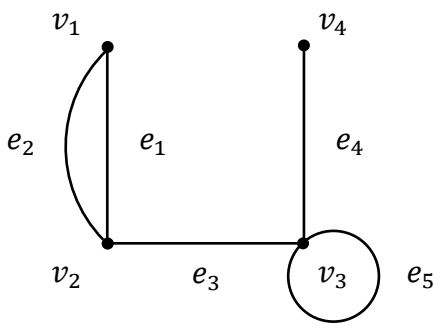


Figure 10.3 A multigraph.

A graph G is a **subgraph** of graph H if and only if the vertex and edge sets of G are respectively subsets of the vertex and edge sets of H. For example, in Figure 10.4 each of the graphs G_1, G_2 and G_3 is a subgraph of graph G.

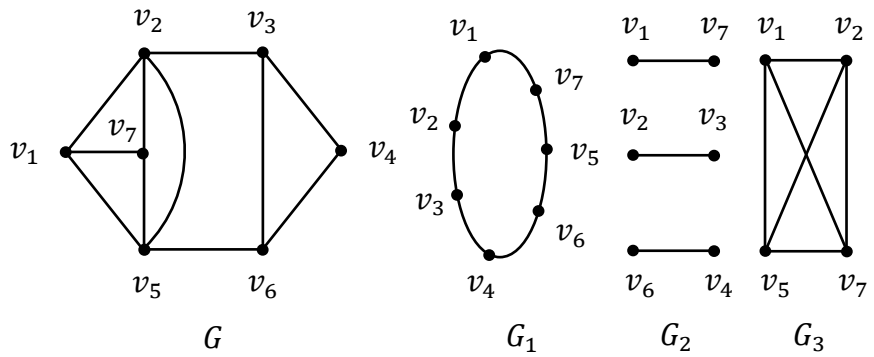
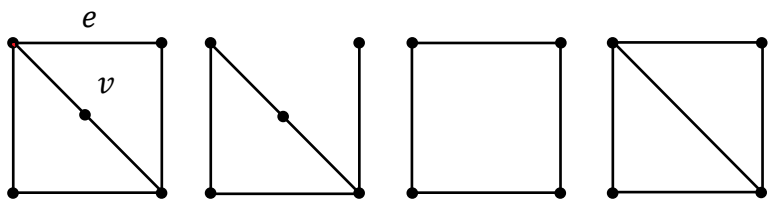


Figure 10.4 A graph and its subgraphs.

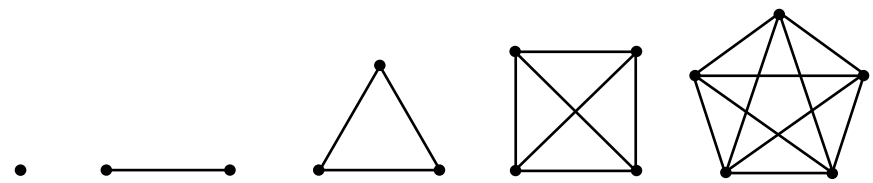
Let us consider four graphs shown in Figure 10.5. Perhaps it is not difficult to observe that among four graphs G_1 and G_2 are subgraphs of G. However, a small effort is good enough to understand why G_3 is not subgraph of G. One reason is that the two vertices of degree 3 in G are not adjacent while their counterparts in G are adjacent.



G $\quad$ $G_1 = G\backslash\{e\}$ $\quad$ $G_2 = G\backslash\{v\}$ $\quad$ G_3

Figure 10.5 Four graphs: G_1 and G_2, but not G_3, are subgraphs of G.

For positive integer n, the **complete graph on n vertices**, denoted K_n, is that graph with n vertices every two of which are adjacent. K_1, K_2, K_3, K_4 and K_5 are shown in Figure 10.6. For all n in K_n $\deg v = n - 1$ and $\sum \deg v = n(n-1)$.

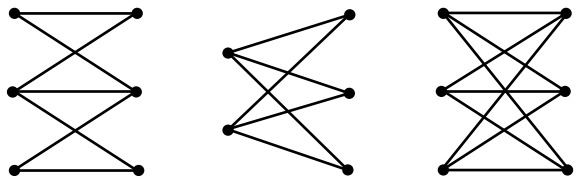


K_1 $\quad$ K_2 $\quad$ K_3 $\quad$ K_4 $\quad$ K_5

Figure 10.6 Complete graphs: K_1 through K_5.

The graphs in Figure 10.7 are not complete, but these graphs have the other interesting properties. In each of them, vertices in the right subset are not adjacent and vertices in the left subset are not adjacent

(they are examples of **bipartite graphs**). In the rightmost two graphs every vertex in the right subset is adjacent to every vertex in the left subset (these two are **complete bipartite graphs**). A **bipartite graph** is the one whose vertices can be partitioned into two disjoint sets V_1 and V_2, called **bipartition sets**, in such a way that every edge joins a vertex in V_1 and a vertex in V_2. (In particular, there are no edges within V_1 nor within V_2). A **complete bipartite graph** is a bipartite graph in which every vertex in V_1 is joined to every vertex in V_2. The complete bipartite graph on bipartition sets of m vertices and n vertices, respectively, is denoted $K_{m,n}$. In Figure 10.7, graphs in the middle and in the right are bipartite graphs with the property $|V_1| = 2, |V_2| = 3$ and $|V_1| = |V_2| = 3$, respectively. These are



$K_{2,3}$ $K_{3,3}$

Figure 10.7 Three bipartite graphs two of which are $K_{2,3}$ and $K_{3,3}$.

complete bipartite graphs $K_{2,3}$ and $K_{3,3}$. Single vertex is a bipartite graph K_1 A graph is bipartite if and only if its vertices can be colored with two colors such that every edge has ends of different colors.

Example 10.1 Let us color vertices of the graph in Figure 10.8 in such a way, the adjacent vertices be different colored. Coloring vertex 1 in red requires vertices 2, 8, 9, 10 be in white. Continuing in the same way, we find that 3, 5, 7 and 11 are red colored while 4 and 6 are white colored. From what we conclude that this is bipartite graph with bipartition sets $V_1 = \{1,3,5,7,11\}$ and $V_2 = \{2,4,6,8,9,10\}$. Meanwhile this is not complete bipartite graph.

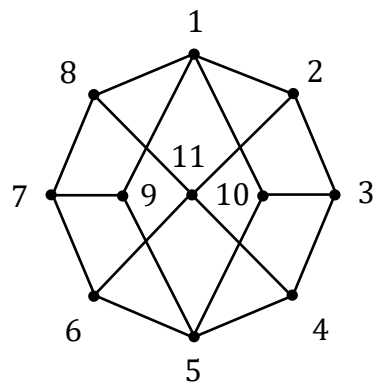


Figure 10.8 Two-colorable graph, which is bipartite but not complete bipartite graph.

Theorem 10.1 The sum of the degrees of the vertices of a pseudograph is an even number equal to twice the number of edges. In symbols, if $G(V, E)$ is a pseudograph, then

$$\sum_{v \in V} \deg(v) = 2|E|.$$

Proof Each edge, which is not a loop, has two ends and therefore contributes twice to the total degree. On the other hand, a loop at a vertex contributes twice to the degree of that vertex and

consequently to the total degree. Thus, sum of the degrees is twice of the number of edges as claimed.

Given pseudograph G, a **degree sequence** $d_1, d_2, \dots, d_n$ is a monotonic nonincreasing sequence of the vertex degrees. Degree sequence of the graph in Figure 10.8 is 4,4,4,3,3,3,3,3,3,3,3.

Corollary The number of odd vertices in a pseudograph is even.

Proof By proposition $\sum_{v \in V} \deg(v) = 2|E|$ is an even number. Since

$$\sum_{v \in V} \deg(v) = \sum_{\substack{v \in V \\ v \text{ even}}} \deg(v) + \sum_{\substack{v \in V \\ v \text{ odd}}} \deg(v)$$

and the first sum on the right, being a sum of even numbers, is even, so also the second sum also must be even. Since the sum of an odd number of odd numbers is odd, the number of terms in the sum here, that is, the number of odd vertices, must be even.

Consider pseudograph in Figure 10.9. Degrees of the vertices are

$$\deg v_1 = 2, \deg v_2 = 4, \deg v_3 = 3, \deg v_4 = 2, \deg v_5 = 1, \deg v_6 = 0.$$

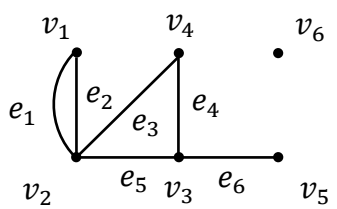


Figure 10.9 A multigraph G.

Degree sequence is given by 4,3,2,2,1,0. We verify that

$$4 + 3 + 2 + 2 + 1 + 0 = 12$$

that is, the sum of the degrees of the vertices is an even number. v_1, v_2, v_4 and v_6 are even vertices while v_3 and v_5 are odd vertices. We again verify that number of odd numbers is even and therefore sum of the odd degrees is even.

10.3 Isomorphism

It is important to know when two graphs are substantially the same and when they are essentially different. When we say graphs are "substantially the same," we mean that they differ only in the way they are labeled or drawn. There should be a one-to-one correspondence between the vertices of the graphs and a one-to-one correspondence between their edges such that corresponding vertices are incident with corresponding edges. The proper word for "substantially the same" is **isomorphic**.

There is a distinction between a graph and its picture. A graph is a set V and a set E of unordered pairs of elements of V. A picture of it consists of dots and lines that can be drawn and arranged in many different ways.

Example 10.2 Figure 10.10 shows two graphs that are not substantially the same. Each graph consists of four vertices and four edges, but G_2 contains a vertex of degree 1, while G_1 has no such vertex.

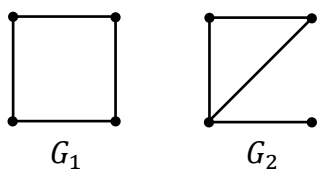


Figure 10.10 Two graphs that are not substantially the same.

Likewise, G_2 has a vertex of degree 3, but G_1 has no such vertex. These observations allow us to conclude that G_1 and G_2 are not substantially the same as we cannot obtain one graph from another by rearranging its components.

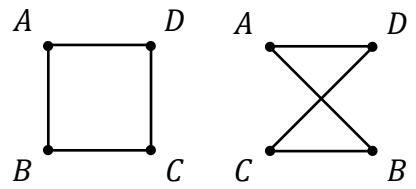


Figure 10.11 Two graphs that are substantially the same.

Example 10.3 The graphs in Figure 10.11 are isomorphic. We label graphs in such way to demonstrate correspondence between vertices in Figure 10.11, the graph in the right can be obtained from one in the left by holding A and D and exchanging B and C. As a result, graph gets "twisted" as shown in the right.

Pause Show that the graphs in Figure 10.12 are isomorphic by assigning the labels A, B, C, and D to appropriate vertices of the graph on the right. (It might help to think of the edges of G_2 as pieces of string knotted at the vertices. How could G_2 be rearranged to look like G_1? Once you see this, it will be easy to do the required labeling.)

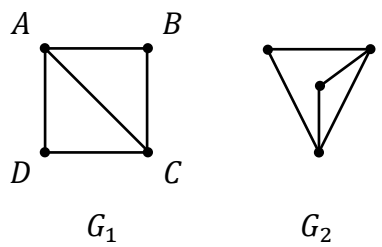


Figure 10.12 Two graphs that are substantially the same.

Definition 10.1 (Graph isomorphism) Given graphs $G_1 = G_1(V_1, E_1)$ and $G_2 = G_2(V_2, E_2)$, we say that G_1 is isomorphic to G_2 and write $G_1 \cong G_2$ if there is a one-to-one function φ from V_1 onto V_2 such that (i) if vw is an edge in E_1, then $\varphi(v)\varphi(w)$ is an edge in E_2, and (ii) every edge in E_2 has the form $\varphi(v)\varphi(w)$ for some edge $vw \in E_1$. We call φ an isomorphism from G_1 to G_2 and, abusing notation, say that $\varphi: G_1 \to G_2$ is an isomorphism.

The definition of isomorphism is symmetric: If G_1 is isomorphic to G_2, then G_2 is isomorphic to G_1. In fact, if $\varphi: G_1 \to G_2$ is an isomorphism then $\varphi^{-1}: G_2 \to G_1$is also isomorphism. Thus, there is no ambiguity if we say that two graphs are isomorphic.

The notion of isomorphism is exceedingly important in mathematics. If the definition looks complicated, the idea is very simple. Isomorphic objects, from a mathematical standpoint, are the same; they differ only in appearance.

We have remarked that the definition of isomorphism is

- **symmetric**: $G_1 \cong G_2$ if and only if $G_2 \cong G_1$
- **reflexive**: $G \cong G$ for any graph G
- **transitive**: If $G_1 \cong G_2$ and $G_2 \cong G_3$ then $G_1 \cong G_3$.

Thus, isomorphism is an equivalence relation on the set of all graphs. The set of all graphs is, therefore partitioned into disjoint equivalence classes called **isomorphism classes**.

Proposition. If G_1 and G_2 are isomorphic graphs, then G_1 and G_2 have same number of vertices, same number of edges, and same degree sequences.

The number of vertices, the number of edges and the degree sequences are invariants in terms of isomorphism.

10.3 Eulerian Circuits

A **walk** in a pseudograph is an alternating sequence of vertices and edges, beginning and ending with a vertex, in which each vertex (except the last) is incident with the edge, which follows and the last edge is incident with the edge, which precedes it. The length of a walk is the number of edges in it. A walk is **closed** if the first vertex is same as the last one and, otherwise, it is **open**. A **trail** is a walk in which all edges are distinct; a **path** is a walk in which all vertices are distinct. A closed trail is a **circuit**. A circuit in which the first vertex appears exactly twice (at the beginning and the end) and in which no other vertex appears more than once is a **cycle**. An ***n*-cycle** is a cycle with n vertices. It is **even** if n is even and **odd** if n is odd.

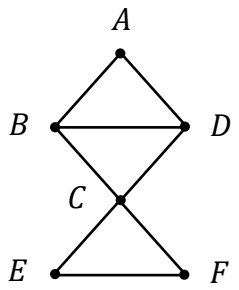


Figure 10.13 A graph.

There are lot of words here and, as always, we cannot emphasize too strongly the importance of coming to grips with their meanings. When thinking about the concepts of path and trail, it is perhaps helpful to note that a path is necessarily a trail: If all the vertices of a walk are different, then all the edges must be different too. As we show in the next example, the converse is not true: A trail need not be a path.

Example 10.4 In the graph shown in Figure 10.13, $ABCEFCBD$ is a walk of length 7 which is neither a trail nor a path; $ABCEFCD$ is a trail, but not a path; $ABCEFCDBA$ is a closed walk which is not a circuit; $BCEFCDB$ is a circuit which is not a cycle; and $BCDB$ is a 3-cycle and hence an odd cycle. The closed walk $CEFCBDC$ is not a cycle because the first and last vertex appears a third time.

Eulerian circuit in a pseudograph is a circuit, which contains every vertex and every edge. **Eulerian pseudograph** is a pseudograph, which contains an Eulerian circuit.

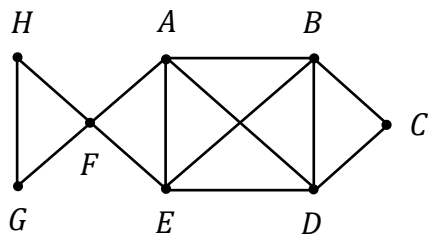


Figure 10.14 A graph.

Example 10.5 A graph shown in Figure 10.14 illustrates the difference between a circuit and a Eulerian circuit. The circuit $ABCDEFGHFA$ is not Eulerian since, while it encompasses all vertices, it omits four edges. The graph is Eulerian, however: $ABCDEFGHFADBEA$ is a Eulerian circuit.

A pseudograph is **connected** if and only if there exists a walk between any two vertices. The above pseudograph is an example to disconnected graph, which has three connected components. Each component is a connected subgraph of the given graph.

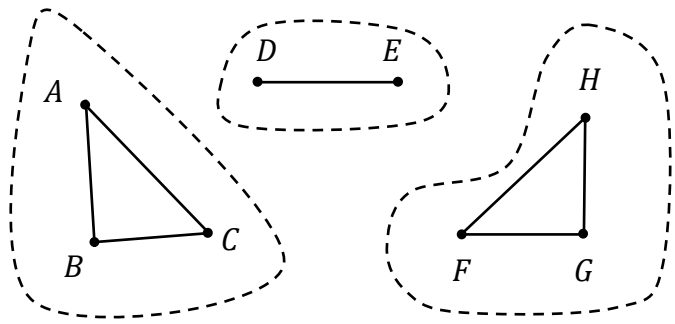


Figure 10.15 A disconnected graph which has three connected components.

Theorem 10.2. A pseudograph (with at least two vertices) is Eulerian if and only if it is connected and every vertex is even.

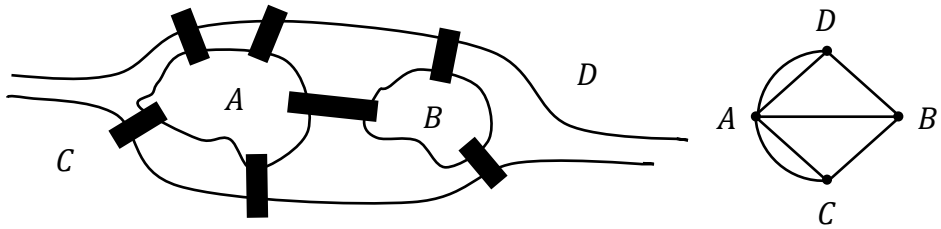


Figure 10.16 Schematic illustration of The Königsberg Bridges Problem (in the left) and its graph model (in the right).

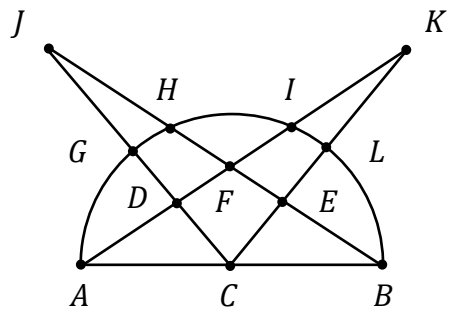


Figure 10.17 A graph which has Eulerian trail.

Eulerian circuits are named after Leonhard Euler, the solver of The Königsberg Bridges Problem. Figure 10.16 shows graph model of The Königsberg Bridges Problem. The degree sequence of the graph is 5, 3, 3, 3. According to Theorem 10.2 this is not Eulerian graph.

Theorem 10.3 A pseudograph G possesses a Eulerian trail between two different vertices u and v if and only if G is connected and all vertices except u and v are even.

Example 10.6 Consider connected graph shown in Figure 10.17. Vertices A and B have degree 3, J and K have degree 2, all the rest vertices have degree 4. According to Theorem 10.3 there exists a Eulerian trail from A to B (or from B to A). Here is an example of Eulerian trail: $AGJHIKLBELIFHGDACDFECB$.

10.4 Hamiltonian Cycles

A Eulerian circuit contains every edge of a graph exactly once. In this section, we discuss circuits that contain each vertex of a graph exactly once. A **Hamiltonian**[10] **cycle** in a graph is a cycle, which

[10] William Rowan Hamilton (1805–1865) was born in Dublin. He is perhaps the most famous Irish scientist. By 10 he had learned Latin, Greek, Hebrew, Italian, French, Arabic, Sanskrit, and Persian. At 18, he entered Trinity College, Dublin. At Trinity, he was a superior student in both the sciences and the classics. Prior to receiving his degree, he was appointed the Astronomer Royal of Ireland. He held this position until his death, living and working at Dunsink Observatory outside of Dublin. Hamilton made important contributions to optics, abstract algebra, and dynamics. In 1857 Hamilton invented "The Icosian Game" based on his work in noncommutative algebra. He sold the idea for 25 pounds to a dealer in games and puzzles. (Because the game never sold well, this turned out to be a bad investment for the dealer.) The "Traveler's Dodecahedron," also

contains every vertex of the graph. A **Hamiltonian graph** is the one with a Hamiltonian cycle. Some authors define a Hamiltonian circuit as a circuit in which every vertex except the first and last appears exactly once. A circuit with no repeated vertices (except the first and the last) is a cycle, so a Hamiltonian circuit is a cycle. Hereby we can use the terms Hamiltonian circuit and Hamiltonian cycle.

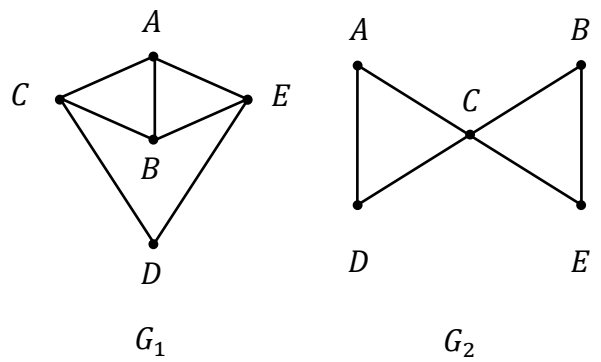


Figure 10.18 Two graphs: $G_1$is Hamiltonian but G_2 is not.

Example 10.7 Graph G_1 shown in Figure 10.18 is Hamiltonian: The cycle $ABCDEA$, for instance, is Hamiltonian. On the other hand, graph G_2 is not Hamiltonian, but how can we convince ourselves of this fact?

While it is possible to decide precisely which graphs possess Eulerian circuits, it is noteworthy that there has not yet been found a way to classify Hamiltonian graphs; in other words, there is no known theorem of the sort, "G is Hamiltonian if and only if . . . ". There are however some properties of cycles which are helpful in

called "A Voyage Round the World," the puzzle described in this textbook, is a variant of that game.

trying to find Hamiltonian cycles. These properties sometimes allow us to conclude whether a graph is Hamiltonian.

Hamiltonian graphs take their name from Sir William Rowan Hamilton, a great Irish mathematician and scientist who invented a game which made use of a wooden regular dodecahedron, that is, a solid with 12 congruent faces, each of which is a regular pentagon as shown in Figure 10.19. The vertices of the dodecahedron were labeled with the names of 20 cities of the world, and the object of the game was to find a route "around the world," along the edges of the solid, which passed through each city exactly once and led back to the city where the tour started.

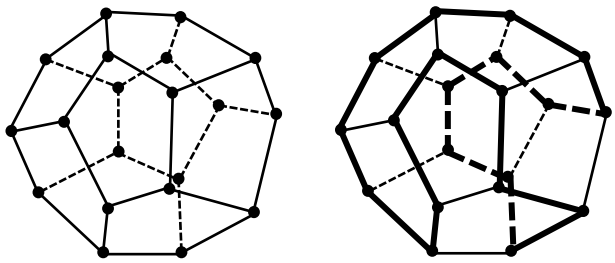


Figure 10.19 A regular dodecahedron has 12 identical pentagonal faces and 20 vertices. Heavy lines show so-called route "around the world" on dodecahedron.

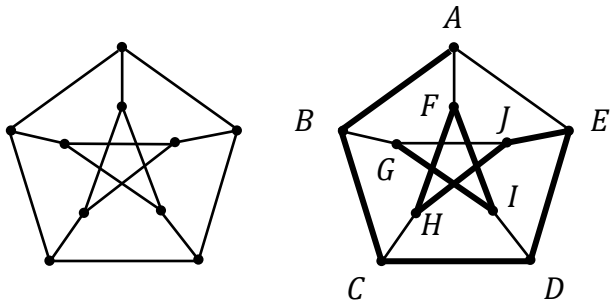


Figure 10.20 Petersen's graph is not Hamiltonian. A route in heavy lines from A to G is an example of unsuccessful attempt to draw Hamiltonian cycle.

Imagine that we stretch the dodecahedron that sits on the pentagon so that the solid collapses until it is flat. The result is a graph in the plane. Hamilton's world tour is possible only in the case if graph contains so- called a Hamiltonian cycle. The graph in fact is Hamiltonian; a Hamiltonian cycle is marked with the heavy lines in Figure 10.9 (the right one).

As a deeper application of the properties of Hamiltonian graphs, we introduce a famous graph named after the Danish mathematician Julius Petersen (1839-1910) which is not Hamiltonian.

There are some graphs in which a Hamiltonian cycle always exists (the complete graph on n vertices, for instance). Assigning the vertices of K_n, the labels $v_1, v_2, \dots, v_n$ then $v_1, v_2, \dots, v_n, v_1$ is a cycle because we are assured of an edge between each pair of vertices. The cycle clearly passes through each vertex (except the first and the last) exactly once, so it is Hamiltonian. It would seem that a graph with "lots" of edges should have a good chance of being Hamiltonian. The following theorem, published by G. A. Dirac in 1952, provides further evidence in support of this idea.

Theorem 10.4. (Dirac[11]) If a graph G has $n \geq 3$ vertices and every vertex has degree at least $n/2$ then G is Hamiltonian.

10.5 Adjacency Matrix

Graphs occur with increasing frequency in modern-day problems. While theoretically, any problem associated with a finite graph is solvable, in practice, the number of cases to consider is often so large and the time needed to deal with each case so great that an exhaustive search of all possibilities is impossible. Accordingly, the discovery of new graph-based algorithms and ways to improve efficiency are flourishing areas of mathematical research today. In order to write an algorithm, which requires the input of a graph, we first must decide how to code the information, which describes a graph. For this purpose, the adjacency matrix is commonly used.

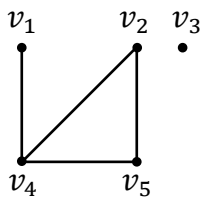
Definition 10.2 (Adjacency matrix) Let G be a graph with n vertices labelled $v_1, \dots, v_n$. For each i and j with $1 \leq i, j \leq n$, define

$$a_{ij} = \begin{cases} 1 & \text{if } v_i v_j \text{ is an edge} \\ 0 & \text{otherwise} \end{cases}$$

[11] Gabriel Andrew Dirac (1925–1984) was born in Budapest. Gabriel A. Dirac entered Cambridge University in 1942, but his studies were interrupted by wartime service in the aviation industry. He obtained his Ph.D. in mathematics in 1951 from the University of London. He held university positions in England, Canada, Austria, Germany, and Denmark, where he spent his last 14 years. Dirac became interested in graph theory early in his career and help raise its status as an important topic of research. He made important contributions to many aspects of graph theory, including graph coloring and Hamilton circuits.

The adjacency matrix of G is then $n \times n$ matrix $A = (a_{ij})$ whose (i, j) entry is a_{ij}.

Example 10.8 Figure 10.21 shows a graph and its adjacency matrix.



$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

Figure 10.21 A graph (in the left) and its adjacency matrix (in the right).

Example 10.9 Below we demonstrate the adjacency matrix A of complete bipartite graph with bipartition sets $V_1 = \{v_1, v_2\}$ and $V_2 = \{v_3, v_4, v_5\}$.

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Let G be a graph with vertices $v_1, v_2, \dots, v_n$ and let $A = [a_{ij}]$ be the adjacency matrix of G. The following properties hold.

- The diagonal entries of A are all 0s; that is, $a_{ii} = 0$ for $i = 1, \dots, n$. This is because an edge from a vertex v_i to v_i is a loop and a graph has no loop. Actually, we can define the adjacency matrix of a pseudograph and show that it has most of the properties given here, but we shall restrict our attention to graphs only.

- The adjacency matrix is a symmetric matrix, that is, $a_{ij} = a_{ji}$ for all i and j. Conversely, given any symmetric matrix A which contains only 0s and 1s and only 0s on its diagonal, there exists a graph G; whose adjacency matrix is A. Thus, there is a one-to-one correspondence between graphs and symmetric 0-1 matrices with only 0's on the main diagonal.

10.6 Trees

A tree is one of often referenced graph types. The name says for it – it is called a tree, because it can be drawn so that be looking like a tree. A **tree** is a connected graph, which contains no circuits. There are several alternative definitions.

- Let G be a graph. Then the following statements are equivalent.
- G is a tree.
- G is connected and acyclic; that is, without cycles.
- Between any two vertices of G there is precisely one path.

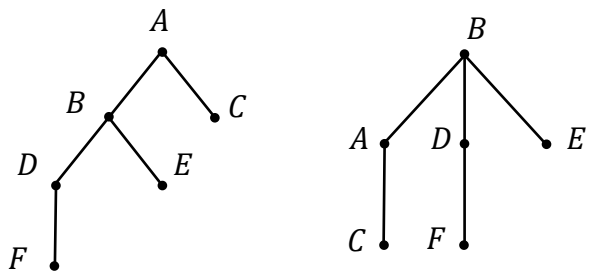


Figure 10.22 Two rooted trees.

A tree is **rooted** if it comes with a specified vertex, called the **root**. The way in which a rooted tree is drawn clearly depends upon which vertex is the root. In Figure 10.22, a tree is drawn in two different ways. A is the root of the tree which is shown in the left while B is the root of the one shown in the right. One can easily check that these two trees are isomorphic.

A tree with more than one vertex must contain a vertex of degree 1. We call such a vertex a **leaf**.

Theorem 10.5 A connected graph with n vertices is a tree if and only if it has $n-1$ edges.

Corollary 1. A tree with more than one vertex has at least two leaves, that is, at least two vertices of degree 1.

Corollary 2. Any edge added to a tree must produce a circuit.

10.7 Spanning Trees

Figure 10.23 represents a map of a province in which we represent towns, roads between towns and distances by vertices, edges and numeric values surrounding corresponding edges, respectively. The province administration wants to pave roads in such a way that one can get between any two towns on pavement. The question here is what roads to pave to minimize the expenditures. The problem asks in the language of graph theory, for so-called a minimum spanning tree of the graph.

Let G be a weighted connected graph. We recall here that G is a **weighted graph** whose edges are assigned certain numeric values called weights. A **spanning tree** of G is a subgraph which is a tree and which includes every vertex of G. A **minimum spanning tree** of a **weighted graph** is a spanning tree having least weight. Stated in

other words, a minimum spanning tree is the one for which the sum of the weights of all edges is the minimum.

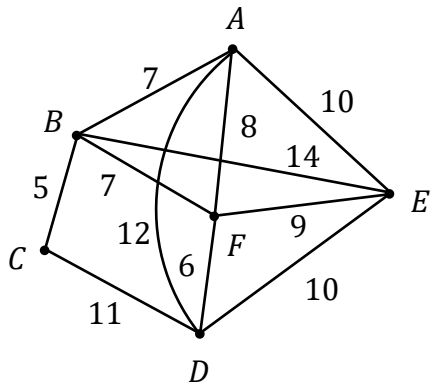


Figure 10.23 A connected weighted graph.

The concept of spanning tree exists only for a connected graph G and, if G has n vertices, any spanning tree must necessarily contain $n-1$ edges.

It is not hard to find a spanning tree in a connected graph G. If G has no circuits, then it is already a tree, so G is a spanning tree for G. If G contains a circuit, then we can delete an edge (without deleting any vertex) so as to remove the circuit but leave the graph connected. By repeating this procedure, we eventually find a connected subgraph without circuits containing all the vertices of G, that is, a spanning tree.

10.8 Minimum Spanning Trees

Now let us find a minimum spanning tree, given a multigraph. Indeed, we could enumerate all the spanning trees of a weighted

graph and simply choose the tree of least weight, but if the graph is at all complicated, this is not an easy procedure. In this section, we discuss two better ways discovered in the 1950s by J. B. Kruskal[12] and R. C. Prim[13]. Both algorithms are in some sense "obvious". They produce a minimum weight spanning tree an edge at a time, at each stage making the best choice of next edge without thought to what has happened already or what will ensure. For this reason, we call them greedy algorithms.

Kruskal's Algorithm

To find a minimum spanning tree in a connected weighted graph with $n > 1$ vertices, carry out the following procedure.

Find an edge of least weight and call this e_1. Set $k = 1$.

While $k < n$

if there exists an edge e such that $\{e\} \cup \{e_1, e_2, \dots, e_k\}$ does not contain a circuit, let

---

[12] Joseph Bernard Kruskal (1928–2010) was born in New York City. Kruskal attended the University of Chicago and received his Ph.D. from Princeton University in 1954. He was an instructor in mathematics at Princeton and at the University of Wisconsin, and later he was an assistant professor at the University of Michigan. In 1959 he became a member of the technical staff at Bell Laboratories, where he worked until his retirement in the late 1990s. Kruskal discovered his algorithm for producing minimum spanning trees when he was a second-year graduate student. His research interests included statistical linguistics and psychometrics.

[13] Robert Clay Prim is an American mathematician and computer scientist. In 1941, Prim received his B.S. in Electrical Engineering from The University of Texas at Austin. Later in 1949, he received his Ph.D. in Mathematics from Princeton University, where he also worked as a research associate from 1948 until 1949. At Bell Laboratories, he served as director of mathematics research from 1958 to 1961. During his career at Bell Laboratories, Robert Prim along with coworker Joseph Kruskal developed two different algorithms for finding a minimum spanning tree in a weighted graph, a basic stumbling block in computer network design.

$e = e_{k+1}$ be such an edge of least weight and replace k by $k + 1$;
else output $e_1, e_2, \ldots, e_{n-1}$ and stop.
end while

Example 10.10 We illustrate with reference to the graph shown in Figure 10.24. Kruskal's algorithm initially picks an edge of lowest weight. There are several edges of weight 2 and none with less weight. We can select any of these edges. Suppose we begin with edge $e_1 = AJ$.

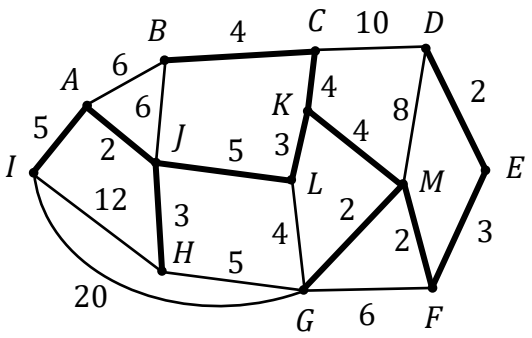


Figure 10.24 A connected weighted graph. Heavy lines show its minimal spanning tree by Kruskal's algorithm.

There still remain edges of weight 2, so we choose any other, say $e_2 = GM$. Among those edges which do not form a circuit with e_1 and e_2 (at this point there are no edges which do), we choose one of lowest weight, say DE, and call this e_3 . Continuing, we might select edges AJ, GM, DE, MF, FE, LK, HJ, KM, CK, BC. At this stage, there remains an edge (GL) of weight 4, but this edge would complete a circuit with LK, KM, GM already selected, and so we choose an edge of weight 5, say JL, and another, IA. Now, any of the remaining edges will complete a circuit with those selected and so we must stop; the algorithm assures us we have the desired minimum

spanning tree. It has weight 39. We illustrate below trace of Kruskal's algorithm for the given example.

step	1	2	3	4	5	6	7	8	9	10	11	12
edge	AJ	GM	DE	MF	FE	LK	HJ	KM	CK	BC	JL	AI
weight	2	2	2	2	3	3	3	4	4	4	5	5

Why does the algorithm work? Suppose G is the given connected graph and T_k is the subgraph produced by Kruskal's algorithm. The algorithm terminates only when the inclusion of any edge of G not in T_k produces a circuit in T_k.

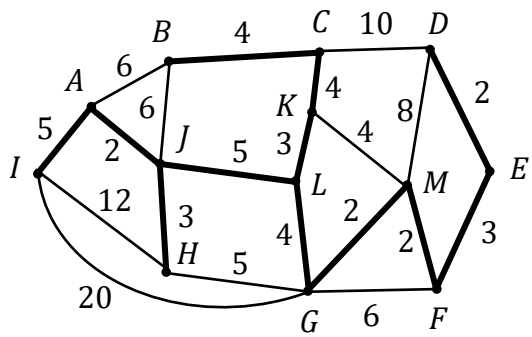


Figure 10.25 A connected weighted graph and its minimal spanning tree by Prim's algorithm.

Prim's Algorithm

To find a minimum spanning tree in a connected weighted graph with $n > 1$ vertices, proceed as follows.

Step 1. Choose any vertex v and let e_1 be an edge of least weight incident with and then set $k = 1$.

Step 2. While $k < n$

if there exists a vertex which is not in the subgraph T whose edges are

$e_1, e_2, \ldots, e_k$
let e_{k+1} be an edge of least weight among all edges of the form ux, where u is a vertex of T and x is a vertex not in T;
replace k by $k + 1$;
else output $e_1, e_2, \ldots, e_k$ and stop
end while

Example 10.10 We illustrate Prim's approach, as it would apply to the same graph shown in the right. Start at any vertex, say K, and select the edge incident with K of least weight. Thus, we select edge KL of weight 3. The next edge chosen is the edge of least weight adjacent to KL. Any of LG, KM, or KC is eligible. Suppose we select LG. Then we select the edge of least weight among those edges forming a tree with the two edges already selected (GM, for instance). We continue to select, in succession, the edge of least weight among those edges, as yet unselected, which form a tree with those already chosen and continue until we have a spanning tree. You should verify that the spanning tree shown in the figure is one, which Prim's algorithm might identify.

step	1	2	3	4	5	6	7	8	9	10	11	12
edge	KL	LG	GM	MF	EF	DE	CK	BC	JL	AJ	JH	AI
weight	3	4	2	2	3	2	4	4	5	2	3	5

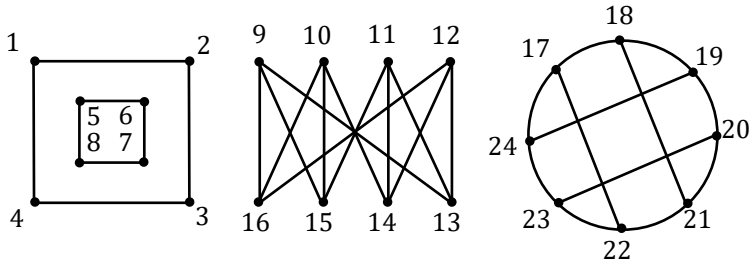
The sum of the weights of the edges in T is 39.

Exercises

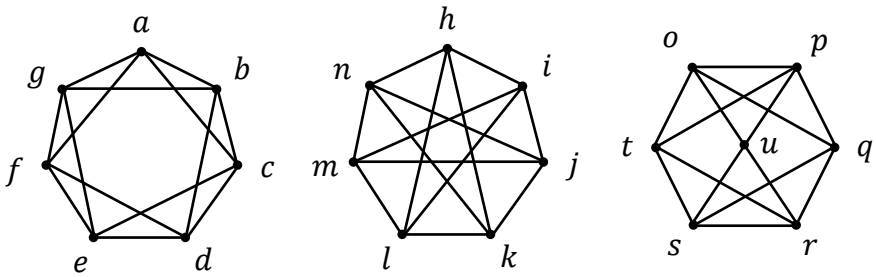
1. Find the minimum number of vertices in a complete bipartite graph, which has 60 edges.

2. Given that $n = m + l$, find the number of edges to be removed from K_n in order to transform it into two disjoint complete graphs: K_m and K_l.
3. Find the number of edges and vertices in $K_{50,100}$.
4. Given that graph K_n has 55 edges. Find the number of its vertices. Determine sum of the degrees of the vertices.
5. How many edges does n-vertex tree have?
6. Given that K_n has 21 edges. Find the number of its vertices.
7. Is it possible to connect 25 telephones with wires so that, each telephone be connected with exactly 7 other telephones.
8. A graph has 56 edges. Find sum of the degrees of its vertices.
9. K_n has 21 edges. Find n.
10. A graph G has 15 edges. Find the number of vertices if G is a complete graph. Find the minimal number of vertices if G is a complete bipartite graph.
11. There are 9 line segments drawn in a plane. Is it possible that each line segment intersects exactly 3 others?
12. There are 25 telephones in certain city. Is it possible to connect them with wires so that each telephone be connected with exactly 7 other telephones.
13. Let $A = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$ be the adjacency matrix of a graph. Draw the graph. Determine whether this graph is a tree. Determine whether this graph is connected graph.

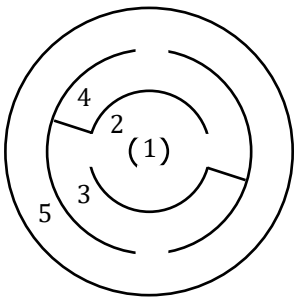
14. Which of the following graphs are isomorphic? If the condition holds, write corresponding isomorphic function(s).



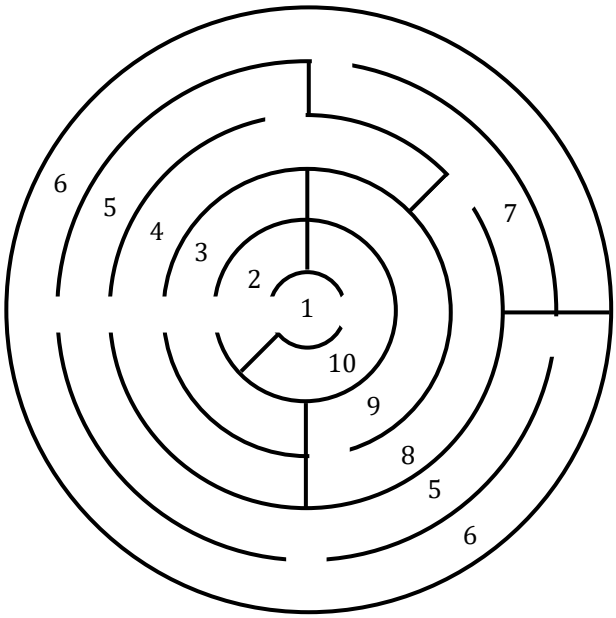
15. Which of the following graphs are isomorphic?



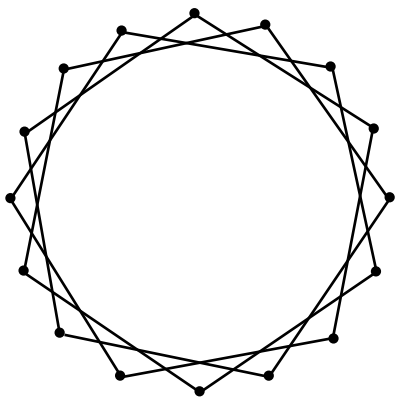
16. You are presently in the room 1 of the following labyrinth. You want to tour all 5 rooms and return back to room 1. When visiting the rooms, you must pass each door exactly once. Is it possible? If so, provide an example of such a tour. Justify your answer based on corresponding facts of graph theory.



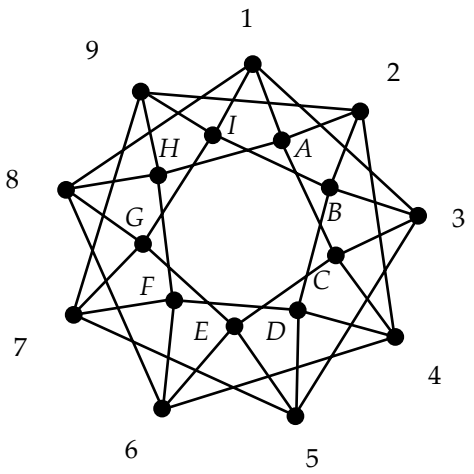
17. In the following 10-room labyrenth, each room has one or more doors. Corresponding pseudograph is obtained by representing rooms and doors by vertices and edges, respectively. Check whether this pseudograph a) is Eulerian; b) is Hamiltonian; c) has a Eulerian trail.



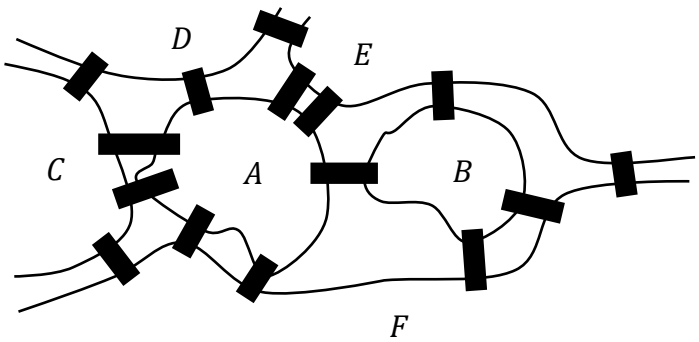
18. Make sure whether the following graph is a) Eulerian, b) has a Eulerian trail and/or c) Hamiltonian.



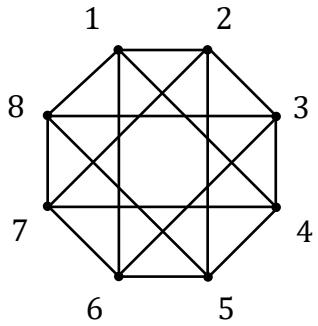
19. Is the following graph Hamiltonian? Is it Eulerian? Does it have a Eulerian trail? Explain your answer.



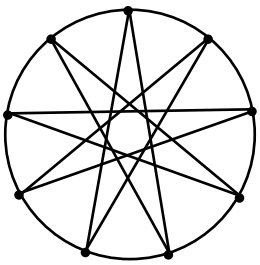
20. Consider a version of The Königsberg Bridges Problem, in which there are 6 land masses connected by 15 bridges as it is shown in the following figure. If 15 bridges can all be traversed in a single trip without doubling back, with the additional requirement, that the trip ends in the same land mass it began? If 15 bridges can all be traversed in a single trip without doubling back, with the additional requirement that the trip does not end in the same land mass it began?



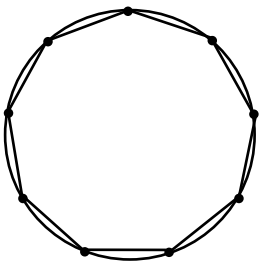
21. A bipartite graph is shown in the figure. Find its bipartition sets.



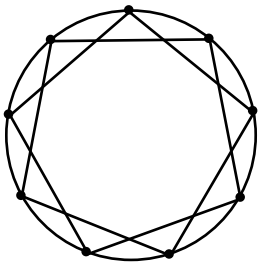
22. Determine which of the following graphs are isomorphic.



G_1

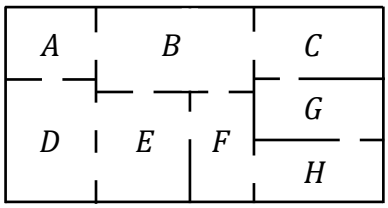


G_2

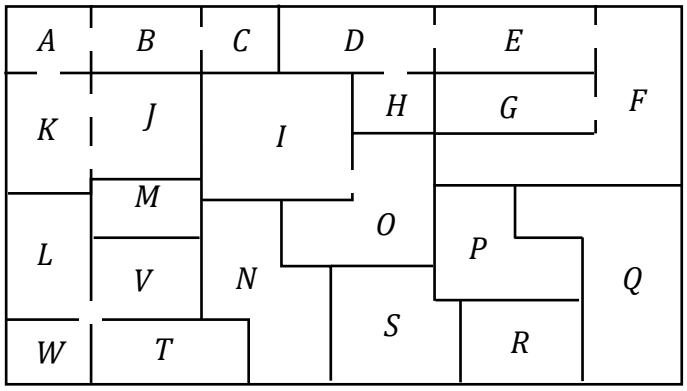


G_3

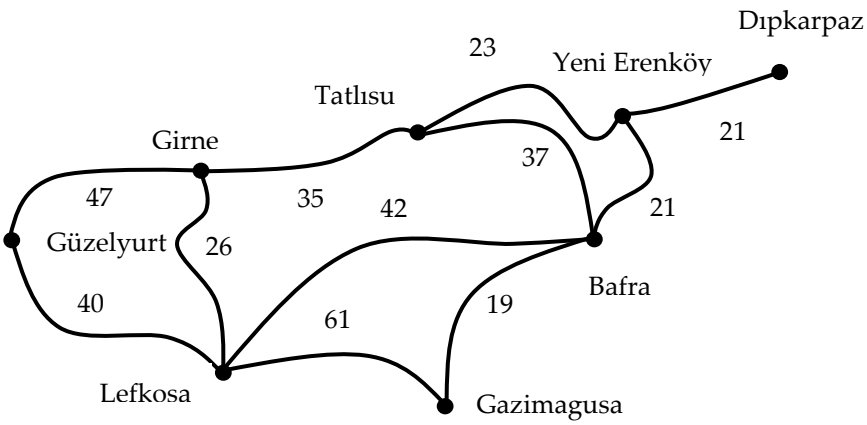
23. The following picture is a floor plan of your brand new 8-room house. You want to tour your house by visiting each room exactly once and be back to the room you will start. Is it possible? If so, provide an example of the tour. What is the name/type of a graph with this property?



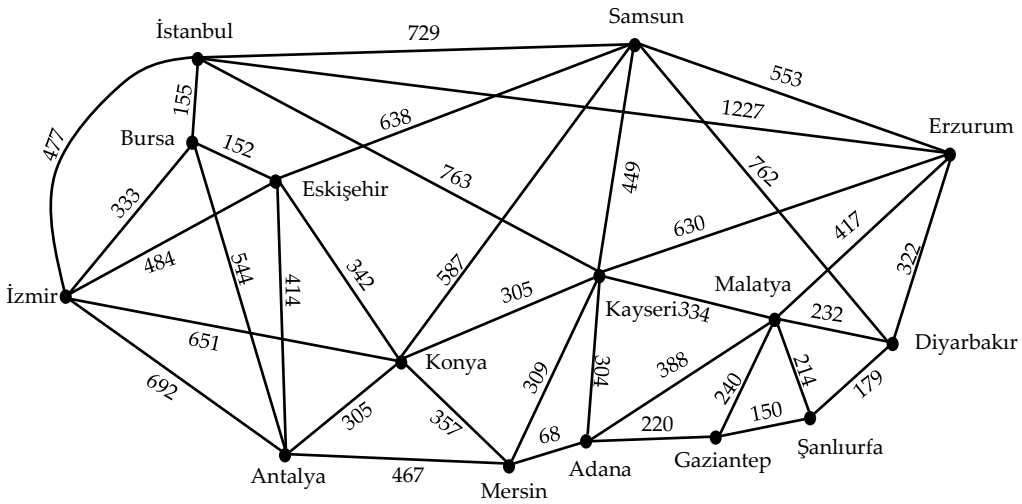
24. The following figure is a floor plan of a 22-room house, in which each room has one or more doors. Corresponding pseudograph is obtained by representing rooms and doors by vertices and edges, respectively. Is it Eulerian graph? If not, make sure whether the graph has an Eulerian trail. Is it Hamiltonian graph?



25. The following weighted graph illustrates distances between 8 urban areas of North Cyprus. Use Prim's algorithm to find minimum spanning tree and find sum of the weights of its edges.



26. The following weighted graph illustrates the road distances between 15 cities of Turkey. Find sum of the weights of the edges in a minimal spanning tree.



Recommended References

Among the resources available for deeper coverage of many topics in discrete mathematics, I recommend the following four books that, as I am sure, can be more useful to our students. The book by Balakrishnan [1] is a concise and compact textbook, which we used as a textbook in discrete mathematics for several years. The book by Rosen [4] is rather comprehensive textbook, which covers many topics, including those not covered by the course outline. For the past decade, we have permanently used the books by Goodaire and Parmenter [2] as textbook for teaching the main body of the course material and McElience, Ash and Ash [3] for teaching logic.

Chapter 1

All four recommended books are easy to read text on sets. Famous set identities can be found in Rosen [4].

Chapter 2

All four recommended books are general references for relations. In addition, comparison property and linear order are discussed in Balakrishnan [1].

Chapter 3

All four books are recommended for functions. In Goodaire and Parmenter [2], a function is defined and then applied in terms of

relations, while in other three books a function is defined in terms of sets. In this textbook, we follow up the style adopted in Goodaire and Parmenter [2].

Chapter 4

General references for propositional logic include Goodaire and Parmenter [2], McElience, Ash and Ash [3], and Rosen [4]. Well-formed propositions and famous theorems of propositional logic are described in McElience, Ash and Ash [3], while application of propositional logic is discussed in Rosen [4].

Chapter 5

For topics in Boolean algebra, the students are referred to McElience, Ash and Ash [3], and Rosen [4].

Chapter 6

For weak form of mathematical induction, the readers are referred to McElience, Ash and Ash [3], while both weak and strong forms are detailed in Balakrishnan [1], Goodaire and Parmenter [2], and Rosen [4].

Chapter 7

For recursive sequences and recurrence relation, the students are referred to Goodaire and Parmenter [2].

Chapter 8

For the principle of inclusion-exclusion, students are referred to Balakrishnan [1], Goodaire and Parmenter [2], and Rosen [4].

Chapter 9

General references for addition and multiplication rules, pigeonhole principle, permutations, combinations, repetitions, binomial expansion and Pascal's triangle include all four recommended books. Permuting indistinguishable objects and derangements are studied in Goodaire and Parmenter [2].

Chapter 10

General references for graph theory include all four recommended books. However, graph isomorphism, Kruskal's minimum spanning tree algorithm are discussed in Balakrishnan [1], Goodaire and Parmenter [2], and Rosen [4].

References

1. V.K. Balakrishnan, Introductory Discrete Mathematics, Dover Publications, Inc., New York, NY, 1996.
2. E.G. Goodaire, M.M. Parmenter, Discrete Mathematics with Graph Theory, 3rd Edition, Prentice Hall, Upper Saddle River, NJ, 2019.

3. R.J. McElience, R.B. Ash, C. Ash, C, Introduction to Discrete Mathematics, Random House Inc, 1989.
4. K.H. Rosen, Discrete Mathematics and Its Applications, 7th Edition, McGraw-Hill, New York, NY, 1999.

Solutions to Exercises

Chapter 1

$|P(P(A))| = 2^{2^2} = 16.$

$|P(P(A))| = 2^{2^3} = 256.$

$C \backslash D = C \cap \bar{D}$. We can prove this equality by comparing corresponding Venn diagrams.

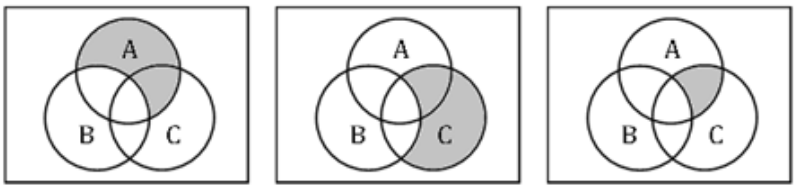
$(A - B) \oplus (B \cap C) = \{3,5\} \oplus \{2,7,8,9\} = \{2,3,5,7,8,9\}.$

$A = (-3,100) \cap (-100,4) = (-3,4)$. $\bar{A} = (-\infty, -3] \cup [4, \infty)$.

$\{\emptyset\} \in \{\emptyset\}$ is wrong claim as $\{\emptyset\}$ is not element of $\{\emptyset\}$.

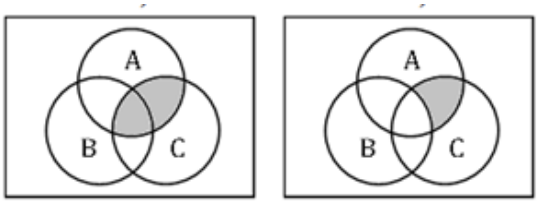
The proof is by step-by-step drawing Venn diagrams for $(A \backslash B) \cap (C \backslash B)$ and $(A \cap C) \backslash B$ and then comparing gray segments in c) and e) (see figure). Since gray segments in c) and e) are matching, this is a correct set identity.

$A \backslash B$ $\quad$ $C \backslash B$ $\quad$ $(A \backslash B) \cap (C \backslash B)$



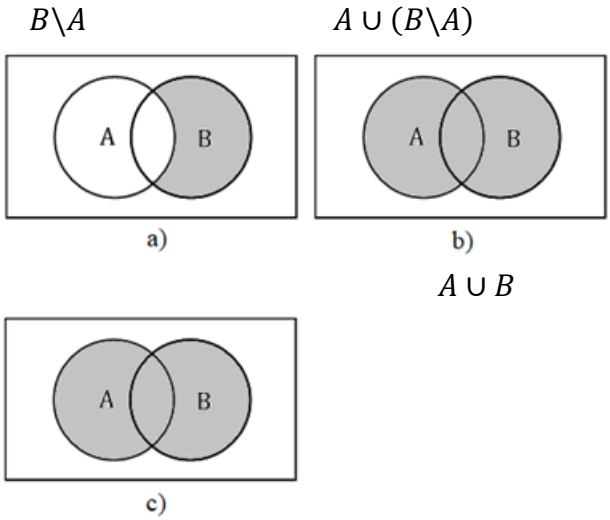
a) $\quad$ b) $\quad$ c)

$(A \cap C)$ $\quad$ $(A \cap C) \backslash B$



d) $\quad$ e)

The proof is by step-by-step drawing Venn diagrams for $A \cup (B \backslash A)$ and $A \cup B$ and then comparing gray segments in b) and c) as shown in the figure. This is a correct set identity because gray segments in b) and c) are matching.



$A \backslash B = \{1,3,5,7,9,11\} \backslash \{2,3,5,7,11\} = \{1,9\}$.

$(A - C) \oplus (B \cap C) = \{1,5\} \oplus \{2,3\} = \{1,2,3,5\} \neq \{1,2,3,5,7\}$.

There exist multiple equivalent solutions. One possible solution is $(A \cap C) \backslash B$.

There exist multiple solutions. One possible solution is $(A \cup B \cup C) \backslash \big((A \cap B) \cup (A \cap C) \cup (B \cap C)\big)$.

Chapter 2

We need to show that R is reflexive, symmetric and transitive relation. R is reflexive, because $a \sim a$ for all residents of the country (trivial pairing for a citizen a). R is symmetric because $a \sim b$ implies $b \sim a$ (if a and b are citizens of the same region then it does not matter how to pair a and b). R is transitive because $a \sim b$ and $b \sim c$ impliy $a \sim c$

(if a and b are citizens of the same region and that if b and c are residents of the same region, then a and c are also citizens of the same region).

R is not reflexive because (d, d) is not in R. R is symmetric because (a, d) and (d, a); (b, e) and (e, b); (c, d) and (d, e) are all elements of R. Because of the same reason R is not antisymmetric. R is not transitive because (d, a) and (a, d) are elements of the relation but (d, d) is not. Hence, R is reflexive and symmetric relation.

The given relation is reflexive, symmetric, antisymmetric and transitive. This is a reflexive relation because (a, a), (b, b) and (c, c) are all in the relation. The relation is symmetric since there is no (x, y), an ordered pair with two distinct first and second elements, to search for its reverse. We make similar deductions with respect to antisymmetric and transitive properties. Since the definitions of antisymmetric and transitive properties are not violated, both properties hold.

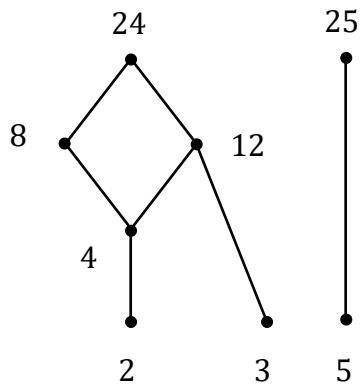
R_1 is reflexive because $x^2 - x^2 = 0 \leq 3$ holds for all $x \in R$. $1^2 - 3^2 = -8 \leq 3$ and therefore $(1,3) \in R_1$, and $(3,1) \notin R_1$ since $3^2 - 1^2 = 8 \nleq 3$. This counter example reveals that R_1 is not symmetric relation. Here is a counter example showing that R_1 is not antisymmetric relation: $(1,2) \in R_1$ because $1^2 - 2^2 = -3 \leq 3$ and $(2,1) \in R_1$ because $2^2 - 1^2 = 3 \leq 3$. On the other hand, $(2,1), (1,0) \in R_1$ and $(2,0) \notin R_1$ is a counter example showing that it is not transitive relation. Thus, R_1 is reflexive but it is not symmetric, antisymmetric or transitive.

None of the given relations is equivalence relation. For instance, relation described in case a) is not reflexive since , $x \ngtr x$ for all $x \in A$. Reflexivity property is violated for the one described in b) as, for example, $0 + 0 \neq 4$. In case c) symmetry property is violated as 2 divides 4, but not vice versa. Finally, the one described in case d) is not reflexive as division by 0 is not defined.

R_1 is reflexive since $x^2 = x^2$ for all $x \in R$. It is a symmetric relation because if $x^2 = y^2$ then $y^2 = x^2$. Finally, R_1 is transitive relation since $x^2 = y^2$ and $y^2 = z^2$ implies $x^2 = z^2$. Here is example, $3^2 = (-3)^2$ and $(-3)^2 = 3^2$ implies $3^2 = 3^2$ is a correct statement. This relation is not antisymmetric relation. Here is a counter example: both (-3,3) and (3,-3) are the elements of R_1.

R is reflexive since $(a, a), (b, b), (c, c), (d, d)$ and (e, e) are all elements of R. It is symmetric relation since (b, e) and (e, b); (d, a) and (a, d) are elements in R. Because of the same reason R is not antisymmetric. It is transitive relation because for (b, e) and (e, b) the resulting ordered pair, (b, b), is also element of R. This is equally true for the couples (e, b) and (b, e); (d, a) and (a, d); (a, d) and (d, a). As a result, R is reflexive, symmetric and transitive relation.

Corresponding Hasse diagram is shown below.



$$\{(a, a), (a, b), (a, c), (a, d), (b, b), (b, c), (b, d), (c, c), (c, d), (d, d)\}$$

$T = \{(a, b) | a + b \leq 3\}$ is the right choice because for $1 + 1 \leq 3$, $1 + 2 \leq 3$, $2 + 1 \leq 3$ and $1 - 1 \leq 3$. Hence, $(1,1), (1,2), (2,1)$ and $(1, -1)$ are all elements of T.

R is reflexive because $a \sim a$ for all citizens a of EU. If a and b are citizens of the same EU country then $a \sim b$ and $b \sim a$ and, therefore, R is symmetric. Because of the same reason R is not antisymmetric. If $a \sim b$ and $b \sim c$ then a, b and c are citizens of the same EU country, from what we conclude that $a \sim c$. Thus, R is transitive. Thus, R is reflexive, symmetric and transitive relation.

$R = \{(1,1), (1,5), (2,2), (2,6), (3,3), (3,7), (4,4), (5,5), (5,1), (6,6), (6,2), (7,7), (7,3)\}$. We scan the relation from left to right and find that, it is reflexive since $(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)$ and $(7,7)$ are present in R. It is symmetric since couples (1,5) and (5,1); (2,6) and (6,2); and (3,7) and (7,3) are all present in R. It is not antisymmetric because of the same observations. Finally, it is transitive because (1,5), (5,1) and (1,1); (2,6), (6,2) and (2,2); (3,7), (7,3) and (3,3) and their reverses are all present in R. From what we conclude that R is equivalence relation, but not partial order.

Because (1,1), (2,2), (3,3) are all present in R, it is reflexive relation. Then we observe that (1,2) and its reverse (2,1) as well as (3,2) and its reverse (2,3) are both present in R. From what we conclude that it is symmetric, but not antisymmetric relation. After that, we find (1,2) and (2,3) are present but the resulting ordered pair (1,3) is not. The above counter example shows that R is not transitive.

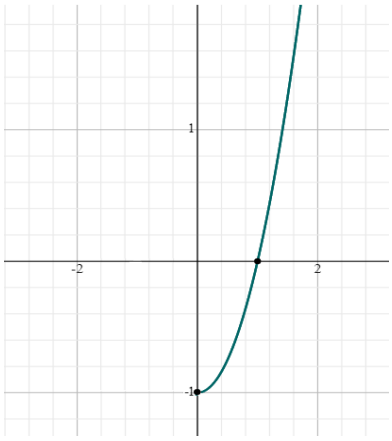
$\{(1,1), (2,2), (3,3)\}$ is the right choice because it is both symmetric and antisymmetric in addition to the fact that it is reflexive and transitive.

Let a, b and c be three arbitrary people. "is a sibling of" is reflexive since trivial pairing (a, a) for all a in the set of all people is present in the relation. It is symmetric, but not antisymmetric, because as far as a and b are siblings both (a, b) and (b, a) are present in the relation. This is transitive relation because if (a, b) and (b, c) are present in the relation then a, b and c are siblings and consequently (a, c) is also present in the relation.

mRn is reflexive because $gcd(n,n) = n > 1$ holds for $n > 2$. It is symmetric, but not antisymmetric, because $gcd(m,n) > 1$ implies $gcd(n,m) > 1$. On the other hand, mRn and nRk do not imply mRk. Here is a counter example, $gcd(4,12) = 4 > 1$, $gcd(12,3) = 3 > 1$ but $gcd(4,3) = 1 \not> 1$.

Chapter 3

rng $f = [-1, \infty)$
$(f \circ g)(x) = f\big(g(x)\big) = (x-2)^2 + 3.$
$(g \circ f)(x) = g\big(f(x)\big) = \frac{2}{(x-2)^2+1}$ and $f^{-1}(x) = x + 2$.
$f^{-1}\big(f(100)\big)$ =100.
$(g \circ f)(x) = g\big(f(x)\big) = \sqrt{4x^2} + 1$; dom $(g \circ f) = [0, \infty)$; cod $(g \circ f) = R$; rng $(g \circ f) = [1, \infty)$.
$s(x) = (x-4)^3$ is one-to-one and onto and, therefore, invertible function from R to R.
A linear function from R to R is injective and surjective and therefore bijective function. Quadratic, ceiling and absolute value functions from R to R are neither injective nor surjective and thus definitely not bijective. Hence, $h(x) = -3x + 4$ is the right choice,
$\{2,4,8\} \neq \{2,4,6,8\}$ and thus rng $A \neq$ codomain. This is not onto function.
$f\big(g(x)\big) = 2(x^2 + 7) + 3 = 2x^2 + 17 = 25$, from which we find that $x = \pm 2$.
Restricting domain of f to the set of nonnegative real numbers and its codomain to $[-1, \infty)$, eliminates the left wing of the parabola and what remains is just its right wing. This function is injective, surjective and thus bijective.



Dom $f = R\backslash\{4\}$, codom $f = R$, rng $f = R\backslash\{0\}$, $f^{-1}(x) = \frac{1}{x} + 4$.
Cubic function from R to R is one-to-one, while quadratic, ceiling and absolute value functions are not. Thus, the right choice is $f(x) = 9x^3 + 1$.
This is a function with restricted domain and codomain. Graph of the function is a fragment of the right wing of concave-up parabola. This function is bijective.

Chapter 4

"If I knew that you know then I would tell him" is the right choice as this is the only declarative sentence among proposed choice.
"$2 + 4 = 5$" is the right choice as it is the only declarative sentence among all choices.
"The only even prime number is 5" is the only declarative sentence among proposed sentences and, thus, this is the right choice.

"$(3+17)/2 > 10$", "Eucalyptus is a tree" and "Milk is black" are declarative sentences and, therefore, each of them is a proposition.
$P \leftrightarrow Q$ is logically equivalent to $\sim P \leftrightarrow \sim Q$.
This is truth table of $\leftrightarrow$ logic connective.
"I understand and I don't understand" is a contradiction.
The proposition can be represented by $(p \to q)(p \to q)$ which is logically equivalent to $(p \leftrightarrow q)$. The latter is neither a tautology nor a contradiction.
Using connectives of logic, the sentence can be expressed as $\big((A \vee \sim A) \wedge A\big) \to A$, which is a tautology.

A	$\sim A$	$(A \vee \sim A)$	$(A \vee \sim A) \wedge A$	$\big((A \vee \sim A) \wedge A\big) \to A$
F	T	T	F	T
T	F	T	T	T

$\big((q \to \sim p) \leftrightarrow (p \leftrightarrow q)\big)$ is WFP.
$(A \wedge B) \vee (A \wedge C)$ is not WFP.
$\big((\sim q \wedge p) \vee (p \vee \sim p)\big)$ is a tautology.
$\sim (p \vee q) \wedge (p \wedge q)$ is a contradiction.

$$(p \vee \sim q) \wedge (q \vee T)$$

Chapter 5

a) 1 b) 1 c) 0 d) 0

$$(x + \bar{y}) \cdot 0$$
$$x + y \cdot 0$$
$$(\bar{x} + 0) \cdot \bar{y}z$$
$$Q = \bar{A}B + AC + \bar{B}C$$
$$Q = AB + (B + C)BC$$
$$f(x,y,z) = (x+y+\bar{z})(\bar{x}+y+z)(\bar{x}+\bar{y}+\bar{z})$$
$$f(x,y,z) = \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy\bar{z}$$
$$\bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy\bar{z}$$

Itemized answers are as follows.

$$\bar{x}\bar{y}\bar{z}w + \bar{x}\bar{y}z\bar{w} + \bar{x}y\bar{z}\bar{w} + \bar{x}y\bar{z}w + \bar{x}yzw + \bar{x}yz\bar{w} + xy\bar{z}\bar{w} + xy\bar{z}w$$
$$+ xyzw + xyz\bar{w} + x\bar{y}\bar{z}\bar{w} + x\bar{y}\bar{z}w + x\bar{y}z\bar{w}$$

	$\bar{z}\bar{w}$	$\bar{z}w$	zw	$z\bar{w}$
$\bar{x}\bar{y}$		1		1
$\bar{x}y$	1	1	1	1
xy	1	1	1	1
$x\bar{y}$	1	1		1

$$y + x\bar{z} + \bar{z}w + z\bar{w}$$

$AB + (B + C)BC$ and $(A + C)B$ are respectively induced by the given logic circuits. By drawing truth tables, one can check that these logic circuits are not equivalent.

By comparing truth-falsity values we can easily see that XNOR gate is equivalent to the given logic diagram.

The simplified functions are as follows:

$$xz$$
$$\bar{y}$$
$$yz$$
$$y$$
$$\bar{y}\bar{z}$$
$$\bar{x}z$$
$$\bar{x}\bar{z}$$
$$z$$
$$\bar{z}$$
$$\bar{x}$$
$$\bar{x}\bar{y}z + xyz$$
$$\bar{x}\bar{y}\bar{z} + x\bar{y}z + \bar{x}yz + xy\bar{z}$$
$$zw + yw + xy\bar{z}$$
$$\bar{x}y + \bar{x}\bar{z} + y\bar{z}\bar{w} + x\bar{y}z\bar{w}$$

$$y\bar{z} + \bar{y}z$$

There exist two solutions: $y\bar{z} + \bar{x}y + \bar{y}w + yz\bar{w}$ and $y\bar{z} + \bar{x}w + \bar{y}w + yz\bar{w}$

Chapter 6

343>27

64>9

We easily handle the base case by observing that for a set with one element, its power set consists of two elements, which are empty set and the set itself. Now assume that $|P(A)| = 2^k$ holds for $|A| = k$. Let us show that $|P(B)| = 2^{k+1}$ holds when $|B| = k + 1$ where $B = A \cup \{a\}$. There are 2^k subsets in B that are also in A. We can form 2^k more subsets by adding a to each subset of A. Hence, $|P(B)| = 2^k + 2^k = 2^{k+1}$.

Base case: By substituting n by 6 in the statement we obtain $5 > 0$, which is correct.

Induction hypothesis: Assume $k^2 - 5k - 1 > 0$ holds.

Induction case: Let us show that then $(k + 1)^2 - 5(k + 1) - 1 > 0$ also holds.

$(k + 1)^2 - 5(k + 1) - 1 > 0$

$k^2 + 2k + 1 - 5k - 5 - 1 > 0$

$(k^2 - 5k - 1) + (2k - 4) > (k^2 - 5k - 1) + 8 > 0$

Due to hypothesis $(k^2 - 5k - 1) > 0$ is correct and, therefore, $(k^2 - 5k - 1) + 8 > 0$ is definitely correct.

Base case: Bu substituting n by 1 in the statement we obtain $1 \leq 3$, which is correct.

Induction hypothesis: Assume that $k \leq 3^k$ for some natural number k.

Induction case: Let us show that then $k + 1 \leq 3^{k+1}$ is also correct.

$k + 1 \leq 3^{k+1}$

$k + 1 \leq 3 \cdot 3^k = 2 \cdot 3^k + 3^k$
$(k + 1) - k \leq (2 \cdot 3^k + 3^k) - 3^k$
After that what remains is $1 \leq 2 \cdot 3^k$, which is correct.
Base case: When $n = 4$ we obtain $4! = 24 > 8$, which is correct.
Induction hypothesis: We suppose that $k! > 2k$ for natural number k.
Induction case: Using hypothesis we will show $(k + 1)! > 2(k + 1)$.
$(k + 1)! > 2(k + 1)$
$(k + 1)k! = k!\,k + k! > 2k + 2$
$(k!\,k + k!) - k! > (2k + 2) - 2k$ (by the hypothesis)
$k!\,k > 2$ holds for $k \geq 4$.
Base case: We substitute n by 1 in $n^3 + 2n$ and obtain 3 which is divisible by 3.
Induction hypothesis: Let $n^3 + 2n$ be divisible by 3 for some natural number k.
Induction case: We will show that then $(n + 1)^3 + 2(n + 1)$ is also divisible by 3.
$(n + 1)^3 + 2(n + 1) = n^3 + 3n^2 + 3n + 1 + 2n + 2 =$
$(n^3 + 2n) + 3(n^2 + n + 1)$.
By hypothesis $(n^3 + 2n)$ is divisible by 3 and $3(n^2 + n + 1)$ is a multiple of 3. Hence, whole expression is divisible by 3.
See solution of Exercise 3.
Base case: We substitute n by 1 in $5^n - 4n - 1$ and obtain 0 which is divisible by 16.
Induction hypothesis: Assume that $5^k - 4k - 1$ is divisible by 16 for some natural number k.
Induction case: Let us show that then $5^{k+1} - 4(k + 1) - 1$ is also divisible by 16.
$5^{k+1} - 4(k + 1) - 1 = 5 \cdot 5^k - 4k - 4 - 1 =$
$(4 + 1) \cdot 5^k - 4k - 4 - 1 =$
$(5^k - 4k - 1) + 4(5^k - 1)$.

By induction hypothesis $(5^k - 4k - 1)$ is divisible by 16 and $(5^k - 1)$ is divisible by 4 (can be shown by induction) and therefore whole expression is divisible by 16.

Base case: We substitute n by 1 in the statement and obtain $3 > 2$ which is correct.
Induction hypothesis: Assume that $3^k > 2k$ for some natural number k.
Induction case: We need to show that then $3^{k+1} > 2(k+1)$ holds.
$3^{k+1} > 2(k+1)$
$3 \cdot 3^k > 2k + 2$
$2 \cdot 3^k + 3^k > 2k + 2$
After taking into account the hypothesis, we obtain $2 \cdot 3^k > 2$ which is correct inequality for natural numbers.
Prove that $2^n \geq n + 10$ for $n \geq 4$.
Base case: When n is 4 the statement turns to $16 \geq 14$ which is just correct.
Induction hypothesis: We suppose that $2^k \geq k + 10$ holds for some natural number k.
Induction case: Let us show that $2^{k+1} \geq k + 11$ holds.
$2^{k+1} \geq k + 11$
$2^k + 2^k \geq (k + 10) + 1$
We subtract 2^k from the left side and $k + 10$ from the right side of the inequality and obtain $2^k \geq 1$, which holds for $k \geq 4$.
Show that $11^n - 4^n$ is divisible by 7 for $n \geq 1$.
Base case: When n is 1, we find that $11^n - 4^n = 7$ which is divisible by 7.
Induction hypothesis: Suppose that $11^k - 4^k$ is divisible by 7 for natural number k.
Induction case: Let us show that then $11^{k+1} - 4^{k+1}$ is divisible by 7.
$11^{k+1} - 4^{k+1}$=$11 \cdot 11^k - 4 \cdot 4^k$=$4(11^k - 4^k) + 7 \cdot 11^k$

$4(11^k - 4^k)$ is divisible by 7 by hypothesis and $7 \cdot 11^k$ is divisible by 7 because it is multiple of 7. By collecting the above two facts we conclude that $11^{k+1} - 4^{k+1}$ is divisible by 7.

Show that $n^2 > n + 1$ for $n \geq 2$.

Base case: When $n = 2$ statement holds as $4 > 3$.

Induction hypothesis: Assume that $k^2 > k + 1$ for some natural k.

Induction case: We must show that then $(k + 1)^2 > k + 2$ also holds.

$(k + 1)^2 > k + 2$

$k^2 + 2k + 1 > k + 2$

$k^2 + (2k + 1) > (k + 1) + 1$

After subtracting k^2 from the left side and $k + 1$ from the right side, we obtain $2k + 1 > 1$ which holds for all natural numbers.

Show that $2^n \geq n^3$ for $n \geq 10$.

Base case: When $n = 10$ we obtain $2^{10} \geq 10^3$, that is $1024 \geq 1000$, which holds.

Induction hypothesis: Assume that $2^k \geq k^3$ holds for some natural number $k \geq 10$.

Induction case: Based on the hypothesis we need to show then $2^{k+1} \geq (k + 1)^3$ also holds. By hypothesis

$2 \cdot 2^k \geq 2 \cdot k^3 = k^3 + k^3 = k^3 + k \cdot k^2 \geq k^3 + 10 \cdot k^2 =$
$k^3 + 3 \cdot k^2 + 3 \cdot k^2 + 4 \cdot k^2 \geq k^3 + 3 \cdot k^2 + 3 \cdot k + 1 =$
$(k + 1)^3$

From what we conclude that $2^{k+1} \geq (k + 1)^3$ is correct inequality.

Fibonacci numbers are generated by the recurrence relation $f_n = f_{n-1} + f_{n-2}$ for $n \geq 3$ where $f_1 = f_2 = 1$. We represent this recurrence relation in more convenient way: $f_{3n} = f_{3n-1} + f_{3n-2}$ for $n \geq 1$ where $f_1 = f_2 = 1$.

Base case: When $n = 1$ we have $f_3 = f_2 + f_1 = 1 + 1 = 2$, that is, the claim holds.

Induction hypothesis: Assume that $f_{3k} = f_{3k-1} + f_{3k-2}$ is even for $k \geq 1$.

Induction case: Based on induction hypothesis we must prove that $f_{3(k+1)} = f_{3(k+1)-1} + f_{3(k+1)-2}$ is even. Since f_{3k} is even, either f_{3k-1} is even and f_{3k-2} is even or both of them are odd. In the first case, proof of the claim is easy. Because f_{3k-1} and f_{3k} are even, $f_{3k+1} = f_{3(k+1)-2}$ is also even as the sum of two even numbers is even. Thereby, $f_{3k+2} = f_{3(k+1)-1}$ is even and, therefore, $f_{3k+3} = f_{3k}$ is even. In the second case, we suppose that f_{3k-1} is odd and f_{3k-2} is odd. Then $f_{3k+1} = f_{3k} + f_{3k-1}$ is odd as the sum of even and odd numbers is odd. Because of the same reason $f_{3k+2} = f_{3k+1} + f_{3k}$ is odd and $f_{3k+3} = f_{3k+2} + f_{3k+1}$ is even as the sum of two even numbers is even. This proves the assertion.

Show hat for $n \in N$

$$\left(1 + \frac{1}{2}\right)^n \geq 1 + \frac{n}{2}.$$

Base case: When $n = 1$ we have $1 + \frac{1}{2} \geq 1 + \frac{1}{2}$ which is true.

Induction hypothesis: Assume that $\left(1 + \frac{1}{2}\right)^k \geq 1 + \frac{k}{2}$ holds for natural number k.

Induction case: Let us show that $\left(1 + \frac{1}{2}\right)^{k+1} \geq 1 + \frac{k+1}{2}$ holds.

$$\left(1 + \frac{1}{2}\right)^{k+1} \geq 1 + \frac{k+1}{2}$$

$$\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{2}\right)^k \geq 1 + \frac{1}{2} + \frac{k}{2}$$

$$\left(1 + \frac{1}{2}\right)^k + \frac{1}{2}\left(1 + \frac{1}{2}\right)^k \geq \left(1 + \frac{k}{2}\right) + \frac{1}{2}$$

We subtract $\left(1 + \frac{1}{2}\right)^k$ from the left side $\left(1 + \frac{k}{2}\right)$ from the right side and obtain

$$\frac{1}{2}\left(1 + \frac{1}{2}\right)^k \geq \frac{1}{2},$$

which is true for natural number k as $\left(1+\frac{1}{2}\right)^k > 1$.

Chapter 7

We are given that $a_2 = a_1 + d = 15$. On the other hand, $a_2 + a_3 + a_4 = (a_1 + d) + (a_1 + 2d) + (a_1 + 3d) = 3a_1 + 6d = 60$. Then we simultaneously solve the system of linear equations

$$\begin{cases} a_1 + d = 15 \\ 3a_1 + 6d = 60 \end{cases}$$

and find that $a_1 = 10, d = 5$.

In this arithmetic sequence $a_1 = 1, d = 1$. $a_{71} = a_1 + 70d = 1 + 70 = 71$. Thus $S_{71} = \frac{a_1+a_{71}}{2}71 = \frac{1+71}{2}71 = 36 \times 71 = 2556$.

$d = a_4 - a_3 = a_5 - a_4 = 7, a_4 = a_1 + 3 \cdot 7 = 10$. Thus $a_1 = -11$, $a_{10} = -11 + 9 \cdot 7 = -11 + 63 = 52$, $S_{10} = \frac{a_1+a_{10}}{2}10 = (-11 + 52)5 = 205$.

$S_5 = \frac{a_1+a_5}{2}5 = \frac{2a_1+4d}{2}5 = \frac{-14+4d}{2}5 = 5$. Then $-14 + 4d = 2$ and $d = 4$.

$S_7 = \frac{a_1+a_7}{2}7 = \frac{2a_1+6d}{2}7 = \frac{2a_1+18}{2}7 = (a_1 + 9)7 = 119$. Then $a_1 = 8$

$a_1 = 2$. $d = a_2 - a_1 = -4$. Then $a_{101} = 2 + 100(-4) = -398$ and $S_{101} = \frac{a_1+a_{101}}{2}101 = \frac{2-398}{2}101 = \frac{-396}{2}101 = -198 \cdot 101 = 19998$.

$$a_n = 2 + 3(n-1) = 401, n = \frac{402}{3} = 134.$$

Find the sum of the first 10 elements of arithmetic sequence whose 1st and 5th elements are -3 and 13, respectively. $a_1 = -3, a_5 = a_1 + 4 \cdot d = -3 + 4 \cdot d = 13$. Hence, $d = 4$. $a_{10} = a_1 + 9 \cdot d = -3 + 9 \cdot 4 = 33$ and $S_{10} = \frac{-3+33}{2}10 = 150$.

The terms of the geometric sequence are in the form of power of 2 as $2^5, 2^8, 2^{11}, \dots, 2^{50} = 2^{5+0\cdot 3}, 2^{5+1\cdot 3}, 2^{5+2\cdot 3}, \dots, 2^{5+k\cdot 3}$. Further we solve $5 + k \cdot 3 = 50$ and find that $k = 15$, thus, there are 16 terms in the given geometric sequence.

$a_1 = -256, r = \frac{128}{-256} = -\frac{1}{2}$,

$$a_7 = a_1 \cdot r^6 = -256 \cdot \left(-\frac{1}{2}\right)^6 = -\frac{2^8}{2^6} = -2^2 = -4.$$

$a_4 = a_1 \cdot r^3 = -2048 \cdot r^3 = 256, r^3 = \frac{256}{-2048} = \frac{2^8}{-2^{11}} = -2^{-3}, r = -2^{-1}$.
The geometric sequence is of the form: $2^5, 2^8, 2^{11}, 2^{14}, \ldots$. We find that $r = 2^3$, $a_{11} = a_1 \cdot r^{10} = 2^5 \cdot 2^{30} = 2^{35}$.
$a_1 = 2^{12}, a_3 = a_1 \cdot r^2 = 2^{12} \cdot r^2 = 2^{10}, r^2 = \frac{2^{10}}{2^{12}} = 2^{-2}, r = 2^{-1}, a_{10} = a_1 \cdot r^9 = 2^{12} \cdot 2^{-9} = 2^3, a_{20} = a_1 \cdot r^{19} = 2^{12} \cdot 2^{-19} = 2^{-7}, S_{12} = \frac{1-2^{-12}}{1-2^{-1}} 2^{12} = (1 - 2^{-12})2^{13} = 2^{13} - 2 = 2(2^{12} - 1)$.

$$r = \frac{a_5}{a_4} = \frac{192}{-96} = -2.$$

$a_1 = 2$, $a_6 = 2r^5 = 2^6, r = 2,\ S_{100} = \frac{1-2^{100}}{1-2} 2 = -2(1 - 2^{100})$.
$a_1 = 4$, $a_2 = a_1 r = 4r = 12$. Thus, $r = 3$. $a_{90} = a_1 r^{89} = 4 \cdot 3^{89}, a_{100} = a_1 r^{99} = 4 \cdot 3^{99}$. $a_{90} a_{100} = 4^2 \cdot 3^{89} \cdot 3^{99} = 4^2 \cdot 3^{188}$.
$r = \frac{a_2}{a_1} = -\frac{5}{10} = -\frac{1}{2}$ and $a_{10} = a_1 r^9 = -5 \cdot \left(-\frac{1}{2}\right)^9 = \frac{5}{512}$.
a) is the right choice. One of two constants in choice b) is a variable; the one represented by choice c) is not linear; recurrence relations represented by choices d) and e) are not second order.
$x^2 + 6x - 5$ is the right choice.
$a_{n-4} = 7a_{n-5} - 11a_{n-6}$ is the right choice.
$a_n = 4a_{n-2}$ is the right choice.
$a_5 = 21$
$a_n = \left(\sqrt{2}\right)^n + \left(-\sqrt{2}\right)^n$.

$$a_n = \frac{1+\sqrt{2}}{2\sqrt{2}}\left(\frac{1+\sqrt{2}}{2}\right)^n - \frac{1-\sqrt{2}}{2\sqrt{2}}\left(\frac{1-\sqrt{2}}{2}\right)^n$$

$$a_n = 3 \cdot 5^n + \frac{2}{5} n \cdot 5^n$$

$$a_n = \frac{5}{3} \cdot 2^n - \frac{1}{3} \cdot 5^n$$

$$\frac{1+\sqrt{5}}{2\sqrt{5}}\left(\sqrt{5}\right)^n-\frac{1-\sqrt{5}}{2\sqrt{5}}\left(-\sqrt{5}\right)^n$$

Chapter 8

$|M| = 60, |Y| = 30, |M \cap Y| = 10,$ $|(M \cup Y)^c| = 100 - |M \cup Y| = 100 - (|M| + |Y| - |M \cap Y|) = 100 - (60 + 30 - 10) = 100 - 80 = 20$.

$|C| = 19, |B| = 11,\ 31 - |C \cup B| = 3,\ |C \cap B| = |C| + |B| - |C \cup B| = 19 + 11 - 28 = 2.$

$|M| = 9, |H| = 9, |P| = 12,\ |M \cap H| = 2, |H \cap P| = 4, |M \cup P| = 3,$ $|M \cap H \cap P| = 1, |U| - |M \cup H \cup P| = 2.$ $|M \cup H \cup P| = (9 + 9 + 12) - (2 + 3 + 4) + 1 = 22, n = |U| = |M \cup H \cup P| + 2 = 24.$

$|F| = 60, |S| = 50, |F \cap S| = 10, |F \cup S| = 60 + 50 - 10 = 100.$

$|W| = 500, |H| = 333, |F| = 200, |W \cap H| = 166, |W \cap F| = 100, |H \cap F| = 66, |W \cap H \cap F| = 33,$ $|W \cup H \cup F| = (500 + 333 + 200) - (166 + 100 + 66) + 33 = 1033 - 332 + 33 = 734.$

$|C| = 120, |A| = 100, |C \cap A| = 70, |C \cup A| = (120 + 100) - 70 = 150,\ |U| - |M \cup H \cup P| = 250 - 150 = 100.$

$|D| = 7, |R| = 8, |L| = 10, |D \cap R| = 3, |D \cap L| = 4, |R \cap L| = 5, |D \cap R \cap L| = 3. |D \cup R \cup L| = (7 + 8 + 10) - (3 + 4 + 5) + 3 = 25 - 12 + 3 = 16, |U| - |D \cup R \cup L| = 20 - 16 = 4.$

$|I| = 166, |E| = 142, |I \cap E| = 23. |I| - |I \cap E| = 166 - 23 = 143. |I \cup E| = (166 + 142) - 23 = 308 - 23 = 285.$

$|H| = 75, |T| = 80.$

$$\begin{aligned} |H^c \cap T^c| = |(H \cup T)^c| &= 100 - |H \cup T| \\ &= 100 - (|H| + |T| - |H \cap T|) \\ &= 100 - (75 + 80 - |H \cap T|) = |H \cap T| - 55. \end{aligned}$$

Since $|H \cap T| \leq |H| = 75,$ $|H^c \cap T^c| = |H \cap T| - 55 \leq 20.$ This number is as large as 20.

Let M, P and C be the sets of students registered for Mathematics, Physics and Chemistry courses, respectively. By the inclusion-exclusion principle,

$|(M \cup P \cup C)^c| = |U| - (|M| + |P| + |C|) + (|M \cap P| + |M \cap C| + |P \cap C|) - |M \cap P \cap C| = 37 - (18 + 16 + 17) + (7 + 6 + 5) - 2 = 37 - 51 + 18 - 2 = 2.$

Chapter 9

The solution is by addition rule. The possible choices are:

Choice 1: 3+1+1

Choice 2: 2+2+1

Choice 3: 2+1+2

Choice 4: 1+2+2

Choice 5: 1+1+3

Choice 6: 1+3+1

Thus, there are six ways to obtain a total of five on three dice.

The solution is by multiplication rule as follows: $26 \times 10 \times 26$.

$\binom{21}{10} = \frac{21!}{11!10!}$.

$$\binom{12}{8}\binom{9}{2} + \binom{12}{9}\binom{9}{1} + \binom{12}{10}.$$

$$\binom{2}{1}\binom{4}{1}\binom{6}{1} = 2 \cdot 4 \cdot 6 = 48.$$

The solution is by combination with repetition formula:

$\binom{5+3-1}{3} = \frac{7!}{4!\,3!}$

$$P(20,2) = \frac{20!}{18!} = 380.$$

The solution is by principle of inclusion-exclusion for two sets and combination formula. It is easier to first count the number of forbidden shuffles. We have two different types of forbidden arrangements. The number of decks with A♡ on top of K♡ is 51!, since we can consider A♡ on top of K♡ as one object and shuffle the 51 different objects. In the same way, we have 51! forbidden decks

involving K♠ on top of A♠. Finally, we need to count the number of elements in the intersection, i.e., decks where both of the forbidden configurations (A♡ on top of K♡ and K♠ on top of A♠) occur. This gives 50! shuffles. The number of forbidden configurations is therefore 51! + 51! − 50!, and the total number of good decks is 52! − 2 × 51! + 50!.

There are $\binom{3}{2}\binom{12}{2}$ combination of with two non-spade queens and exactly two spades of other kinds; $\binom{3}{1}\binom{12}{2}$ combinations of queen of spades, a non-spade queen and two spades of other types; $\binom{3}{1}\binom{12}{1}\binom{36}{1}$ combinations of queen of spades, a non-spade queen and a non-spade cards. There are altogether $\binom{3}{2}\binom{12}{2}+\binom{3}{1}\binom{12}{2}+\binom{3}{1}\binom{12}{1}\binom{36}{1}$ combinations of 4 cards of desired type.

The solution is by multiplication rule and combination formula:

$$\binom{12}{3}\binom{40}{2}=\frac{12!}{9!\,3!}\cdot\frac{40!}{38!\,2!}$$

The solution is by combination with repetition formula:

$$\binom{4+3-1}{3}=\frac{6!}{3!\,3!}=20.$$

The solution is by pigeonhole principle: $= 101, n = 3 \cdot 101 + 1 = 304$

.

$P(40,11) = \frac{40!}{29!}$.

The solution is by combination formula and addition rule:

$$\binom{6}{3}+\binom{7}{3}+\binom{8}{3}=\frac{6!}{3!\,3!}+\frac{7!}{4!\,3!}+\frac{8!}{5!\,3!}=20+35+56=111.$$

$$|A\cup K|=|A|+|K|-|A\cap K|=\binom{4}{3}\binom{48}{2}+\binom{4}{2}\binom{48}{3}-\binom{4}{3}\binom{4}{2}.$$

The solution is by r-permutation formula:

$$P(5,5)=5!=120$$
$$P(4,4)=4!=24$$
$$P(4,4)=4!=24$$
$$3!\cdot 2=12$$

$$4! = 24$$

$$\binom{7}{1}\binom{5}{2} + \binom{7}{2}\binom{5}{1} + \binom{7}{3} = \frac{7!}{6!\,1!} \cdot \frac{5!}{3!\,2!} + \frac{7!}{5!\,2!} \cdot \frac{5!}{4!\,1!} + \frac{7!}{4!\,3!}$$
$$= 7 \cdot 10 + 21 \cdot 5 + 35 = 70 + 105 + 35 = 210.$$

The solution is by multiplication rule: $9 \cdot 8 \cdot 7 = 504$.

The solution is by permuting indistinguishable objects formula: $\frac{6!}{2!2!2!} = 90$.

The solution is by strong form of pigeonhole principle:

$m = 7, n = 68, \left\lceil \frac{68}{7} \right\rceil = 10.$

It is addition rule that we use to find the answer. We show below all possible cases.

$2 + 6 = 8$
$3 + 5 = 8$
$4 + 4 = 8$
$5 + 3 = 8$
$6 + 2 = 8$

The answer is 5.

We observe that there are 6 choices for the first position (4 to 9 numbers) of a 3-digit integer to be filled by a number. For each of these choices there are 9 more choices for the second position (0 to 9 numbers excluding number used in the first position) to be filled. Finally, there are 8 more choices for the last position (0 to 9 numbers excluding numbers used in the first and second positions). According to multiplication rule, there are $6 \cdot 9 \cdot 8 = 432$ such numbers.

According to strong form of the pigeonhole principle $m = 101$ and $n = 3 \cdot 101 + 1 = 304$.

$$m = 366, \left\lceil \frac{1473}{366} \right\rceil = 5$$
$$m = 2, n = 16 + 2 = 18.$$

It is combinations and multiplication rule that we use find the answer.

$\binom{12}{1}\binom{7}{1}\binom{5}{1} = 12 \cdot 7 \cdot 5 = 420.$

We apply pigeonhole principle as follows to find the solution.

$m = 6, n = 2 \cdot 6 + 1 \cdot 5 + 1 = 18.$

It is combination with repetition formula, which we apply to find the solution.

$\binom{n+r-1}{r} = \frac{(n+r-1)!}{(n-1)!r!}.$

It is combination with repetition formula when $n = 3$ and $r = 6$, which we apply to find the solution.

$\binom{3+6-1}{6} = \frac{8!}{2!6!} = 28.$

It is combination with repetition formula when $n = 5$ and $r = 3$, which we apply to find the solution.

$\binom{5+3-1}{3} = \frac{7!}{4!3!} = 35.$

Let A be the set of teams of size 11 that include one of the boys and let B be the set of teams of size 11 that include the second of these boys. Using the principle of inclusion-exclusion we find the number of teams that include one or none of these two boys as follows:

$$|A \cup B| - |A \cap B| = |A| + |B| - 2|A \cap B| = 2\binom{39}{10} - 2\binom{38}{9}.$$

It is pigeonhole principle that we use find the solution.

$m = 5, n = 3 \cdot 5 + 1 = 16.$

The solution is by multiplication rule and derangement formula as follows:

$\binom{5}{2} D_3 = 10 \cdot 2 = 20$

We need to find the number of ordered samples with repetition. There are 5 choices for the first marble to be put and for each of these choices there are 5 more choices the second marble to be put and so for the third marble, and thus $5 \cdot 5 \cdot 5 = 125.$

The solution is by combination with repetition formula as follows:

$\frac{6!}{3!\,3!} = 20$

It is permutation of indistinguishable objects formula, which we use to find the solution.

$\binom{10}{5,1,3,1} = \frac{10!}{5!\,1!\,3!\,1!}$

It is permutation of indistinguishable objects formula, which we use to find the solution.

$\binom{8}{2,3,1,2} = \frac{8!}{2!\,3!\,1!\,2!}$

The solution is by permutation of indistinguishable objects formula as follows

$\binom{7}{2,3,1,2} = \frac{7!}{2!\,3!\,1!\,2!}$

The solution is by permutation of indistinguishable objects formula as follows

$\binom{8}{2,2,2,2} = \frac{8!}{2!\,2!\,2!\,2!}$

It is combination formula, derangement formula and multiplication rule, that we use to find the solution.

$\binom{5}{2} \cdot D_3 = 10 \cdot 2 = 20.$

The solution is due to combination formula, derangement formula and multiplication rule as follows:

$\binom{10}{4} \cdot D_6 = 210 \cdot 265 = 55650.$

99

$$\binom{15}{8} 3^7 4^8 = \frac{15!}{7!\,8!} 3^7 4^8$$

$\binom{5}{2} (x^3)^3 (2x^{-1})^2 = 40x^7$. The coefficient of x^7 is 40.

$\binom{20}{15} (x^2)^5 (2y^3)^{15} = \binom{20}{15} 2^{15} x^{10} y^{45}$. The coefficient of $x^{10}y^{45}$ is $\binom{20}{15} 2^{15}$.

$x = \binom{10}{5} = 252.$

Chapter 10

There are six complete bipartite graphs having 60 edges: $K_{1,60}$, $K_{2,30}$, $K_{3,20}$, $K_{4,15}$, $K_{5,12}$, $K_{6,10}$, that are composed of 61, 32, 23, 19, 17, 16 vertices, respectively. Thus, the right answer choice is 16.

$\frac{1}{2}[n(n-1)-k(k-1)-l(l-1)]=\frac{1}{2}[(k+l)(k+l-1)-k(k-1)-l(l-1)]=kl.$

$K_{50,100}$ is complete bipartite graph with $|V_1|=50, |V_2|=100$ and, thus, $|V_1|+|V_2|=150$ and $|E|=|V_1|\cdot|V_2|=5000$.

In complete graph K_n, $|E|=\frac{(n-1)n}{2}$. Further, we solve $(n-1)n=110$ and find that $n=|V|=11$. Sum of degrees of all vertices is twice more than number of edges, that is, $2|E|=2\cdot 55=110$.

$n-1$

In complete graph K_n, $|E|=\frac{(n-1)n}{2}$. Further, we solve $(n-1)n=42$ and find that $n=|V|=7$.

Let us suppose that such an arrangement is possible. This can be viewed as a graph in which telephones are represented using vertices and wires using the edges. Now we have 25 vertices in this graph. The degree of each vertex in the graph is 7. We need to understand that an edge connects two vertices. Thus, the sum of degrees of all the vertices is equal to twice the number of edges. Therefore, $25\times 7=2|E|$ from where we obtain $|E|=87{,}5$ which is not possible. Hence, it is not possible.

$|E|=56, \sum_{v\in V}\deg(v)=2|E|=112.$

$n(n-1)=42, n=7$

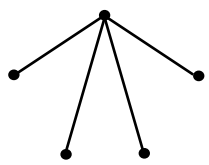
If G is a complete graph, then $n(n-1)=30, n=6$. If G is a complete bipartite graph, then $|E|=n\cdot m=3\cdot 5=15, |V|=n+m=8$.

This problem seems very difficult initially. If we try to approach this problem by using line segments as edges of a graph, we seem to

reach nowhere, which sounds confusing initially. Here we need to consider a graph where each line segment is represented as a vertex. Now two vertices of this graph are connected if the corresponding line segments intersect. Now this graph has 9 vertices. The degree of each vertex is 3. We know that for a graph sum of degrees of all vertices is twice of number of edges in the graph. Since the sum of degrees of vertices in the above problem is $9 \times 3 = 27$ i.e odd, such an arrangement is not possible.

Let us suppose that such an arrangement is possible. This can be viewed as a graph in which telephones are represented using vertices and wires using the edges. Now we have 25 vertices in this graph. The degree of each vertex in the graph is 7. We need to understand that an edge connects two vertices. So, the sum of degrees of all the vertices is equal to twice the number of edges. Therefore, $25 \times 7 = 2 \times |E|$ and $|E| = 25 \times 7 / 2 = 87.5$. This is not an integer. As a result, we can conclude that our supposition is wrong and such an arrangement is not possible.

This a connected graph and a tree as shown below.



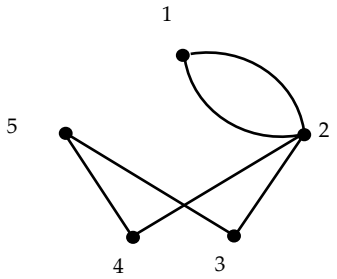
In the leftmost graph, deg $v = 2$, while in other two graphs, deg $v = 3$. The leftmost graph is not isomorphic to remaining graphs, while the remaining two graphs are isomorphic. Here is example of isomorphic mapping.

$\varphi(9) = 17, \varphi(10) = 23, \varphi(11) = 21, \varphi(12) = 19,$
$\varphi(13) = 18, \varphi(14) = 20, \varphi(15) = 22, \varphi(16) = 24.$

Let us show that the leftmost graph is not isomorphic with the graph in the middle. We match triangles in the former graph with

corresponding triangles in the latter one. Image of the triangle consisting of a, b, g and corresponding incident edges could be triangle consisting of $h, k,\ l$ and edges incident with these vertices. Image of the second triangle with a in it is a, g, f together with corresponding incident edges. Image of the second triangle could be triangle formed by h, k, n. The third triangle with a in it is the one composed of a, b, c whose image is the triangle formed by h, l, i. Notice that there is one more mapping of three triangles of one graph into their counterparts in another graph, but these two mappings are symmetrical and thus second mapping adds nothing new to already described procedure. We observe that f and c are adjacent with the same vertex, which is e, while their counterparts i and n are not adjacent with the same vertex. This is contradiction, which reveals that the leftmost graph and the graph in the middle are not isomorphic. Using similar technique, which based on mapping of triangles, we can show that none of the given graphs is isomorphic to any other.

Below we illustrate corresponding pseudograph. The problem is about whether the graph is Hamiltonian. Vertex 1 is connected to the subgraph composed of vertices 3,4 and 5 through vertex 2 and, therefore, when touring all vertices 1 and 2 must be visited two times. This is violation of Hamiltonian condition and, thereby, this pseudograph is not Hamiltonian.



All rooms have even number of doors, meaning that every vertex in corresponding pseudograph is even. Due to Euler theorem on Eulerian circles, this is a Eulerian pseudograph and therefore has no Eulerian trail. On the other hand, this pseudograph has Hamiltonian cycle. Here is an example of Hamiltonian cycle:

1-2-3-4-5-6-7-8-9-10-1

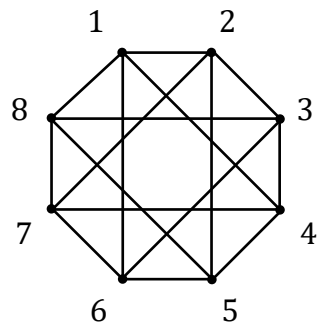
The graph has a Eulerian circuit because its vertices are even and because of the same reason the graph has no Eulerian trail. In the given circular arrangement of vertices, vertex v is adjacent to $next(next(next(v)))$ in clockwise (or anticlockwise) direction. A tour in any direction visits all the vertices and ends up at the vertex the tour stated with and therefore the graph has a Hamiltonian cycle. Due to the theorem on Eulerian circuits, this is a Eulerian graph because each vertex is even. This is a Hamiltonian graph. Here is example of Hamiltonian cycle:

1-*A* -2-*B* -3-*C* -4-*D* -5-*E* -6-*F* -7-*G* -8-*H* -9-*I* -1

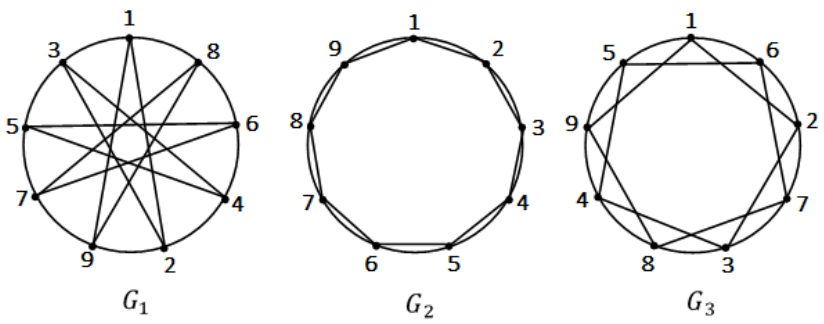
The solution is due to two theorems by Euler. The degree sequence is 8, 6, 5, 4, 4, 3. There does not exist a Eulerian circuit, but there is an Eulerian trail because the graph has exactly two odd vertices, that are D and E. An example of Eulerian trail:

D-C-A-C-F-A-F-B-F-E-B-A-D-E-A-E

We observe that this is 2-colorable graph. If 1 is red then 2,4,6 and 8 are blue and continuing this logic we find that bipartition sets are $V_1 = \{1, 3, 5, 7\}$ and $V_2 = \{2, 4, 6, 8\}$.



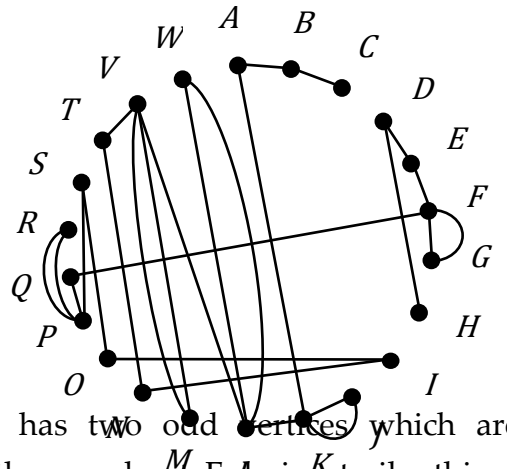
These are graphs on 9 vertices and 18 edges. None of these graphs is isomorphic to another. In each graph, starting vertex 1 we can define two Hamiltonian cycles: one over cycle while another by visiting the vertices in the order 1,2,3, ... ,9. If any two of the graphs were isomorphic both Hamiltonian cycles on one graph would be matching with Hamiltonian cycles of another graph. Let us now try to match Hamiltonian cycles on which vertices are visited in the order 1,2,3, ... ,9, that is, we choose the function $\varphi(k) = k$ for $k = 1, \ldots, 9$ to relate the vertices of any two graph. When we relate vertices in this way, exactly one Hamiltonian cycle of each graph matches with that of another, but not the second ones. This approves the fact that these graphs are not isomorphic.



The question is about whether graph is Hamiltonian. Corresponding graph is Hamiltonian. Here is an example of a Hamiltonian cycle:

A-B-C-G-H-F-E-D-A

Corresponding pseudograph is shown below.



This pseudograph has two odd vertices which are C and H. According to the therem about Eulerian trails, this pseudograph contains an Eulerian trail. Here is example of an Eulerian trail:

C-B-A-K-J-K-L-W-L-V-M-V-T-N-I-O-S-P-R-P-Q-F-G-F-E-D-H

Since the preudograph contains odd vertices it is not an Eulerian pseudograph. On the other hand, this pseudograph does not contain a Hamiltonian cycle. This is because the pseudograph has two vertices of degree 1, which are C and H, and any Hamiltonian tour on the prseudograph that starts at a vertex different than C and H must end up at one of these two vertices; and a tour which starts at C or H cannot end up at the same vertex. Consequently, this is not Hamiltonian pseudograph,

Below we provide trace of Prim's algorithm for the given graph.

1	Gazimagusa – Bafra	19
2	Yeni Erenköy – Dıpkarpaz	21
3	Bafra – Yeni Erenköy	21

4	Tatlısu – Yeni Erenköy	23
5	Girne – Lefkoşa	26
6	Girne – Tatlısu	35
7	Lefkoşa – Güzelyurt	40
Sum of the weights		185

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